

Supplemental Appendix for *Accuracy Limits of Causal Trees for Individualized Treatment Effects*

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SA-1 Overview and Omitted Details

This supplement contains the proofs for the results in the main paper, together with additional theoretical results and omitted details. We start with regression estimation in Section SA-2, showing that the standard CART decision tree estimator of a constant conditional mean suffers from slow uniform convergence rates. In Section SA-3, we study the causal effect estimators discussed in the main paper: the inverse probability weighting (IPW), difference in means (DIM), and sum of squared errors (SSE) estimators are considered in Sections SA-3.1, SA-3.2, and SA-3.3, respectively. The main paper results are connected to this supplement as follows.

- **Proof of Theorem 1:** The conclusions follow from Corollary SA-10, Corollary SA-12, Theorem SA-18, Theorem SA-20, Theorem SA-27, and Theorem SA-29. The deeper NSS-DIM and NSS-SSE conclusions use the depth condition stated in Theorem 1 of the main paper; the IPW, honest, and depth one source results do not. The sharper probability constant $1/e$ for no sample splitting is available for the IPW transformed outcome tree and for the depth one DIM result. For SSE, the split location result has the same $1/e$ split index constant, but transferring a large bivariate SSE split fluctuation to the terminal CATE contrast uses an additional directional transfer constant. For deeper NSS-DIM and NSS-SSE trees, later recursive refinements of the selected root child introduce treatment fraction remainder terms, so the final theorem records a generic positive probability constant for those cases.
- **Proof of Theorem 2:** The conclusions follow from Corollary SA-11, Corollary SA-13, Theorem SA-19, Theorem SA-21, Theorem SA-28, and Theorem SA-30.
- **Correction to Eicker (1979):** The correction to Eicker [1979, Theorem 5], including the corrected Darling–Erdős result used in the proofs, is stated and proved in Section SA-4.

In Theorem 1 of the main paper, the constants for no sample splitting are denoted by C_1, C_2 and the honest constants by C_3, C_4 . The statements below use local constant notation for each estimator, often again denoted C_1, C_2 ; when those statements concern honest estimators, their local constants correspond to the constants C_3, C_4 in the main paper. The only cases in which we retain the sharper $1/e$ probability factor for the CATE error lower bound are the transformed outcome IPW tree and the depth one DIM statement. For SSE, the split index theorem has the sharp $1/e$ probability factor, but the CATE error lower bound uses a generic positive directional transfer factor. The deeper NSS-DIM and NSS-SSE results use the refinement transfer with random arm weights in Lemma SA-40, together with the depth condition stated in Theorem 1 of the main paper.

In addition, this supplement presents the following additional results and omitted details.

- Probability analogues of the integrated mean squared error bounds in Theorem 2; see Section SA-1.3.
- Technical details on the proof strategy for Theorem 1, regularization and small cells, α regularity and small subgroups, the JTPA resampling design, and companion numerical figures; see Sections SA-1.2, SA-1.4, SA-1.5, SA-1.6, and SA-1.7.
- Regression and causal tree unbiasedness results, together with squared T statistic estimators; see Sections SA-2.3, SA-3.4, and SA-3.5.

SA-1.1 Notation

The following notation is used throughout the supplement.

- **Sets.** $\mathbb{R}$ is the set of real numbers and $\mathbb{N}$ the positive integers. For $n \in \mathbb{N}$ we write $[n] = \{1, \dots, n\}$.
- **Vectors and matrices.** Boldface lowercase letters (e.g. $\mathbf{x}$) denote column vectors, and boldface uppercase letters (e.g. $\mathbf{A}$) denote matrices. For a vector $\mathbf{x}$, its i -th component is x_i ; for a matrix $\mathbf{A}$, its (i, j) -th entry is A_{ij} . Denote by $\mathbf{e}_j$ the j -th unit vector.
- **Norms.** For $\mathbf{x} \in \mathbb{R}^d$, define $\|\mathbf{x}\| = (\sum_{i=1}^d x_i^2)^{1/2}$, and $\|\mathbf{x}\|_\infty = \max_{i \leq d} |x_i|$. For a matrix $A \in \mathbb{R}^{m \times n}$, the operator norm is $\|A\| = \sup_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$, and the max norm is $\|A\|_{\max} = \max_{1 \leq i \leq m, 1 \leq j \leq n} |A_{ij}|$. For a bounded measurable function g , $\|g\|_\infty = \sup_x |g(x)|$. For the covariate vector $\mathbf{x}_i$, let P_X denote its marginal distribution and let F_X denote its distribution function. For a random variable Z with distribution P_Z , denote the population L_2 norm by $\|Z\| = (\int \|z\|^2 dP_Z(z))^{1/2}$; and given a random sample $\mathcal{D} = \{Z_1, \dots, Z_n\}$, denote the empirical L_2 norm by $\|Z\|_{\mathcal{D}} = (n^{-1} \sum_{i=1}^n \|Z_i\|^2)^{1/2}$.

- **Asymptotics.** For real sequences, $a_n \ll b_n$ (or $a_n = o(b_n)$) means $\limsup_{n \rightarrow \infty} |a_n|/|b_n| = 0$, while $|a_n| \lesssim |b_n|$ (or $a_n = O(b_n)$) means there exist constants C and $N > 0$ such that $|a_n| \leq C|b_n|$ for all $n > N$. For sequences of random variables, $a_n = o_{\mathbb{P}}(b_n)$ means $\text{plim}_{n \rightarrow \infty} |a_n|/|b_n| = 0$, while $|a_n| \lesssim_{\mathbb{P}} |b_n|$ means $\limsup_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(|a_n/b_n| \geq M) = 0$. Throughout the supplement, the covariate dimension p is fixed. Generic positive constants may therefore depend on p , in addition to the distributional quantities explicitly listed in each statement; sharp constants such as the displayed $1/e$ factors are stated separately.
- **Other.** $\mathbf{1}(\cdot)$ denotes the indicator function. For two random variables X and Y , $X \perp\!\!\!\perp Y$ means X and Y are independent. For $x \in \mathbb{R}$, $\lfloor x \rfloor$ and $\lceil x \rceil$ denote the floor and ceiling of x , respectively. $\mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\text{Beta}(\alpha, \beta)$ denotes the Beta distribution with parameters (α, β) . A stochastic process $\{B(t), 0 \leq t \leq 1\}$ is a Brownian bridge if B is a continuous Gaussian process with $\mathbb{E}[B(t)] = 0$ and $\mathbb{E}[B(t)B(s)] = \min\{t, s\} - ts$. When real sequences such as r_n or s_n are used as split index cutoffs, inequalities such as $r_n \leq k \leq n - r_n$ mean $\lceil r_n \rceil \leq k \leq n - \lceil r_n \rceil$; this harmless rounding convention is suppressed below. When a split index optimization is written as an optimization over $k \in [n]$, the endpoint $k = n$ is not a valid split and the optimization is understood over the valid split indices $1 \leq k < n$, together with any validity restrictions specific to the estimator.
- **Boundary sequences.** Given $a \in (0, 1)$ and a deterministic sequence $\eta_n \downarrow 0$, define

$$\mathcal{X}_n(a, \eta_n) = \{\mathbf{x} \in [0, 1]^p : x_j \leq \eta_n n^{a-1} \text{ or } 1 - x_j \leq \eta_n n^{a-1} \text{ for some } j \in [p]\}.$$

SA-1.2 Proof Strategy of Theorem 1

The key idea behind the proof is that greedy recursive partitioning tends to select highly imbalanced splits with probability bounded away from zero. In particular, even in the simplest case of a decision stump (a tree of depth one), the optimal split often occurs near the boundary of the covariate space, producing child nodes with very small sample sizes. These small cells lead to large estimation variance in some regions of $\mathcal{X}$, which ultimately prevents causal tree estimators from achieving polynomial uniform convergence rates.

Underlying our theoretical results are several technical properties of decision stumps, and hence trees of depth one. For each tree splitting criterion and sample splitting design, we first analyze the probabilistic behavior of the split location at the root node. This analysis characterizes the regions of the covariate space $\mathcal{X}$ where the first split is most likely to occur and determines the effective sample sizes of the resulting child nodes.

Our results show that, with probability bounded away from zero, the optimal split concentrates near the boundary of the parent node (a cell in the partition of $\mathcal{X}$). As a consequence, one of the child nodes may contain only a very small number of observations. This phenomenon arises at the very first step of the recursive tree construction and ultimately drives the slow uniform convergence rate. More precisely, let $\hat{i} = n(\mathbf{t}_L)$ and $\hat{j}$ be the CART split index and split variable at the root node, respectively, for $l \in \{\text{IPW}, \text{DIM}, \text{SSE}\}$. The construction without sample splitting and the honest construction use the same root split criterion; in the honest case, that criterion is computed on the construction fold, whose sample size is comparable to n . For each $a, b \in (0, 1)$ with $a < b$ and $j \in \{1, 2, \dots, p\}$, we establish that

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i} \leq n^b, \hat{j} = j) \geq \frac{b-a}{2pe}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i} \leq n - n^a, \hat{j} = j) \geq \frac{b-a}{2pe}. \quad (\text{SA-1})$$

Thus the left and right boundary split regions each have probability bounded away from zero for every coordinate.

The slow uniform convergence rate of the decision stump estimator arises because the optimal split tends to concentrate near the boundary of the support, producing highly imbalanced partitions. In such cases, one child node contains only a small number of construction observations, making the corresponding local average estimator highly variable. Relation (SA-1) quantifies this phenomenon: for each coordinate $j = 1, \dots, p$ and each $b \in (0, 1)$, summing the two boundary events and letting $a \downarrow 0$ gives probability at least $b/(pe)$ that one of the child cells $\{\mathbf{x} \in \mathcal{X} : x_j \leq \hat{\xi}\}$ or $\{\mathbf{x} \in \mathcal{X} : x_j > \hat{\xi}\}$ is highly anisotropic and contains at most n^b construction observations. Consequently, with probability bounded away from zero, the estimator exhibits arbitrarily slow convergence in some region of $\mathcal{X}$. For deeper NSS trees, these insights extend through the terminal descendants of the imbalanced root child. For ordinary regression and IPW transformed outcome trees, the relevant root child average is exactly a convex combination of terminal descendant averages. For deeper NSS-DIM and NSS-SSE trees, a refinement transfer argument based on the random treatment-arm weights shows that the same rate is inherited by a terminal descendant under the moderate depth condition in Theorem 1 of the main paper.

For honest trees, the selected construction-fold small cell is evaluated on an independent estimation fold; an occupancy argument and finite-arm anti-concentration transfer the construction-fold small-cell event to the final honest estimator.

The core of the proof studies the tree construction as the maximizer of the split criteria in (SA-7) and (SA-8), indexed by the split location and the covariate coordinate. The analysis relies on nonasymptotic high-dimensional central limit theorems, Gaussian comparison inequalities, Gaussian process embeddings, Darling–Erdős-type extreme value theory, and empirical process techniques [El-Yaniv and Pechyony, 2009, Petrov, 2007, Shorack and Smythe, 1976, Skorski, 2023]. The argument proceeds in four main steps.

Step 1: Split Criterion Approximation. Using empirical process techniques, we establish an asymptotic equivalence between the split criterion underlying each causal tree estimator and the split criterion of a standard regression tree employing CART. For $l = \text{DIM}$ and $l = \text{IPW}$, this corresponds to a regression tree applied to the transformed outcomes $y_i \frac{d_i - \xi}{\xi(1 - \xi)}$. For $l = \text{SSE}$, after centering within treatment arms, the approximating process is the sum of two independent split criterion processes, one based on $\frac{d_i}{\xi} \varepsilon_i(1)$ for treated units and one based on $\frac{1 - d_i}{1 - \xi} \varepsilon_i(0)$ for control units; the arm-specific constants cancel from the split contrasts within each treatment arm. A truncation argument removes extremely small or large split indices where empirical process approximations are less reliable [Csörgő and Horváth, 1997, Theorem A.4.1].

Step 2: Conditional Gaussian Approximation. Conditional on the ordering of the covariates, the square root of the split criterion process can be approximated by a Gaussian process with the same conditional covariance structure. For $l = \text{IPW}$ and $l = \text{DIM}$, the split criterion can be written as a sum of i.i.d. high-dimensional random vectors indexed by split location and coordinate. Applying the high-dimensional central limit theorem of [Chernozhukov et al., 2017, Theorem 2.1], we obtain a Gaussian approximation conditional on the ordering. Because of the structure of the split criterion, a high-dimensional central limit theory over hyperrectangles suffices. For $l = \text{SSE}$, the treated and control components are stacked into a higher-dimensional vector and a central limit theory for convex sets [Chernozhukov et al., 2017, Proposition 3.1] is employed.

Step 3: Unconditional Gaussian Approximation. When $p > 1$, different covariate coordinates induce different orderings of the observations. We therefore show that the conditional Gaussian process from Step 2 is close to an unconditional Gaussian process in which splits across different coordinates are asymptotically uncorrelated. This implies asymptotic independence of the corresponding subprocesses and reduces the problem to studying the maximization of the split criterion along a single coordinate. The approximation is established using a Gaussian comparison inequality [Chernozhukov et al., 2022, Proposition 2.1] together with bounds on the difference between the conditional and unconditional covariance matrices. For $l = \text{IPW}$ and $l = \text{DIM}$, the argument follows directly from a high-dimensional central limit theory for hyperrectangles. For $l = \text{SSE}$, additional approximation error is controlled using Nazarov’s inequality [Nazarov, 2003].

Step 4: Lower bound on imbalanced split probability. The unconditional Gaussian processes obtained in Step 3 correspond to the squared norm of a univariate ($l \in \{\text{IPW}, \text{DIM}\}$) or a weighted quadratic form of a bivariate ($l = \text{SSE}$) Ornstein–Uhlenbeck process, with a one-to-one transformation between split index for the tree and time for the O-U process [Csörgő and Révész, 1981, Göing-Jaeschke and Yor, 2003]. The Darling–Erdős theorem [Eicker, 1979, Horváth, 1993] and its weighted quadratic form analogue [Zhdanov, 2022] then characterize the distribution of the maximum of this process over an interval. Combining this result with the Gaussian correlation inequality [Latała and Matlak, 2017, Remark 3(i)] yields the lower bound in (SA-1), which in turn determines the effective sample sizes of the child nodes.

The remaining arguments use these insights either by tracking terminal descendants of the imbalanced root child for NSS estimators or, for honest estimators, by conditioning on the selected partition and using the independent estimation fold. The descendant-tracking step is exact for regression and transformed outcome IPW averages, and it uses the random treatment-arm weights for deeper NSS-DIM and NSS-SSE trees.

SA-1.3 Probability Bounds for Integrated Error

Theorem 2 in the main text reports expectation bounds for integrated mean squared error. The probability analogues are stated estimator by estimator in the results below and hold under the same assumptions as Theorem 2; in particular, K is the same depth upper bound as in Theorem 2 and the constants do not depend on K or n . For no sample splitting, Corollary SA-11, Theorem SA-19, and Theorem SA-28 imply that, for

$l \in \{\text{IPW}, \text{DIM}, \text{SSE}\},$

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} |\hat{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C_l \frac{2^K \log^4(n) \log(np)}{n} \right) = 0,$$

where C_l depends only on the distributional quantities appearing in the corresponding result. For honest sample splitting, Corollary SA-13, Theorem SA-21, and Theorem SA-30 imply that

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} |\check{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C_l \frac{2^K \log^5(n)}{n} \right) = 0,$$

provided that $\rho \leq n_{\tau}/n_{\tau} \leq \rho^{-1}$ for some $\rho \in (0, 1)$, as in Theorem 2; in this honest case, C_l may also depend on ρ . Consequently, the bounds imply integrated L_2 consistency along any sequence $K = K_n$ satisfying

$$\frac{2^{K_n} \log^4(n) \log(np)}{n} \rightarrow 0 \quad \text{for no sample splitting,} \quad \frac{2^{K_n} \log^5(n)}{n} \rightarrow 0 \quad \text{for honest sample splitting.}$$

SA-1.4 Regularization and Small Cells

Minimum node-size constraints and related penalties can reduce the variance created by small cells, but they can also change the adaptive behavior of the procedure. Such constraints are common in causal tree implementations [Athey and Imbens, 2016, Online Appendix] and related software [Athey and Imbens, 2026, Xu et al., 2026]. Adaptive trees create small cells for two reasons. First, local refinement can be useful when the conditional mean or treatment effect has sharp structure. Second, sampling noise can generate highly imbalanced splits even when the target function is locally flat. In applications, these two mechanisms are difficult to distinguish from the data alone.

For example, the `causalTree` documentation defines `minsize` as requiring at least `minsize` treated and `minsize` control observations in each leaf, with default `minsize=2` and example code setting `minsize=20` [Athey and Imbens, 2016, Online Appendix]; see also Athey and Imbens [2026]. The `htetree` package uses default `minsize=20` in its high-level causal tree, IPW, matching, and forest wrappers [Xu et al., 2026]. These implementation choices illustrate how minimum node-size regularization enters applied causal tree procedures through user-facing tuning parameters.

The distinction matters because a rule that removes small cells may also remove splits that help the tree recover useful structure after an earlier poor split. For example, consider the threshold CATE function

$$\tau(\mathbf{x}) = \mathbf{1}(x_1 \leq 1/2).$$

Suppose the tree first makes a poor split on an irrelevant coordinate, say $x_2 = 0.05$. This split carries no information about the treatment effect, but it leaves most observations in one descendant node, including the signal region $\{x_1 \leq 1/2\}$. The algorithm can still split this large descendant node and recover the threshold in x_1 . A minimum node-size requirement can force the first poor split to divide the sample more evenly, leaving less sample size on every downstream branch. Thus, imbalanced splits can be harmful for pointwise estimation and still play a useful algorithmic role in deeper trees [Ishwaran, 2015].

SA-1.5 α Regularity and Small Subgroups

α regularity requires every split to leave at least an α fraction of the parent node in each child node. This condition is used in positive analyses of honest causal trees and forests [Wager and Athey, 2018] and related forest methods [Meinshausen, 2006, Athey et al., 2019, Behr et al., 2022]. It rules out the small cells that drive the lower bounds in the main text, but it can also limit the ability of a tree to isolate small subgroups.

To see this, consider

$$\tau(\mathbf{x}) = \mathbf{1}(x_1 \leq \eta), \quad \eta \in (0, 1),$$

with $\mathbf{x} = (x_1, \dots, x_p)^\top$ uniformly distributed on $[0, 1]^p$. At the population level, an unrestricted CART procedure can recover this function with a single split on x_1 at $x_1 = \eta$. If $\eta < \alpha$, an α -regular tree cannot make that split at the root. It must instead approximate the threshold through a sequence of finer splits, requiring at least

$$\left\lceil \log_{1/\alpha}(1/\eta) \right\rceil$$

successive splits along x_1 . This reduces parsimony and weakens variable selection adaptivity. In finite samples, especially when p is large relative to n , repeated refinement along one coordinate creates additional chances for spurious splits on irrelevant covariates. Balanced trees may still approximate such structures in integrated loss, but the resulting partitions need not recover the underlying subgroup itself.

SA-1.6 JTPA Resampling Design

The empirical resampling exercise in the main text uses data from the National Job Training Partnership Act Study. Applicants were randomly assigned either to a program group eligible for JTPA services or to a control group that was not eligible for those services for 18 months after assignment, with later employment and earnings measured using followup surveys and administrative records [Bloom et al., 1997].

The resampling exercise preserves empirical features of the JTPA covariates but removes the original covariate-outcome association within each treatment arm. We first preprocess the covariate support to reduce the influence of large mass points in preprogram earnings. After restricting attention to the target range of X_1 , we group observations by their exact X_1 value and randomly thin each mass point without replacement. In the one-covariate design, if a mass point contains n_g observations, we retain $\min\{\max(1, \lfloor 0.5n_g \rfloor), 30\}$ units. In the two-covariate design, we retain $\min\{\max(1, \lfloor 0.2n_g \rfloor), 200\}$ units.

For the Monte Carlo resampling step, outcomes Y and treatment assignments D are held fixed, and covariates are permuted without replacement within each treatment arm. In the one-covariate design, X_1 is permuted separately among treated and control units. In the two-covariate design, the pair (X_1, X_2) is permuted jointly within each treatment arm. This construction preserves the within-arm marginal distribution of X_1 in the one-covariate design and the within-arm joint distribution of (X_1, X_2) in the two-covariate design. The permutation step removes the original association between covariates and outcomes within each treatment arm.

SA-1.7 Companion One-Dimensional Numerical Figures

Figures 1 and 2 report the one-dimensional companion designs for the numerical evidence in the main text. They display the same qualitative pattern as the bivariate figures: pointwise RMSE is lowest in the interior of the support and rises near regions where adaptive splitting creates terminal cells with limited empirical support.

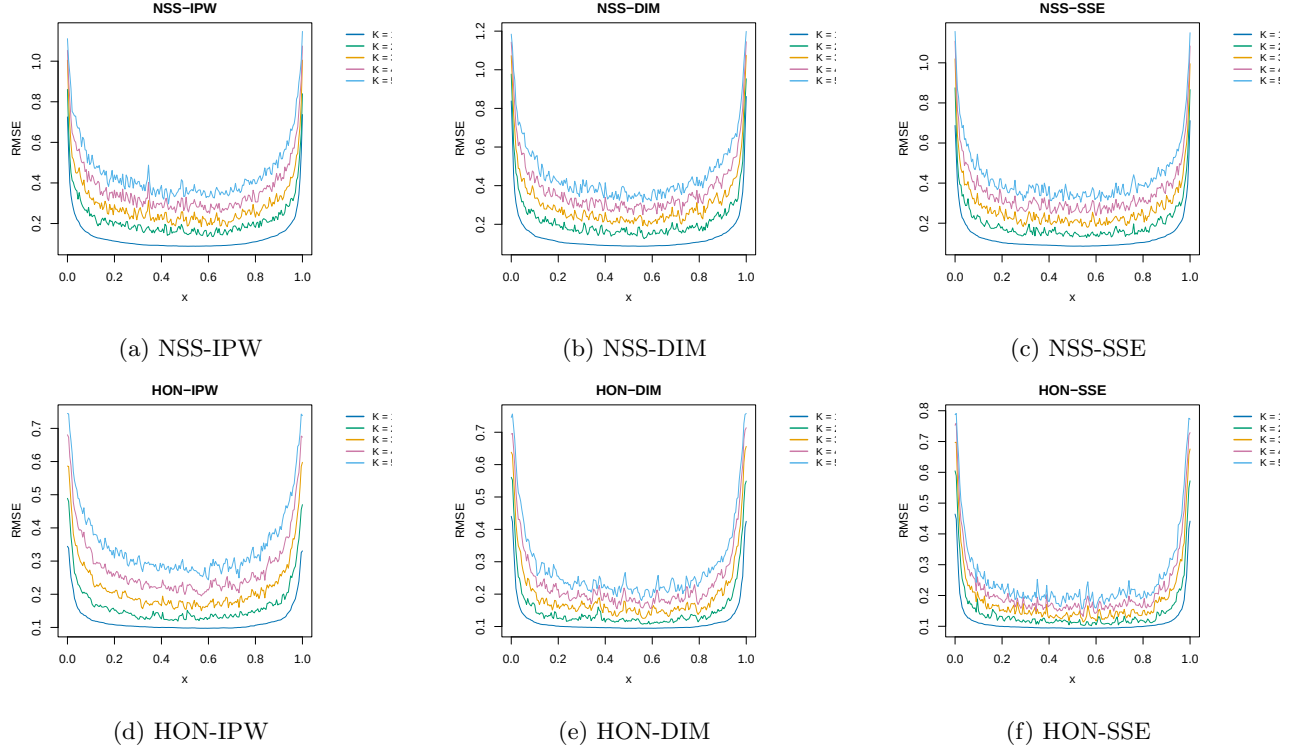


Figure 1: Synthetic Monte Carlo evidence for the univariate design ($p = 1$). Each panel reports pointwise root mean squared error (RMSE) over the covariate support $[0, 1]$ for tree depths $K = 1, 2, \dots, 5$. Rows compare estimators without sample splitting (NSS) and with honest sample splitting (HON); columns compare inverse probability weighting (IPW), difference in means (DIM), and squared error (SSE) splitting criteria. RMSE is smallest in the interior of the support and rises near the boundary, matching the small cell mechanism in the theory. Results are based on 2,000 Monte Carlo replications.

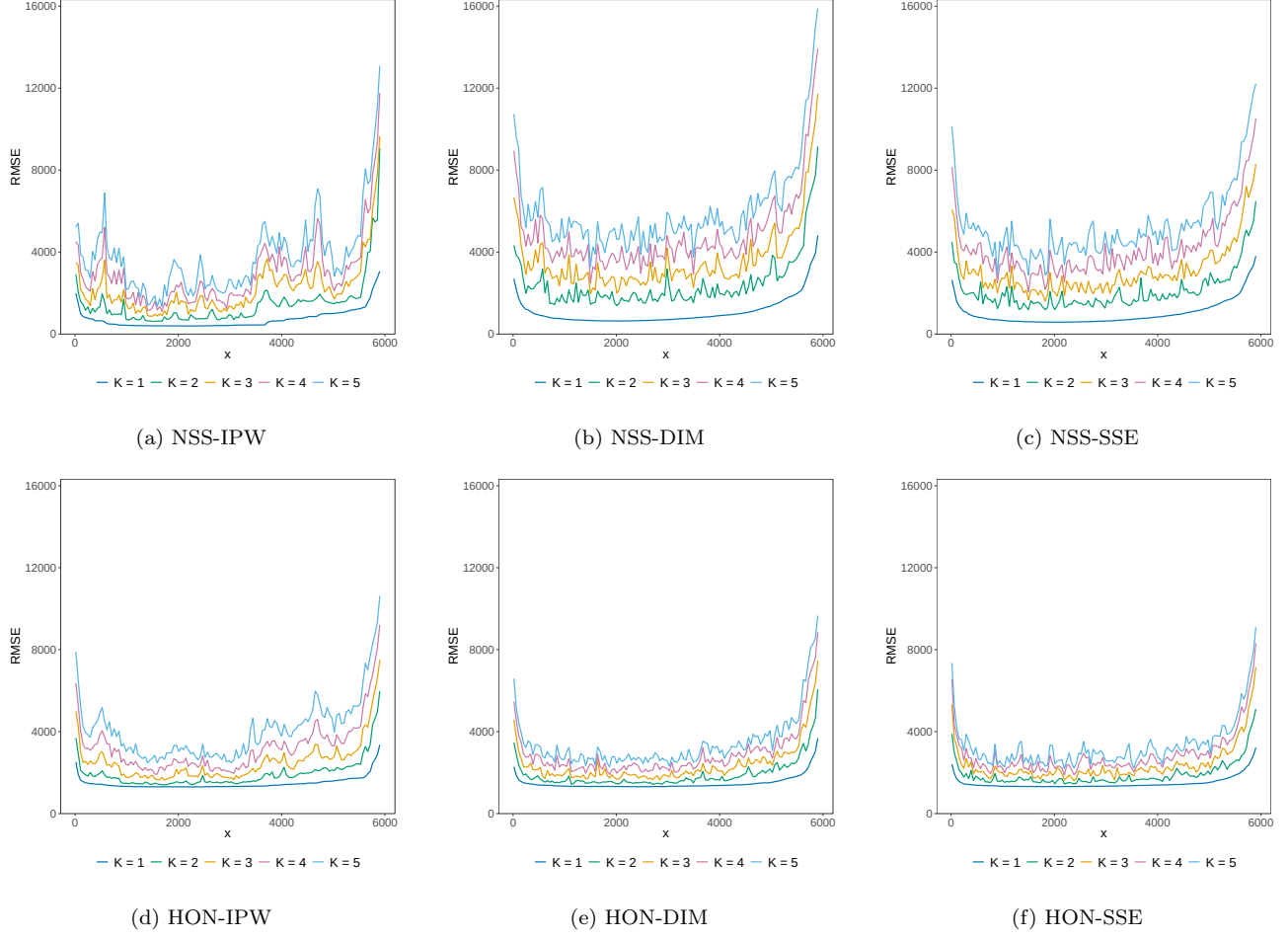


Figure 2: Empirical JTPA resampling evidence for the one-covariate design ($p = 1$). Panels report pointwise RMSE over the empirical covariate support after permuting the covariate within treatment arms, for depths $K = 1, \dots, 5$. Rows compare NSS and HON; columns compare IPW, DIM, and SSE. RMSE rises in regions with limited empirical support, matching the small-cell pattern in the simulations.

SA-2 Regression Estimation

This section is self contained, and substantially improves on the results reported in [Cattaneo et al. \[2022\]](#). The results presented herein are of independent interest in regression estimation settings, and they also offer a gentle introduction to the more technically involved results discussed in [Section SA-3](#).

These regression results also clarify the relationship between our framework and earlier large-sample analyses of CART decision stumps. In the univariate setting, [Bühlmann and Yu \[2002\]](#) and [Banerjee and McKeague \[2007\]](#) studied empirical split minimizers and established cube-root type behavior under smoothness and identification conditions on the regression function. Our lower bounds concern a different, but complementary, regime: the constant regression function belongs to standard smoothness classes, but it removes the population-level signal that identifies an interior split. In that no-signal case, the empirical split criterion is driven by stochastic fluctuations, and the selected split can concentrate near the boundary with nonvanishing probability. Thus the results below do not contradict the classical cube-root analyses; instead, they show why the identification conditions in those analyses are essential and why uniform guarantees over broad function classes cannot be inferred from pointwise analyses away from flat cases.

Consider the canonical regression model where the observed data $\{(y_i, \mathbf{x}_i^T) : i = 1, 2, \dots, n\}$ is a random sample satisfying

$$y_i = \mu(\mathbf{x}_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | \mathbf{x}_i] = 0, \quad \mathbb{E}[\varepsilon_i^2 | \mathbf{x}_i] = \sigma^2(\mathbf{x}_i), \quad (\text{SA-2})$$

with $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T$ a vector of p covariates taking values on some support set $\mathcal{X}$.

Assumption SA-1 (Location Regression Model). $\mathcal{D} = \{(y_i, \mathbf{x}_i^T) : 1 \leq i \leq n\}$ is generated by i.i.d. latent pairs $(\mathbf{x}_i, \varepsilon_i)$ satisfying Equation (SA-2) and the following conditions for all $i = 1, 2, \dots, n$:

1. $y_i = \mu(\mathbf{x}_i) + \varepsilon_i$, with $\mathbb{E}[\varepsilon_i | \mathbf{x}_i] = 0$ and $\mathbf{x}_i \perp \varepsilon_i$.
2. $\mu(\mathbf{x}) = c$ for all $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^p$, where c is some constant.
3. $x_{i1}, \dots, x_{ip}$ are independent and continuously distributed.
4. There exists $\alpha > 0$ such that $\mathbb{E}[\exp(\lambda \varepsilon_i)] < \infty$ for all $|\lambda| < 1/\alpha$ and $\sigma^2 = \mathbb{E}[\varepsilon_i^2] > 0$.

Because recursive tree splits are invariant to strictly increasing coordinatewise transformations, statements and proofs involving $\mathcal{X} = [0, 1]^p$ or the lower corner $\mathbf{0}$ are understood after applying the marginal probability integral transforms to the covariates. Thus, without loss of generality, the covariates may be treated as uniformly distributed on $[0, 1]^p$ for those arguments. Throughout the asymptotic statements below, the covariate dimension p is treated as fixed. We retain the $\log(np)$ factors in the upper bounds to record the dependence on the number of candidate split coordinates.

To illustrate the CART estimation strategy, given any tree $\mathbf{T}$, the CART estimator is as follows:

Definition SA-1 (CART Estimate). Suppose $\mathbf{T}$ is the tree used, and $\mathcal{D}_\mu = \{(y_i, \mathbf{x}_i^T) : i = 1, 2, \dots, n_\mu\}$, with $n_\mu \leq n$, is the dataset used. Let $\mathbf{t}$ be the unique terminal node in $\mathbf{T}$ containing $\mathbf{x} \in \mathcal{X}$. The CART estimator is

$$\hat{\mu}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\mu) = \frac{1}{n(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} y_i,$$

where $n(\mathbf{t}) = \sum_{i=1}^{n_\mu} \mathbf{1}(\mathbf{x}_i \in \mathbf{t})$ is the “local” sample size. In case $n(\mathbf{t}) = 0$, take $\hat{\mu}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\mu) = 0$.

Definition SA-2 (Tree Construction). Given a dataset $\mathcal{D}_\mathbf{T} = \{(y_i, \mathbf{x}_i^T) : i = 1, 2, \dots, n_\mathbf{T}\}$, with $n_\mathbf{T} \leq n$, a parent node $\mathbf{t}$ in the tree (i.e., a region in $\mathcal{X}$) is divided into two child nodes, $\mathbf{t}_\mathbf{L}$ and $\mathbf{t}_\mathbf{R}$, by minimizing the sum of squared errors (SSE),

$$\min_{1 \leq j \leq p} \min_{\beta_\mathbf{L}, \beta_\mathbf{R}, \varsigma \in \mathbb{R}} \sum_{\mathbf{x}_i \in \mathbf{t}} (y_i - \beta_\mathbf{L} \mathbf{1}(x_{ij} \leq \varsigma) - \beta_\mathbf{R} \mathbf{1}(x_{ij} > \varsigma))^2, \quad (\text{SA-3})$$

where $(\beta_\mathbf{L}, \beta_\mathbf{R}, \varsigma, j)$ denote the two child-node outputs, split point, and split direction, respectively. A candidate split is valid only if both child nodes contain at least one construction-sample observation; invalid candidates are omitted from the optimization. If a parent node has no valid split, it is left terminal. Ties are resolved by a fixed deterministic rule. The resulting CART tree is denoted by $\mathbf{T}(\mathcal{D}_\mathbf{T})$.

Definition SA-3 (Sample Splitting). Recall Definition SA-1 and Definition SA-2, and that $\mathcal{D} = \{(y_i, \mathbf{x}_i^T) : i = 1, 2, \dots, n\}$ is the available random sample.

- **No Sample Splitting (NSS)**: The dataset $\mathcal{D}$ is used for both the tree construction and the treatment effect estimation, that is, $\mathcal{D}_\mathbf{T} = \mathcal{D}$ and $\mathcal{D}_\mu = \mathcal{D}$. The CART tree estimator is

$$\hat{\mu}^{\text{NSS}}(\mathbf{x}) = \hat{\mu}(\mathbf{x}; \mathbf{T}(\mathcal{D}), \mathcal{D}).$$

- **Honesty (HON)**: The dataset $\mathcal{D}$ is divided into two independent datasets $\mathcal{D}_\mathbf{T}$ and $\mathcal{D}_\mu$ with sample sizes $n_\mathbf{T}$ and n_μ , respectively, and satisfying $n \lesssim n_\mathbf{T}, n_\mu \lesssim n$. The CART tree estimator is

$$\hat{\mu}^{\text{HON}}(\mathbf{x}) = \hat{\mu}(\mathbf{x}; \mathbf{T}(\mathcal{D}_\mathbf{T}), \mathcal{D}_\mu).$$

SA-2.1 No Sample Splitting

We start from the no sample splitting (NSS) case, and characterize the location of the first split.

Decision Stumps

For each variable $j = 1, 2, \dots, p$, let π_j be the permutation such that $x_{\pi_j(i),j}$ is nondecreasing in the index $i = 1, 2, \dots, n$. Then, minimizing Equation (SA-3) can equivalently be recast as maximizing the *impurity gain*:

$$\begin{aligned} & \sum_{\mathbf{x}_i \in \mathbf{t}} (y_i - \bar{y}_{\mathbf{t}})^2 - \sum_{\mathbf{x}_i \in \mathbf{t}} (y_i - \bar{y}_{\mathbf{t}_L} \mathbf{1}(\mathbf{x}_i \in \mathbf{t}_L) - \bar{y}_{\mathbf{t}_R} \mathbf{1}(\mathbf{x}_i \in \mathbf{t}_R))^2 \\ &= \frac{\left(\frac{1}{\sqrt{n(\mathbf{t})}} \sum_{\mathbf{x}_i \in \mathbf{t}_L} (y_i - \mu) - \frac{n(\mathbf{t}_L)}{n(\mathbf{t})} \frac{1}{\sqrt{n(\mathbf{t})}} \sum_{\mathbf{x}_i \in \mathbf{t}} (y_i - \mu) \right)^2}{(n(\mathbf{t}_L)/n(\mathbf{t}))(1 - n(\mathbf{t}_L)/n(\mathbf{t}))}, \end{aligned}$$

where $\bar{y}_{\mathbf{t}} = n(\mathbf{t})^{-1} \sum_{\mathbf{x}_i \in \mathbf{t}} y_i \mathbf{1}(\mathbf{x}_i \in \mathbf{t})$. This is also equivalent to maximizing the *conditional variance given the split*:

$$\frac{n(\mathbf{t}_L)n(\mathbf{t}_R)}{n(\mathbf{t})} (\bar{y}_{\mathbf{t}_L} - \bar{y}_{\mathbf{t}_R})^2.$$

We start by considering the case when the tree is depth one ($K = 1$), i.e., a decision stump. Then optimization objectives are equivalent to choosing a splitting coordinate $\hat{j}$, and a splitting index $\hat{i}$ such that

$$\mathbf{t}_L = \{\mathbf{u} \in \mathcal{X} : \mathbf{u}_{\hat{j}} \leq x_{\pi_{\hat{j}}(\hat{i}),\hat{j}}\}, \quad \mathbf{t}_R = \{\mathbf{u} \in \mathcal{X} : \mathbf{u}_{\hat{j}} > x_{\pi_{\hat{j}}(\hat{i}),\hat{j}}\}.$$

The tree output can then be written as

$$\hat{\mu}^{\text{stump}}(\mathbf{x}) = \begin{cases} \bar{y}_{\mathbf{t}_L}, & \mathbf{x} \in \mathbf{t}_L \\ \bar{y}_{\mathbf{t}_R}, & \mathbf{x} \in \mathbf{t}_R \end{cases},$$

where $x_{\hat{j}}$ denotes the value of the $\hat{j}$ -th component of $\mathbf{x}$.

The following theorem characterizes the regions of the support $\mathcal{X}$ where the first CART split index $\hat{i}$, at the root node, has probability bounded away from zero. As a consequence, the theorem also characterizes the effective sample size of the resulting cells (recall the data is ordered so that the split point is $x_{\pi_{\hat{j}}(\hat{i}),\hat{j}}$ and hence $\hat{i} = \#\{\mathbf{x}_i : x_{i\hat{j}} \leq x_{\pi_{\hat{j}}(\hat{i}),\hat{j}}\}$).

Theorem SA-1 (Imbalanced Splits). *Suppose Assumption SA-1 holds, and let $(\hat{i}, \hat{j})$ be the CART split index and split direction at the root node. For each $a, b \in (0, 1)$ with $a < b$, and $\ell \in [p]$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i} \leq n^b, \hat{j} = \ell) \geq \frac{b-a}{2pe}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i} \leq n - n^a, \hat{j} = \ell) \geq \frac{b-a}{2pe}, \quad (\text{SA-4})$$

which implies

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i} \leq n^b) \geq \frac{b-a}{2e}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i} \leq n - n^a) \geq \frac{b-a}{2e}.$$

Theorem SA-2 (Convergence Rates for Decision Stumps). *Suppose Assumption SA-1 holds. Suppose the CART tree has depth $K = 1$. Then for any $a, b \in (0, 1)$ with $a < b$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\mu}^{\text{stump}}(\mathbf{x}) - \mu| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)}\right) \geq \frac{b}{e}, \quad (\text{SA-5})$$

and, under the probability integral transform normalization described after Assumption SA-1, for any deterministic sequence $\eta_n \downarrow 0$,

$$\liminf_{n \rightarrow \infty} \inf_{\mathbf{x} \in \mathcal{X}_n(a, \eta_n)} \mathbb{P}\left(|\hat{\mu}^{\text{stump}}(\mathbf{x}) - \mu| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)}\right) \geq \frac{b-a}{2pe}. \quad (\text{SA-6})$$

Deep Trees

We will show that the imbalanced split issue is inherited from decision stumps by any finite recursive tree with at least one split; the depth may vary with n .

Theorem SA-3 (Convergence Rates for Deep Trees). *Suppose Assumption SA-1 holds, and the CART tree has depth at least one. Then for any $b \in (0, 1)$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\mu}^{\text{NSS}}(\mathbf{x}) - \mu| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)} \right) \geq b/e.$$

Therefore, decision trees grown with CART methodology need not converge faster than any polynomial in n , when uniformity over the full support of the data $\mathcal{X}$, and over possible data generating processes, is of interest.

However, for the L_2 risk we still have the following positive result. This is because the small cells that lead to issues in uniform consistency will have small P_X measure.

Theorem SA-4 (L_2 Convergence Rate for NSS). *Suppose Assumption SA-1 holds. Then for any tree with depth at most K (possibly depending on n),*

$$\mathbb{E} \left[\int_{\mathcal{X}} (\hat{\mu}^{\text{NSS}}(\mathbf{x}) - \mu)^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C \frac{2^K \log^4(n) \log(np)}{n},$$

where C is a positive constant that only depends on the distribution of ε_i . Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} (\hat{\mu}^{\text{NSS}}(\mathbf{x}) - \mu)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^4(n) \log(np)}{n} \right) = 0,$$

where C' is a positive constant that only depends on the distribution of ε_i .

SA-2.2 Sample Splitting

For the sample splitting strategy, we also present a lower bound on uniform consistency and an upper bound on integrated L_2 risk.

Theorem SA-5. *Suppose Assumption SA-1 holds, and the construction fold honest CART tree has at least one split. Then for any $b \in (0, 1)$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\mu}^{\text{HON}}(\mathbf{x}) - \mu| \geq \frac{C_1 \mathbb{E}[|y_i - \mu|]}{n^{b/2}} \right) \geq C_2 \frac{\mathbb{E}[|y_i - \mu|^2]}{\mathbb{V}[y_i]} b,$$

where C_1 and C_2 are positive constants depending only on the distribution of y_i and on the lower and upper limiting ratios of $n_{\top}/n_{\mu}$.

Theorem SA-6 (L_2 Convergence Rate for HON). *Suppose Assumption SA-1 holds. Then for any regression tree with depth at most K (possibly depending on n),*

$$\mathbb{E} \left[\int_{\mathcal{X}} (\hat{\mu}^{\text{HON}}(\mathbf{x}) - \mu)^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C \frac{2^K \log^5(n)}{n},$$

provided $\rho \leq \frac{n_{\top}}{n_{\mu}} \leq \rho^{-1}$ for some $\rho \in (0, 1)$, and C is a positive constant that only depends on the distribution of y_i and ρ . Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} (\hat{\mu}^{\text{HON}}(\mathbf{x}) - \mu)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^5(n)}{n} \right) = 0,$$

where C' is some constant only depending on ρ and the distribution of y_i .

Compared to Theorem SA-3, the lower bound on the LHS of Theorem SA-5 that we characterize has one less $\sqrt{(2 + o(1)) \log \log(n)}$. Compared to Theorem SA-4, the upper bound on the RHS of Theorem SA-6 has $\log(np)$ replaced by $\log(n)$. These changes are due to the sample splitting strategy. In particular, the L_2 convergence rate bounds imply integrated L_2 consistency whenever $2^K \log^4(n) \log(np)/n \rightarrow 0$ for no sample splitting, and whenever $2^K \log^5(n)/n \rightarrow 0$ for honest sample splitting.

SA-2.3 Unbiasedness

The regression CART estimator is unbiased under honest estimation because the terminal averages are computed on outcomes that are independent of the selected partition. For no sample splitting, the same conclusion requires a symmetry condition on the full residual vector, since the partition is selected using the same residuals that enter the terminal average.

Lemma SA-7 (Regression unbiasedness). *Suppose Assumption SA-1 holds, and fix $\mathbf{x} \in \mathcal{X}$. Let $N_{\text{HON}}(\mathbf{x})$ denote the final estimation sample size in the terminal node containing $\mathbf{x}$ for the honest CART estimator. Then*

$$\mathbb{E}[\hat{\mu}^{\text{HON}}(\mathbf{x})] = \mu - \mu \mathbb{P}(N_{\text{HON}}(\mathbf{x}) = 0).$$

For the estimator without sample splitting, if in addition the residual distribution is symmetric about zero, $\varepsilon_i \stackrel{d}{=} -\varepsilon_i$, then

$$\mathbb{E}[\hat{\mu}^{\text{NSS}}(\mathbf{x})] = \mu.$$

SA-3 Causal Effect Estimation

In this section, we consider the heterogeneous causal effect estimation problem from the main paper. The assumptions on the data generating process and the definitions of causal trees are the same as in the main paper. For completeness, we include them here:

Assumption SA-2 (Data Generating Process). $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : 1 \leq i \leq n\}$ is generated by i.i.d. latent vectors $(\mathbf{x}_i, d_i, \varepsilon_i(0), \varepsilon_i(1))$, where $y_i = d_i y_i(1) + (1 - d_i) y_i(0)$, $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})^\top$, and the following conditions hold for all $d = 0, 1$ and $i = 1, 2, \dots, n$.

1. $d_i \perp (\mathbf{x}_i, \varepsilon_i(0), \varepsilon_i(1))$, and $\xi = \mathbb{P}(d_i = 1) \in (0, 1)$.
2. $y_i(d) = \mu_d(\mathbf{x}_i) + \varepsilon_i(d)$, with $\mathbb{E}[\varepsilon_i(d)] = 0$ and $\mathbf{x}_i \perp (\varepsilon_i(0), \varepsilon_i(1))$.
3. $\mu_d(\mathbf{x}) = c_d$ for all $\mathbf{x} \in \mathcal{X}$, where c_d is some constant, and $\mathcal{X}$ is the support of $\mathbf{x}_i$.
4. $x_{i,1}, \dots, x_{i,p}$ are independent and continuously distributed.
5. There exists $\alpha > 0$ such that $\mathbb{E}[\exp(\lambda \varepsilon_i(d))] < \infty$ for all $|\lambda| < 1/\alpha$ and $\mathbb{E}[\varepsilon_i^2(d)] > 0$.

Throughout this causal section, the same coordinatewise marginal probability integral transform normalization described in Section SA-2 is used whenever statements involve $\mathcal{X} = [0, 1]^p$ or the boundary sets $\mathcal{X}_n(a, \eta_m)$. Because treatment assignment and potential outcome errors are independent of the covariates under Assumption SA-2, this monotone reparametrization preserves the tree construction problem and does not change the treatment, outcome error, or CATE components of the model.

The causal tree estimators are constructed as follows.

Definition SA-4 (CATE Estimators). *Suppose T is the tree used, and $\mathcal{D}_\tau = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n_\tau\}$, with $n_\tau \leq n$, is the dataset used. Let $\mathbf{t}$ be the unique terminal node in T containing $\mathbf{x} \in \mathcal{X}$.*

- The Inverse Probability Weighting (IPW) estimator is

$$\hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathsf{T}, \mathcal{D}_\tau) = \frac{1}{n(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} \frac{d_i - \xi}{\xi(1 - \xi)} y_i,$$

where $n(\mathbf{t}) = n_0(\mathbf{t}) + n_1(\mathbf{t}) = \sum_{i=1}^{n_\tau} \mathbf{1}(\mathbf{x}_i \in \mathbf{t})$ is the “local” sample size. In case $n(\mathbf{t}) = 0$, take $\hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathsf{T}, \mathcal{D}_\tau) = 0$.

- The Difference in Means (DIM) estimator is

$$\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathsf{T}, \mathcal{D}_\tau) = \frac{1}{n_1(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} d_i y_i - \frac{1}{n_0(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} (1 - d_i) y_i,$$

where $n_d(\mathbf{t}) = \sum_{i=1}^{n_\tau} \mathbf{1}(\mathbf{x}_i \in \mathbf{t}, d_i = d)$, for $d = 0, 1$, are the “local” sample sizes. In case $n_0(\mathbf{t}) = 0$ or $n_1(\mathbf{t}) = 0$, take $\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathsf{T}, \mathcal{D}_\tau) = 0$.

Definition SA-5 (Tree Construction). Suppose $\mathcal{D}_T = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n_T\}$, with $n_T \leq n$, is the dataset used to construct the tree T . Throughout this definition, a candidate split is valid only if both child nodes contain at least one construction-sample observation. When a split criterion uses treatment-arm means or node-specific treatment coefficients, the candidate is valid only if the corresponding treated and control denominators in both child nodes are positive; otherwise the candidate is omitted from the optimization. If a parent node has no valid split, it is left terminal. Ties are resolved by a fixed deterministic rule.

- *Variance Maximization*: A parent node $\mathbf{t}$ (i.e., a terminal node partitioning $\mathcal{X}$) in a previous tree T' is divided into two child nodes, $\mathbf{t}_L$ and $\mathbf{t}_R$, forming the new tree T , by maximizing

$$\frac{n(\mathbf{t}_L)n(\mathbf{t}_R)}{n(\mathbf{t})} \left(\hat{\eta}_l(\mathbf{t}_L; T, \mathcal{D}_T) - \hat{\eta}_l(\mathbf{t}_R; T, \mathcal{D}_T) \right)^2, \quad l \in \{\text{IPW}, \text{DIM}\}. \quad (\text{SA-7})$$

The resulting causal trees are denoted by $T_{\text{IPW}}(\mathcal{D}_T)$ and $T_{\text{DIM}}(\mathcal{D}_T)$, respectively, for $l \in \{\text{IPW}, \text{DIM}\}$.

- *SSE Minimization*: A parent node $\mathbf{t}$ (i.e., a terminal node partitioning $\mathcal{X}$) in the previous tree T' is divided into two child nodes, $\mathbf{t}_L$ and $\mathbf{t}_R$, forming the next tree T , by solving

$$\min_{a_L, b_L, a_R, b_R \in \mathbb{R}} \sum_{\mathbf{x}_i \in \mathbf{t}_L} (y_i - a_L - b_L d_i)^2 + \sum_{\mathbf{x}_i \in \mathbf{t}_R} (y_i - a_R - b_R d_i)^2, \quad (\text{SA-8})$$

where only the data $\mathcal{D}_T$ is used. The resulting causal tree is denoted by $T_{\text{SSE}}(\mathcal{D}_T)$.

Definition SA-6 (Sample Splitting and Estimators). Recall Definition SA-4 and Definition SA-5, and that $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n\}$ is the available random sample.

- *No Sample Splitting (NSS)*: The dataset $\mathcal{D}$ is used for both the tree construction and the treatment effect estimation, that is, $\mathcal{D}_T = \mathcal{D}$ and $\mathcal{D}_\tau = \mathcal{D}$. The causal tree estimators are

$$\begin{aligned} \hat{\eta}_{\text{IPW}}(\mathbf{x}) &= \hat{\eta}_{\text{IPW}}(\mathbf{x}; T_{\text{IPW}}(\mathcal{D}), \mathcal{D}), \\ \hat{\eta}_{\text{DIM}}(\mathbf{x}) &= \hat{\eta}_{\text{DIM}}(\mathbf{x}; T_{\text{DIM}}(\mathcal{D}), \mathcal{D}), \quad \text{and} \\ \hat{\eta}_{\text{SSE}}(\mathbf{x}) &= \hat{\eta}_{\text{DIM}}(\mathbf{x}; T_{\text{SSE}}(\mathcal{D}), \mathcal{D}), \end{aligned}$$

- *Honesty (HON)*: The dataset $\mathcal{D}$ is divided into two independent datasets $\mathcal{D}_T$ and $\mathcal{D}_\tau$ with sample sizes n_T and n_τ , respectively, and satisfying $n \lesssim n_T, n_\tau \lesssim n$. The causal tree estimators are

$$\begin{aligned} \check{\eta}_{\text{IPW}}(\mathbf{x}) &= \hat{\eta}_{\text{IPW}}(\mathbf{x}; T_{\text{IPW}}(\mathcal{D}_T), \mathcal{D}_\tau), \\ \check{\eta}_{\text{DIM}}(\mathbf{x}) &= \hat{\eta}_{\text{DIM}}(\mathbf{x}; T_{\text{DIM}}(\mathcal{D}_T), \mathcal{D}_\tau), \quad \text{and} \\ \check{\eta}_{\text{SSE}}(\mathbf{x}) &= \hat{\eta}_{\text{DIM}}(\mathbf{x}; T_{\text{SSE}}(\mathcal{D}_T), \mathcal{D}_\tau). \end{aligned}$$

While the estimators $\hat{\eta}_l(\mathbf{x})$ and $\check{\eta}_l(\mathbf{x})$, $l \in \{\text{IPW}, \text{DIM}, \text{SSE}\}$, depend on the depth of the tree construction used, our notation keeps this dependence implicit except in the results whose conclusions require an explicit depth restriction. The integrated risk statements keep the depth K explicit.

For later use, whenever a sample rectangle or tree node $\mathbf{t}$ satisfies $n_0(\mathbf{t}) \wedge n_1(\mathbf{t}) > 0$, write

$$\bar{\varepsilon}_{1,\mathbf{t}} = \frac{1}{n_1(\mathbf{t})} \sum_{\mathbf{x}_i \in \mathbf{t}} d_i \varepsilon_i(1), \quad \bar{\varepsilon}_{0,\mathbf{t}} = \frac{1}{n_0(\mathbf{t})} \sum_{\mathbf{x}_i \in \mathbf{t}} (1 - d_i) \varepsilon_i(0),$$

and

$$\Delta(\mathbf{t}) = \bar{\varepsilon}_{1,\mathbf{t}} - \bar{\varepsilon}_{0,\mathbf{t}}.$$

SA-3.1 IPW Estimator

The transformed outcomes $y_i \frac{d_i - \xi}{\xi(1 - \xi)}$, $1 \leq i \leq n$, are i.i.d. with

$$\mathbb{E} \left[y_i \frac{d_i - \xi}{\xi(1 - \xi)} \middle| \mathbf{x}_i \right] = \mathbb{E}[y_i(1) - y_i(0) | \mathbf{x}_i] = c_1 - c_0,$$

and

$$\tilde{\varepsilon}_i = y_i \frac{d_i - \xi}{\xi(1 - \xi)} - (c_1 - c_0) = (c_1 + \varepsilon_i(1)) \frac{d_i}{\xi} - (c_0 + \varepsilon_i(0)) \frac{1 - d_i}{1 - \xi} - (c_1 - c_0) \perp \mathbf{x}_i.$$

Assumption SA-2 implies $\mathbb{E}[\exp(\lambda \tilde{\varepsilon}_i)] < \infty$ for all $|\lambda| \leq 1/\beta$ with β only depending on ξ and α , and $\mathbb{E}[\tilde{\varepsilon}_i^2] > 0$. Hence the following results are immediate corollaries from the results in Section SA-2.

SA-3.1.1 No Sample Splitting

Corollary SA-8 (Imbalanced Split). *Suppose Assumption SA-2 holds. Then for each $a, b \in (0, 1)$ with $a < b$, for every $\ell \in [p]$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i} \leq n^b, \hat{j} = \ell) \geq \frac{b - a}{2pe}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i} \leq n - n^a, \hat{j} = \ell) \geq \frac{b - a}{2pe}.$$

Corollary SA-9 (Uniform Rates for Decision Stumps). *Suppose Assumption SA-2 holds, and the tree has depth $K = 1$. Then for any $a, b \in (0, 1)$ with $a < b$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{IPW}}(\mathbf{x}) - \tau| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)}\right) \geq \frac{b}{e},$$

where $\sigma^2 = \mathbb{V}\left[\frac{d_i y_i(1)}{\xi} - \frac{(1 - d_i) y_i(0)}{1 - \xi}\right]$. Moreover, under the marginal probability integral transform normalization described after Assumption SA-2, for any deterministic sequence $\eta_n \downarrow 0$,

$$\liminf_{n \rightarrow \infty} \inf_{\mathbf{x} \in \mathcal{X}_n(a, \eta_n)} \mathbb{P}\left(|\hat{\tau}_{\text{IPW}}(\mathbf{x}) - \tau| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)}\right) \geq \frac{b - a}{2pe}.$$

Corollary SA-10 (Uniform Rates for Deep Trees). *Suppose Assumption SA-2 holds. Then for any $b \in (0, 1)$ and any NSS-IPW transformed outcome tree with at least one split, there exists a positive constant C depending only on the distribution of $(y_i(0), y_i(1), d_i)$ such that*

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{IPW}}(\mathbf{x}) - \tau| \geq C n^{-b/2} \sqrt{\log \log(n)}\right) \geq \frac{b}{e}.$$

Corollary SA-11 (L_2 Convergence Rate for NSS). *Suppose Assumption SA-2 holds. Then the following bounds hold uniformly over any data-dependent axis-aligned causal tree with depth at most K (possibly depending on n) when the displayed NSS-IPW terminal estimator is computed on that tree:*

$$\mathbb{E}\left[\int_{\mathcal{X}} (\hat{\tau}_{\text{IPW}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x})\right] \leq C \frac{2^K \log^4(n) \log(np)}{n},$$

where C is a positive constant that only depends on the distribution of $\tilde{\varepsilon}_i = y_i \frac{d_i - \xi}{\xi(1 - \xi)} - \tau$. Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P}\left(\int_{\mathcal{X}} (\hat{\tau}_{\text{IPW}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^4(n) \log(np)}{n}\right) = 0,$$

where C' is a positive constant that only depends on the distribution of $\tilde{\varepsilon}_i$.

SA-3.1.2 Sample Splitting

Corollary SA-12 (Uniform Rates for HON). *Suppose Assumption SA-2 holds, and the construction fold HON-IPW tree has at least one split. Then for any $b \in (0, 1)$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\tilde{\tau}_{\text{IPW}}(\mathbf{x}) - \tau| \geq \frac{C_1 \mathbb{E}[|\tilde{\varepsilon}_i|]}{n^{b/2}}\right) \geq C_2 \frac{\mathbb{E}[|\tilde{\varepsilon}_i|]^2}{\mathbb{V}[\tilde{\varepsilon}_i]} b,$$

where C_1 and C_2 are positive constants only depending on the distribution of $\tilde{\varepsilon}_i = y_i \frac{d_i - \xi}{\xi(1 - \xi)} - \tau$ and on the lower and upper limiting ratios of $n_{\top}/n_{\tau}$.

Corollary SA-13 (L_2 Convergence Rate for HON). *Suppose Assumption SA-2 holds. Then the following bounds hold uniformly over any data-dependent axis-aligned causal tree with depth at most K (possibly depending on n) when the displayed honest IPW terminal estimator is computed on that tree:*

$$\mathbb{E} \left[\int_{\mathcal{X}} (\tilde{\tau}_{\text{IPW}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C \frac{2^K \log^5(n)}{n},$$

provided $\rho \leq \frac{n\tau}{n_\tau} \leq \rho^{-1}$ for some $\rho \in (0, 1)$, and C is some constant only depending on the distribution of $\tilde{\varepsilon}_i = y_i \frac{d_i - \xi}{\xi(1-\xi)} - \tau$ and ρ . Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} (\tilde{\tau}_{\text{IPW}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^5(n)}{n} \right) = 0,$$

where C' is some constant only depending on the distribution of $\tilde{\varepsilon}_i$ and ρ .

SA-3.2 DIM Estimator

The DIM estimator is not itself a CART regression tree on a transformed outcome. We connect it to the IPW transformed outcome tree by comparing the two split criteria uniformly over split indices and coordinates.

SA-3.2.1 No Sample Splitting

Approximation Results on Decision Stumps

Denote by π_ℓ permutation of index $[n]$ such that $x_{\pi_\ell(1), \ell} \leq x_{\pi_\ell(2), \ell} \leq \dots \leq x_{\pi_\ell(n), \ell}$, $1 \leq \ell \leq p$. Consider the split criterion for the regression and ipw trees when splitting at the root node when $\#\{\mathbf{x}_{\pi_\ell(i)} \in t_L\} = k$: For $1 \leq \ell \leq p$, $1 \leq k \leq n$, consider

$$\begin{aligned} \mathcal{J}^{\text{DIM}}(k, \ell) &= \frac{k(n-k)}{n} \left(\hat{\tau}_L^{\text{DIM}}(k, \ell) - \hat{\tau}_R^{\text{DIM}}(k, \ell) \right)^2, \\ \mathcal{J}^{\text{IPW}}(k, \ell) &= \frac{k(n-k)}{n} \left(\bar{\tau}_L^{\text{IPW}}(k, \ell) - \bar{\tau}_R^{\text{IPW}}(k, \ell) \right)^2, \end{aligned}$$

where

$$\begin{aligned} \hat{\tau}_L^{\text{DIM}}(k, \ell) &= \frac{\sum_{i=1}^k d_{\pi_\ell(i)} y_{\pi_\ell(i)}}{\sum_{i=1}^k d_{\pi_\ell(i)}} - \frac{\sum_{i=1}^k (1 - d_{\pi_\ell(i)}) y_{\pi_\ell(i)}}{\sum_{i=1}^k (1 - d_{\pi_\ell(i)})}, \\ \hat{\tau}_R^{\text{DIM}}(k, \ell) &= \frac{\sum_{i=k+1}^n d_{\pi_\ell(i)} y_{\pi_\ell(i)}}{\sum_{i=k+1}^n d_{\pi_\ell(i)}} - \frac{\sum_{i=k+1}^n (1 - d_{\pi_\ell(i)}) y_{\pi_\ell(i)}}{\sum_{i=k+1}^n (1 - d_{\pi_\ell(i)})}, \\ \bar{\tau}_L^{\text{IPW}}(k, \ell) &= \frac{1}{k} \sum_{i=1}^k \frac{d_{\pi_\ell(i)}}{\xi} \varepsilon_{\pi_\ell(i)}(1) - \frac{1}{k} \sum_{i=1}^k \frac{1 - d_{\pi_\ell(i)}}{1 - \xi} \varepsilon_{\pi_\ell(i)}(0), \\ \bar{\tau}_R^{\text{IPW}}(k, \ell) &= \frac{1}{n-k} \sum_{i=k+1}^n \frac{d_{\pi_\ell(i)}}{\xi} \varepsilon_{\pi_\ell(i)}(1) - \frac{1}{n-k} \sum_{i=k+1}^n \frac{1 - d_{\pi_\ell(i)}}{1 - \xi} \varepsilon_{\pi_\ell(i)}(0). \end{aligned}$$

Replacing $\varepsilon_{\pi_\ell(i)}$ by $y_{\pi_\ell(i)}$ would give $\hat{\tau}_L^{\text{IPW}}$ (or $\hat{\tau}_R^{\text{IPW}}$) instead of $\bar{\tau}_L^{\text{IPW}}$ (or $\bar{\tau}_R^{\text{IPW}}$). Using $\varepsilon_{\pi_\ell(i)}$ here lets us approximate the $\mathcal{J}^{\text{DIM}}(\cdot, \ell)$ processes.

The optimization objective based on Definition SA-5 for the regression-based estimator with variance maximization is equivalent to choosing a splitting coordinate $\hat{j}_{\text{reg}}$, and a splitting index $\hat{i}_{\text{reg}}$ such that

$$\mathbf{t}_L = \{\mathbf{u} \in \mathcal{X} : \mathbf{u}_{\hat{j}_{\text{reg}}} \leq x_{\pi_{\hat{j}_{\text{reg}}}(\hat{i}_{\text{reg}}), \hat{j}_{\text{reg}}}\}, \quad \mathbf{t}_R = \{\mathbf{u} \in \mathcal{X} : \mathbf{u}_{\hat{j}_{\text{reg}}} > x_{\pi_{\hat{j}_{\text{reg}}}(\hat{i}_{\text{reg}}), \hat{j}_{\text{reg}}}\},$$

that maximizes

$$\frac{n(t_L)n(t_R)}{n(\mathbf{t})} \left(\hat{\tau}_{\text{reg}}(t_L) - \hat{\tau}_{\text{reg}}(t_R) \right)^2,$$

that is,

$$(\hat{i}_{\text{reg}}, \hat{j}_{\text{reg}}) = \arg \max_{k, \ell} \mathcal{J}^{\text{DIM}}(k, \ell).$$

A technical aspect is to control for fluctuations of objects of the form $\frac{\sum_{i=1}^k d_{\pi_\ell(i)} y_{\pi_\ell(i)}}{\sum_{i=1}^k d_{\pi_\ell(i)}}$, for which we will use a truncation argument that requires $\sum_{i=1}^k d_{\pi_\ell(i)} \geq r_n$ with $r_n \rightarrow \infty$. Let $\mathcal{V}_{\text{DIM}}$ denote the set of candidate splits for which all treated and control denominators in the left and right child nodes are positive. In the approximation lemmas below, maxima are taken over this set of valid candidates, equivalently over the candidates retained by Definition SA-5. On the balanced ranges used below, the event that all required denominators are positive has probability tending to one. This gives the following lemma:

Lemma SA-14 (Approximation Error). *Suppose Assumption SA-2 holds. Let $(r_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that $r_n \rightarrow \infty$. Then*

$$\max_{\substack{1 \leq \ell \leq p, \\ (k, \ell) \in \mathcal{V}_{\text{DIM}}}} \left| \mathcal{J}^{\text{DIM}}(k, \ell) - \bar{\mathcal{J}}^{\text{IPW}}(k, \ell) \right| = O_{\mathbb{P}} \left(\frac{\log \log(n)}{\sqrt{r_n}} \right).$$

We also control for the truncation error:

Lemma SA-15 (Truncation Error). *Suppose Assumption SA-2 holds. Let ρ_n be a sequence taking values in $(0, 1)$ such that $\rho_n \rightarrow 0$ and $\rho_n \log \log(n) \rightarrow \infty$, and take $s_n = \exp((\log n)^{\rho_n})$. Then*

$$\max_{\substack{1 \leq \ell \leq p, \\ (k, \ell) \in \mathcal{V}_{\text{DIM}} \\ 1 \leq k \leq s_n \text{ or } n - s_n \leq k \leq n}} \left| \mathcal{J}^{\text{DIM}}(k, \ell) - \bar{\mathcal{J}}^{\text{IPW}}(k, \ell) \right| = O_{\mathbb{P}} \left(\rho_n \log \log(n) + \frac{s_n}{n - s_n} \log \log(n) \right).$$

Rates for Decision Stumps

The previous two lemmas reduce the $\arg \max$ of $\mathcal{J}^{\text{DIM}}$ to the $\arg \max$ of $\bar{\mathcal{J}}^{\text{IPW}}$. The latter is the split criterion based on CART with *transformed outcome* $\frac{d_i}{\xi} \varepsilon_i(1) - \frac{1-d_i}{1-\xi} \varepsilon_i(0)$, and results from Section SA-2 can be applied.

Theorem SA-16 (Imbalanced Split). *Suppose Assumption SA-2 holds. Then for each $a, b \in (0, 1)$ with $a < b$, for every $\ell \in [p]$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i}_{\text{DIM}} \leq n^b, \hat{j}_{\text{DIM}} = \ell) \geq \frac{b-a}{2pe}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i}_{\text{DIM}} \leq n - n^a, \hat{j}_{\text{DIM}} = \ell) \geq \frac{b-a}{2pe}.$$

The issue of imbalanced cells gives rise to the slow uniform convergence rate.

Theorem SA-17 (Uniform Rates for Decision Stumps). *Suppose Assumption SA-2 holds, and the tree has depth $K = 1$. Then for any $a, b \in (0, 1)$ with $a < b$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)} \right) \geq \frac{b}{e},$$

where $\sigma^2 = \mathbb{V}[\tilde{\varepsilon}_i]$, with $\tilde{\varepsilon}_i = \frac{d_i}{\xi} \varepsilon_i(1) - \frac{1-d_i}{1-\xi} \varepsilon_i(0)$. Under the marginal probability integral transform normalization described after Assumption SA-2, for any deterministic sequence $\eta_n \downarrow 0$,

$$\liminf_{n \rightarrow \infty} \inf_{\mathbf{x} \in \mathcal{X}_n(a, \eta_n)} \mathbb{P} \left(|\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)} \right) \geq \frac{b-a}{2pe}.$$

Deeper Trees

We generalize the above results on decision stumps to recursive trees whose depth may vary with n , subject to the leaf count condition below.

Theorem SA-18 (Uniform Rates for Deep Trees). *Suppose Assumption SA-2 holds. Fix $b \in (0, 1)$, and suppose the NSS-DIM tree has depth at least one and at most $K = K_n$, where*

$$2^K \log^2 n = o\left(n^{b/4} \sqrt{\log \log n}\right).$$

Then there exist positive constants c_{DIM} and q_{DIM} , depending only on the distribution of $(y_i(0), y_i(1), d_i)$ and on the fixed covariate dimension p , and not on b, K , or n , such that

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq c_{\text{DIM}} n^{-b/2} \sqrt{\log \log(n)}\right) \geq q_{\text{DIM}} b.$$

For integrated L_2 loss, we also have the following upper bound.

Theorem SA-19 (L_2 Convergence Rate for NSS). *Suppose Assumption SA-2 holds. Then the following bounds hold uniformly over any data-dependent axis-aligned causal tree with depth at most K (possibly depending on n) when the displayed NSS-DIM terminal estimator is computed on that tree:*

$$\mathbb{E}\left[\int_{\mathcal{X}} (\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x})\right] \leq C \frac{2^K \log^4(n) \log(np)}{n},$$

where C is a positive constant that only depends on the distribution of $(y_i(0), y_i(1), d_i)$. Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P}\left(\int_{\mathcal{X}} (\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^4(n) \log(np)}{n}\right) = 0,$$

where C' is a positive constant that only depends on the distribution of $(y_i(0), y_i(1), d_i)$.

SA-3.2.2 Sample Splitting

For honest sample splitting, the same structure yields a lower bound for uniform accuracy and an upper bound for integrated L_2 loss. The rates differ from the case without sample splitting because final estimation is performed on an independent estimation sample.

Theorem SA-20 (Uniform Rates for HON). *Suppose Assumption SA-2 holds, and the construction fold HON-DIM tree has at least one split. Then for any $b \in (0, 1)$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\check{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq C_1 n^{-b/2}\right) \geq C_2 b.$$

where C_1 and C_2 are positive constants only depending on the distribution of $(y_i(0), y_i(1), d_i)$ and on the lower and upper limiting ratios of $n_{\text{T}}/n_{\text{T}}$. In particular, the fixed treatment-arm positivity factor involving ξ is absorbed into C_2 .

Theorem SA-21 (L_2 Convergence Rate for HON). *Suppose Assumption SA-2 holds. Then the following bounds hold uniformly over any data-dependent axis-aligned causal tree with depth at most K (possibly depending on n) when the displayed honest DIM terminal estimator is computed on that tree:*

$$\mathbb{E}\left[\int_{\mathcal{X}} (\check{\tau}_{\text{DIM}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x})\right] \leq C \frac{2^K \log^5(n)}{n},$$

provided $\rho \leq \frac{n_{\text{T}}}{n_{\text{T}}} \leq \rho^{-1}$ for some $\rho \in (0, 1)$, and C is a positive constant that only depends on ρ and the distribution of $(y_i(0), y_i(1), d_i)$. Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P}\left(\int_{\mathcal{X}} (\check{\tau}_{\text{DIM}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^5(n)}{n}\right) = 0,$$

where C' is a positive constant that only depends on ρ and the distribution of $(y_i(0), y_i(1), d_i)$.

SA-3.3 SSE Estimator

Throughout, **SSE** names the split selection rule: after the SSE tree is selected, terminal node treatment effects are estimated by the DIM estimator. Thus, while the CATE estimators given the tree of the **SSE** strategy coincide with the DIM strategy, the tree construction methods differ. The analysis again proceeds by approximating the split criterion. For **SSE**, the criterion can be approximated by the sum of two transformed outcome regression criteria, one for treated units and one for control units. A careful high-dimensional Gaussian approximation with respect to the geometry of simple convex sets then enables us to characterize the limiting distribution of splitting indices.

For a fixed valid candidate split of a parent node $\mathbf{t}$ into $\mathbf{t}_L$ and $\mathbf{t}_R$, the objective in (SA-8) can be profiled over the node-specific intercepts and treatment coefficients. Up to terms that do not depend on the candidate split, minimizing the residual sum of squares is equivalent to maximizing the treated and control variance gains

$$\begin{aligned} & \frac{n_1(\mathbf{t}_L)n_1(\mathbf{t}_R)}{n_1(\mathbf{t})} \left(\frac{1}{n_1(\mathbf{t}_L)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_L} d_i y_i - \frac{1}{n_1(\mathbf{t}_R)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_R} d_i y_i \right)^2 \\ & + \frac{n_0(\mathbf{t}_L)n_0(\mathbf{t}_R)}{n_0(\mathbf{t})} \left(\frac{1}{n_0(\mathbf{t}_L)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_L} (1-d_i) y_i - \frac{1}{n_0(\mathbf{t}_R)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_R} (1-d_i) y_i \right)^2. \end{aligned}$$

Thus the SSE rule can be viewed as selecting the split with the largest combined improvement in separate treated and control outcome fits.

SA-3.3.1 No Sample Splitting

Decision Stump

For each variable $j = 1, 2, \dots, p$, the data $\{x_{ij} : \mathbf{x}_i \in \mathbf{t}\}$ is relabeled so that x_{ij} is increasing in the index $i = 1, 2, \dots, n(\mathbf{t})$, where $n(\mathbf{t}) = \#\{\mathbf{x}_i \in \mathbf{t}\}$. The fit-based objective is to minimize

$$\min_{a_L, b_L, a_R, b_R \in \mathbb{R}} \sum_{\mathbf{x}_i \in \mathbf{t}_L} (y_i - a_{t_L} - b_{t_L} d_i)^2 + \sum_{\mathbf{x}_i \in \mathbf{t}_R} (y_i - a_{t_R} - b_{t_R} d_i)^2 \quad (\text{SA-9})$$

with respect to the index i and variable j . Again, the maximizers are denoted by $(\hat{i}_{\text{SSE}}, \hat{j}_{\text{SSE}})$, and the optimal split point $\hat{\zeta}_{\text{SSE}}$ that maximizes (SA-9) can be expressed as $x_{\pi_{\hat{j}_{\text{SSE}}}(\hat{i}_{\text{SSE}}), \hat{j}_{\text{SSE}}}$.

All ratios in this subsection are evaluated only on valid candidate splits in the sense of Definition SA-5. To break down the criterion (SA-9), denote

$$\begin{aligned} \hat{\mu}_{L,0}(k, \ell) &= \frac{\sum_{i=1}^k (1 - d_{\pi_\ell(i)}) y_{\pi_\ell(i)}}{\sum_{i=1}^k (1 - d_{\pi_\ell(i)})}, & \hat{\mu}_{L,1}(k, \ell) &= \frac{\sum_{i=1}^k d_{\pi_\ell(i)} y_{\pi_\ell(i)}}{\sum_{i=1}^k d_{\pi_\ell(i)}}, \\ \hat{\mu}_{R,0}(k, \ell) &= \frac{\sum_{i=k+1}^n (1 - d_{\pi_\ell(i)}) y_{\pi_\ell(i)}}{\sum_{i=k+1}^n (1 - d_{\pi_\ell(i)})}, & \hat{\mu}_{R,1}(k, \ell) &= \frac{\sum_{i=k+1}^n d_{\pi_\ell(i)} y_{\pi_\ell(i)}}{\sum_{i=k+1}^n d_{\pi_\ell(i)}}. \end{aligned}$$

To denote the counts compactly, set $n_0 = \sum_{i=1}^n (1 - d_i)$, $n_{L,0}(k) = \sum_{i=1}^k (1 - d_{\pi_\ell(i)})$, $n_{R,0}(k) = \sum_{i=k+1}^n (1 - d_{\pi_\ell(i)})$, and $n_1 = \sum_{i=1}^n d_i$, $n_{L,1}(k) = \sum_{i=1}^k d_{\pi_\ell(i)}$, $n_{R,1}(k) = \sum_{i=k+1}^n d_{\pi_\ell(i)}$. Then maximizing Equation (SA-9) is equivalent to maximizing

$$\mathcal{J}^{\text{SSE}}(k, \ell) = \frac{n_{L,0} n_{R,0}}{n_0} (\hat{\mu}_{L,0}(k, \ell) - \hat{\mu}_{R,0}(k, \ell))^2 + \frac{n_{L,1} n_{R,1}}{n_1} (\hat{\mu}_{L,1}(k, \ell) - \hat{\mu}_{R,1}(k, \ell))^2.$$

Since μ_0 and μ_1 are constant in this subsection, the arm-specific means cancel from all split contrasts within each treatment arm. We therefore approximate this empirical process by the residual centered proxy

$$\mathcal{J}^{\text{prox}}(k, \ell) = (1 - \xi) \frac{k(n-k)}{n} (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 + \xi \frac{k(n-k)}{n} (\bar{\mu}_{L,1}(k, \ell) - \bar{\mu}_{R,1}(k, \ell))^2,$$

with

$$\begin{aligned} \bar{\mu}_{L,0}(k, \ell) &= \frac{1}{k} \sum_{i \leq k} \frac{1 - d_{\pi_\ell(i)}}{1 - \xi} \varepsilon_{\pi_\ell(i)}(0), & \bar{\mu}_{L,1}(k, \ell) &= \frac{1}{k} \sum_{i \leq k} \frac{d_{\pi_\ell(i)}}{\xi} \varepsilon_{\pi_\ell(i)}(1), \\ \bar{\mu}_{R,0}(k, \ell) &= \frac{1}{n-k} \sum_{i > k} \frac{1 - d_{\pi_\ell(i)}}{1 - \xi} \varepsilon_{\pi_\ell(i)}(0), & \bar{\mu}_{R,1}(k, \ell) &= \frac{1}{n-k} \sum_{i > k} \frac{d_{\pi_\ell(i)}}{\xi} \varepsilon_{\pi_\ell(i)}(1). \end{aligned}$$

The latter can be approximated by the squared norm of a bivariate time-transformed O-U process, for fixed coordinate $\ell \in [p]$. Let $\mathcal{V}_{\text{SSE}}$ denote the set of valid candidates for the SSE criterion, where both treatment-arm denominators are positive in both child nodes. Maxima in the following approximation lemmas are taken over $\mathcal{V}_{\text{SSE}}$. The following lemmas formalize the approximation.

Lemma SA-22 (Approximation Error). *Suppose Assumption SA-2 holds. Let $(r_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that $r_n \rightarrow \infty$. Then*

$$\max_{\substack{1 \leq \ell \leq p, \\ r_n \leq k < n - r_n \\ (k, \ell) \in \mathcal{V}_{\text{SSE}}}} \left| \mathcal{J}^{\text{SSE}}(k, \ell) - \mathcal{J}^{\text{prox}}(k, \ell) \right| = O_{\mathbb{P}} \left(\frac{(\log \log n)^{3/2}}{\sqrt{r_n}} \right).$$

Lemma SA-23 (Truncation Error). *Suppose Assumption SA-2 holds. Let ρ_n be a sequence taking values in $(0, 1)$ such that $\rho_n \rightarrow 0$ and $\rho_n \log \log(n) \rightarrow \infty$, and take $s_n = \exp((\log n)^{\rho_n})$. Then*

$$\max_{\substack{1 \leq \ell \leq p, \\ (k, \ell) \in \mathcal{V}_{\text{SSE}} \\ 1 \leq k \leq s_n \text{ or } n - s_n \leq k \leq n}} \left| \mathcal{J}^{\text{SSE}}(k, \ell) - \mathcal{J}^{\text{prox}}(k, \ell) \right| = O_{\mathbb{P}} \left(\rho_n \log \log(n) + \frac{s_n}{n - s_n} \log \log(n) \right).$$

Theorem SA-24 (Imbalanced Split for SSE). *Suppose Assumption SA-2 holds. Then for each $a, b \in (0, 1)$ with $a < b$, for every $\ell \in [p]$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i}_{\text{SSE}} \leq n^b, \hat{j}_{\text{SSE}} = \ell) \geq \frac{b - a}{2pe}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i}_{\text{SSE}} \leq n - n^a, \hat{j}_{\text{SSE}} = \ell) \geq \frac{b - a}{2pe}.$$

Remark 1. When the treated and control residual variances differ, the bivariate O-U approximation in the proof becomes a weighted quadratic form rather than a squared Euclidean norm. Lemma SA-38 supplies the needed extreme value approximation; it reduces to Lemma SA-32 after rescaling when the two variances are equal. $\square$

Lemma SA-25 (Directional transfer for SSE stumps). *Suppose Assumption SA-2 holds. For every $0 < a < b < 1$ and every $\ell \in [p]$, there exist positive constants c and q , depending only on the distribution of $(y_i(0), y_i(1), d_i)$, such that, with $\mathbf{t}_L$ denoting the selected left root child,*

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i}_{\text{SSE}} \leq n^b, \hat{j}_{\text{SSE}} = \ell, n(\mathbf{t}_L) \Delta(\mathbf{t}_L)^2 \geq c^2 \log \log n) \geq q(b - a).$$

The same conclusion holds with $\mathbf{t}_R$ in place of $\mathbf{t}_L$ for the right boundary event $n - n^b \leq \hat{i}_{\text{SSE}} \leq n - n^a$. Consequently, the displayed event implies the corresponding depth one CATE error lower bound at rate $n^{-b/2} \sqrt{\log \log n}$.

Although the SSE splitting criterion differs from the DIM variance maximization criterion, once a tree is fixed the terminal node CATE estimator is the same. Hence the following results follow from Theorem SA-24, Lemma SA-25, and the corresponding arguments for the DIM estimator.

Theorem SA-26 (Uniform Rates for Decision Stumps). *Suppose Assumption SA-2 holds. For any $a, b \in (0, 1)$ with $a < b$, we have*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau| \geq C_1 n^{-b/2} \sqrt{\log \log(n)} \right) \geq C_2 b,$$

where C_1 and C_2 are positive constants depending only on the distribution of $(y_i(0), y_i(1), d_i)$. Moreover, under the marginal probability integral transform normalization described after Assumption SA-2, for any deterministic sequence $\eta_n \downarrow 0$,

$$\liminf_{n \rightarrow \infty} \inf_{\mathbf{x} \in \mathcal{X}_n(a, \eta_n)} \mathbb{P} \left(|\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau| \geq C_1 n^{-b/2} \sqrt{\log \log(n)} \right) \geq C_2(b - a).$$

Deeper Trees

Theorem SA-27 (Uniform Rates for Deep Trees). *Suppose Assumption SA-2 holds. Fix $b \in (0, 1)$, and suppose the NSS-SSE tree has depth at most $K = K_n$, where*

$$2^K \log^2 n = o \left(n^{b/4} \sqrt{\log \log n} \right).$$

Then there exist positive constants c_{SSE} and q_{SSE} , depending only on the distribution of $(y_i(0), y_i(1), d_i)$ and on the fixed covariate dimension p , and not on b , K , or n , such that

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau| \geq c_{\text{SSE}} n^{-b/2} \sqrt{\log \log(n)} \right) \geq q_{\text{SSE}} b.$$

Theorem SA-28 (L_2 Convergence Rate for NSS). *Suppose Assumption SA-2 holds. Then for any NSS-SSE tree with depth at most K (possibly depending on n),*

$$\mathbb{E} \left[\int_{\mathcal{X}} (\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C \frac{2^K \log^4(n) \log(np)}{n},$$

where C is a positive constant that only depends on the distribution of $(y_i(0), y_i(1), d_i)$. Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} (\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^4(n) \log(np)}{n} \right) = 0,$$

where C' is a positive constant that only depends on the distribution of $(y_i(0), y_i(1), d_i)$.

SA-3.3.2 Sample Splitting

Theorem SA-29 (Uniform Rates for HON). *Suppose Assumption SA-2 holds. Then for any $b \in (0, 1)$ and any HON-SSE tree with at least one split,*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\tilde{\tau}_{\text{SSE}}(\mathbf{x}) - \tau| \geq C_1 n^{-b/2} \right) \geq C_2 b.$$

where C_1 and C_2 are positive constants only depending on the distribution of $(y_i(0), y_i(1), d_i)$ and on the lower and upper limiting ratios of $n_{\top}/n_{\tau}$. In particular, the fixed treatment-arm positivity factor involving ξ is absorbed into C_2 .

Theorem SA-30 (L_2 Convergence Rate for HON). *Suppose Assumption SA-2 holds. Then for any HON-SSE tree with depth at most K (possibly depending on n),*

$$\mathbb{E} \left[\int_{\mathcal{X}} (\tilde{\tau}_{\text{SSE}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C \frac{2^K \log^5(n)}{n},$$

provided $\rho \leq \frac{n_{\top}}{n_{\tau}} \leq \rho^{-1}$ for some $\rho \in (0, 1)$, and C is a positive constant that only depends on ρ and the distribution of $(y_i(0), y_i(1), d_i)$. Moreover,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\int_{\mathcal{X}} (\tilde{\tau}_{\text{SSE}}(\mathbf{x}) - \tau)^2 dF_{\mathbf{X}}(\mathbf{x}) \geq C' \frac{2^K \log^5(n)}{n} \right) = 0,$$

where C' is a positive constant that only depends on ρ and the distribution of $(y_i(0), y_i(1), d_i)$.

SA-3.4 Squared T Statistic Estimators

The fourth method proposed by [Athey and Imbens \[2016\]](#) is the squared T statistic tree. Consider the version formed from a scalar pseudo outcome τ_i , with left and right averages $\hat{\tau}_L(k, \ell)$ and $\hat{\tau}_R(k, \ell)$ computed after sorting along coordinate ℓ . At the root node the index and coordinate to split $(\hat{i}, \hat{j})$ are chosen so that the *squared T statistics metric* is maximized over nondegenerate candidates, that is,

$$(\hat{i}, \hat{j}) = \arg \max_{1 \leq k < n, \ell \in [p]} n \frac{(\hat{\tau}_L(k, \ell) - \hat{\tau}_R(k, \ell))^2}{S^2(k, \ell)/k + S^2(k, \ell)/(n - k)},$$

where $S^2(k, \ell) > 0$ is the conditional sample variance given the split,

$$\begin{aligned} S^2(k, \ell) &= \frac{1}{n-2} \sum_{i \leq k} (\tau_i - \hat{\tau}_L(k, \ell))^2 + \frac{1}{n-2} \sum_{i > k} (\tau_i - \hat{\tau}_R(k, \ell))^2 \\ &= \frac{1}{n-2} \left[\sum_{i=1}^n (\tau_i - n^{-1} \sum_{j=1}^n \tau_j)^2 - \frac{k(n-k)}{n} (\hat{\tau}_L(k, \ell) - \hat{\tau}_R(k, \ell))^2 \right]. \end{aligned}$$

Let

$$A(k, \ell) = \frac{k(n-k)}{n} (\hat{\tau}_L(k, \ell) - \hat{\tau}_R(k, \ell))^2, \quad T_0 = \sum_{i=1}^n (\tau_i - n^{-1} \sum_{j=1}^n \tau_j)^2.$$

Then, for candidates with $T_0 > A(k, \ell)$,

$$\begin{aligned} & n \frac{(\hat{\tau}_L(k, \ell) - \hat{\tau}_R(k, \ell))^2}{S^2(k, \ell)/k + S^2(k, \ell)/(n-k)} \\ &= \frac{n(n-2)A(k, \ell)}{T_0 - A(k, \ell)}. \end{aligned}$$

The map $A \mapsto n(n-2)A/(T_0 - A)$ is strictly increasing on $[0, T_0]$. Hence, on the nondegenerate candidates retained by the rule, the squared T statistic split is the same as the variance maximization split based on the same scalar pseudo outcomes. To avoid ambiguity, the squared T statistic rule is understood to optimize over candidates with positive pooled residual variance, equivalently $T_0 > A(k, \ell)$. If no such candidate exists, a fixed deterministic valid split, or no split if none is valid, is chosen. This degenerate fallback is irrelevant for the equivalence above and for the stochastic lower bound arguments, which operate on nondegenerate candidates with probability tending to one.

This equivalence is specific to the pooled scalar pseudo outcome statistic displayed above. A split rule that studentizes treatment and control means separately, uses heteroskedastic or sandwich standard errors, or otherwise changes the denominator specific to the candidate need not induce the same ordering as variance maximization.

SA-3.5 Unbiasedness

Lemma SA-31 (Unbiasedness). *Suppose Assumption SA-2 holds. If $\mathbf{t}_{\text{HON}}(\mathbf{x})$ denotes the terminal node used for honest final estimation at $\mathbf{x}$,*

$$\begin{aligned} \mathbb{E}[\tilde{\tau}_{\text{IPW}}(\mathbf{x}; K)] &= \tau - \tau \mathbb{P}(n(\mathbf{t}_{\text{HON}}(\mathbf{x})) = 0), \\ \mathbb{E}[\tilde{\tau}_l(\mathbf{x}; K)] &= \tau - \tau \mathbb{P}(n_0(\mathbf{t}_{\text{HON}}(\mathbf{x})) = 0 \text{ or } n_1(\mathbf{t}_{\text{HON}}(\mathbf{x})) = 0), \quad l \in \{\text{DIM}, \text{SSE}\}. \end{aligned}$$

For $q = \text{NSS}$, under deterministic tie breaking, the DIM and SSE conclusions hold if the potential outcome error vector is jointly centrally symmetric,

$$(\varepsilon_i(0), \varepsilon_i(1)) \stackrel{d}{=} -(\varepsilon_i(0), \varepsilon_i(1)).$$

Because observations are i.i.d. and the potential outcome errors are independent of covariates and treatment assignments, this pairwise central symmetry implies central symmetry of the full vector $(\varepsilon_i(0), \varepsilon_i(1))_{i=1}^n$ conditional on $(\mathbf{X}, \mathbf{d})$. The IPW conclusion holds under deterministic tie breaking if, with

$$\tilde{\varepsilon}_i = y_i \frac{d_i - \xi}{\xi(1 - \xi)} - \tau,$$

the transformed residual vector $(\tilde{\varepsilon}_1, \dots, \tilde{\varepsilon}_n)$ is centrally symmetric conditional on the treatment assignments. The displayed empty cell terms are the finite-sample bias formulas. If, in addition, the relevant terminal node denominators are positive almost surely,

$$\mathbb{E}[\hat{\tau}_l(\mathbf{x}; K)] = \tau, \quad l \in \{\text{DIM}, \text{SSE}\},$$

under the potential outcome error symmetry condition, and the same exact unbiasedness statement holds for $l = \text{IPW}$ under the transformed residual symmetry condition.

Remark 2 (A simple sufficient condition for IPW symmetry). The transformed residual symmetry condition in Lemma SA-31 is implied by a transparent centering condition. If the potential outcome error vector is jointly centrally symmetric and

$$(1 - \xi)c_1 + \xi c_0 = 0,$$

then, conditional on the treatment assignments,

$$\tilde{\varepsilon}_i = \begin{cases} \varepsilon_i(1)/\xi, & d_i = 1, \\ -\varepsilon_i(0)/(1 - \xi), & d_i = 0, \end{cases}$$

and the transformed residual vector is centrally symmetric. In the balanced experiment case $\xi = 1/2$, this centering condition reduces to $c_1 + c_0 = 0$. $\square$

SA-4 Correction to Eicker (1979)

As part of the technical arguments used in this paper, we correct a statement concerning the limiting distribution of the maximum absolute value of an Ornstein–Uhlenbeck (O-U) process in Eicker [1979, Theorem 5], and of the maximum norm in the vector valued version used below. Specifically, the term $\log(c)$ should appear in place of $2\log(c)$, where $c > 0$. This matters because replacing $\log(c)$ by $2\log(c)$ changes the factor $e^{-(z-\log(c))} = ce^{-z}$ to $e^{-(z-2\log(c))} = c^2e^{-z}$, so the limiting exponent depends quadratically rather than linearly on the window length. The problem is that this is incompatible with the Gaussian correlation inequality argument we used for adjacent windows. That is, for two adjacent windows of lengths $c_1 \log n$ and $c_2 \log n$, the sublevel event for the supremum of the absolute value, or of the norm in the vector case, over the combined window is the intersection of the corresponding symmetric convex sublevel events on the two pieces. The Gaussian correlation inequality implies that the probability for the combined window must be at least the product of the two individual window probabilities. With the correct term $\log(c)$, this is perfectly consistent, because the limiting probability for a window of length $(c_1 + c_2) \log n$ is

$$\exp(-(c_1 + c_2)e^{-z}) = \exp(-c_1e^{-z})\exp(-c_2e^{-z}),$$

so the dependence on window length adds in exactly the way required by the inequality. If one instead used $2\log(c)$, then the same reasoning would force

$$\exp(-(c_1 + c_2)^2e^{-z}) \geq \exp(-(c_1^2 + c_2^2)e^{-z}),$$

because the left hand side would be the limiting probability for the combined window. The right hand side would come from the product lower bound for the two adjacent pieces. But this inequality is false, since $(c_1 + c_2)^2 > c_1^2 + c_2^2$ whenever $c_1, c_2 > 0$. Thus the extra factor 2 leads to a direct contradiction with the probability inequality implied by the Gaussian correlation argument. For completeness, we state below a corrected version of the result in a slightly more general form, allowing for the maximum of the norm of a possibly multivariate O-U process. Let $\Gamma(\cdot)$ denote the Euler gamma function.

Lemma SA-32 (Vector Valued Markov Type Darling–Erdős). *Let $\{(V_1(t), \dots, V_d(t)) : 0 \leq t < \infty\}$ be d independent identically distributed Ornstein–Uhlenbeck processes with $\mathbb{E}[V_i(t)] = 0$ and $\mathbb{E}[V_i(t)V_i(s)] = \exp(-|t - s|/2)$, $1 \leq i \leq d$. Define*

$$N(t) = \left(\sum_{1 \leq i \leq d} V_i^2(t) \right)^{1/2}.$$

For any $c > 0$, $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(a(\log(n)) \sup_{0 \leq t \leq c \log(n)} N(t) - b_d(\log(n)) \leq z \right) = \exp \left(-e^{-(z-\log(c))} \right),$$

where $a(t) = (2 \log(t))^{1/2}$ and $b_d(t) = 2 \log(t) + \frac{d}{2} \log \log(t) - \log \Gamma(d/2)$.

Proof of Lemma SA-32

Taking $T = c \log(n)$ in Horváth [1993, Lemma 2.1], we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq t \leq c \log(n)} N(t) \leq \frac{z + b_d(c \log(n))}{a(c \log(n))} \right) = \exp \left(-e^{-z} \right).$$

Expand the term $\frac{z + b_d(c \log(n))}{a(c \log(n))}$. For notational simplicity, denote

$$L = \log \log n, \quad A = \log c, \quad L \rightarrow \infty \quad (n \rightarrow \infty).$$

First, we present some elementary expansions,

$$\begin{aligned} \sqrt{2(A+L)} &= \sqrt{2L} \sqrt{1 + \frac{A}{L}} = \sqrt{2L} \left(1 + \frac{A}{2L} - \frac{A^2}{8L^2} + O(L^{-3}) \right), \\ \frac{1}{\sqrt{2(A+L)}} &= \frac{1}{\sqrt{2L}} \left(1 - \frac{A}{2L} + \frac{3A^2}{8L^2} + O(L^{-3}) \right), \\ \log(L+A) &= \log L + \frac{A}{L} - \frac{A^2}{2L^2} + O(L^{-3}). \end{aligned}$$

Expanding the numerator $b_d(c \log(n))$ gives

$$\begin{aligned} N_1 &= z + 2A + 2L + \frac{d}{2} \log(\log(c \log n)) - \log \Gamma(d/2), \\ N_2 &= z + 2A + 2L + \frac{d}{2} \log L - \log \Gamma(d/2), \\ N_3 &= z + A + 2L + \frac{d}{2} \log L - \log \Gamma(d/2). \end{aligned}$$

Then

$$\begin{aligned} &\frac{z + b_d(c \log n)}{a(c \log(n))} - \frac{z + \log(c) + b_d(\log n)}{a(\log(n))} \\ &= \frac{N_1}{\sqrt{2(A+L)}} - \frac{N_3}{\sqrt{2L}} \\ &= N_1 \left(\frac{1}{\sqrt{2(A+L)}} - \frac{1}{\sqrt{2L}} \right) + \frac{1}{\sqrt{2L}} (N_1 - N_3) \\ &= N_1 \frac{1}{\sqrt{2L}} \left(-\frac{A}{2L} + \frac{3A^2}{8L^2} + O(L^{-3}) \right) + \frac{1}{\sqrt{2L}} \left(\frac{d}{2} \left(\frac{A}{L} - \frac{A^2}{2L^2} + O(L^{-3}) \right) + A \right). \end{aligned}$$

Since $N_1 = 2L + O(\log L)$, the preceding display gives the bound

$$\frac{z + b_d(c \log n)}{a(c \log(n))} - \frac{z + \log(c) + b_d(\log n)}{a(\log(n))} = O \left(\frac{1 + \log L}{L^{3/2}} \right) = o(L^{-1/2}).$$

Since $a(\log(n)) = \Theta(L^{1/2})$, we have

$$\begin{aligned} &\mathbb{P} \left(\sup_{0 \leq t \leq c \log(n)} N(t) \leq \frac{z + \log(c) + b_d(\log(n))}{a(\log(n))} \right) \\ &= \mathbb{P} \left(\sup_{0 \leq t \leq c \log(n)} N(t) \leq \frac{z + o(1) + b_d(c \log(n))}{a(c \log(n))} \right) \\ &= \mathbb{P} \left(a(c \log(n)) \sup_{0 \leq t \leq c \log(n)} N(t) - b_d(c \log(n)) \leq z + o(1) \right) \rightarrow \exp(-e^{-z}) \text{ as } n \rightarrow \infty, \end{aligned}$$

where the last line follows from convergence in distribution of $a(c \log(n)) \sup_{0 \leq t \leq c \log(n)} N(t) - b_d(c \log(n))$ to a continuous distribution and Slutsky's Theorem. Replacing z in the preceding display by $z - \log(c)$ gives the stated normalization with threshold $(z + b_d(\log n))/a(\log n)$ and limiting probability $\exp(-e^{-(z - \log(c))})$.

SA-5 Proofs

SA-5.1 Technical Lemmas

We first record standard external probability tools used below. We then give two auxiliary lemmas tailored to the SSE argument.

Lemma SA-33 (High Dimensional CLT over hyperrectangles; Chernozhukov et al. [2017], Theorem 2.1). *Let $X_1, \dots, X_n$ be independent centered random vectors in $\mathbb{R}^m$, $m \geq 3$, and let $Y_1, \dots, Y_n$ be independent centered Gaussian random vectors such that $Y_i \sim \mathcal{N}(0, \mathbb{E}[X_i X_i^\top])$. Define*

$$S_n^X = n^{-1/2} \sum_{i=1}^n X_i, \quad S_n^Y = n^{-1/2} \sum_{i=1}^n Y_i,$$

and, for a class $\mathcal{A}$ of Borel subsets of $\mathbb{R}^m$,

$$\rho_n(\mathcal{A}) = \sup_{A \in \mathcal{A}} |\mathbb{P}(S_n^X \in A) - \mathbb{P}(S_n^Y \in A)|.$$

Let $\mathcal{A}_m^{\text{re}}$ be the class of all hyperrectangles in $\mathbb{R}^m$. Set

$$L_n = \max_{1 \leq j \leq m} \frac{1}{n} \sum_{i=1}^n \mathbb{E}[|X_{ij}|^3],$$

and, for $\phi \geq 1$,

$$M_{n,X}(\phi) = \frac{1}{n} \sum_{i=1}^n \mathbb{E} \left[\max_{1 \leq j \leq m} |X_{ij}|^3 \mathbf{1} \left(\max_{1 \leq j \leq m} |X_{ij}| > \frac{\sqrt{n}}{4\phi \log m} \right) \right],$$

with $M_{n,Y}(\phi)$ defined analogously, and let $M_n(\phi) = M_{n,X}(\phi) + M_{n,Y}(\phi)$. If

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_{ij}^2] \geq \underline{\sigma}^2 > 0, \quad j = 1, \dots, m,$$

then there exist constants $K_1, K_2 > 0$, depending only on $\underline{\sigma}$, such that for every $\bar{L}_n \geq L_n$,

$$\rho_n(\mathcal{A}_m^{\text{re}}) \leq K_1 \left[\left(\frac{\bar{L}_n^2 \log^7 m}{n} \right)^{1/6} + \frac{M_n(\phi_n)}{\bar{L}_n} \right], \quad \phi_n = K_2 \left(\frac{\bar{L}_n^2 \log^4 m}{n} \right)^{-1/6}.$$

Lemma SA-34 (High Dimensional CLT over simple convex sets; Chernozhukov et al. [2017], Proposition 3.1). *Let X_i, Y_i, S_n^X, S_n^Y , and $\rho_n(\mathcal{A})$ be as in Lemma SA-33. Fix constants $a, d, \underline{\sigma} > 0$. For a convex polytope $A^M \subset \mathbb{R}^m$ generated by at most M halfspaces, write*

$$A^M = \bigcap_{v \in \mathcal{V}(A^M)} \{w \in \mathbb{R}^m : v^\top w \leq \mathcal{S}_{A^M}(v)\},$$

where $\mathcal{V}(A^M)$ is the set of outward unit normals to the facets and $\mathcal{S}_{A^M}$ is the support function. For $\epsilon > 0$, define

$$A^{M,\epsilon} = \bigcap_{v \in \mathcal{V}(A^M)} \{w \in \mathbb{R}^m : v^\top w \leq \mathcal{S}_{A^M}(v) + \epsilon\}.$$

Suppose that every $A \in \mathcal{A}$ admits an approximation $A^M \subset A \subset A^{M,a/n}$ with $M \leq (mn)^d$, and that for every such A and every $v \in \mathcal{V}(A^M)$,

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}[(v^\top X_i)^2] \geq \underline{\sigma}^2, \quad \frac{1}{n} \sum_{i=1}^n \mathbb{E}[|v^\top X_i|^{2+k}] \leq B_n^k, \quad k = 1, 2,$$

and

$$\mathbb{E}[\exp(|v^\top X_i|/B_n)] \leq 2, \quad i = 1, \dots, n,$$

for some $B_n \geq 1$. Then

$$\rho_n(\mathcal{A}) \leq C \left(\frac{B_n^2 \log^7(mn)}{n} \right)^{1/6},$$

where C depends only on a, d , and $\underline{\sigma}$.

Lemma SA-35 (Gaussian to Gaussian comparison; Chernozhukov et al. [2022], Proposition 2.1). *Let Z_1 and Z_2 be centered Gaussian random vectors in $\mathbb{R}^m$ with covariance matrices Σ^1 and Σ^2 . Suppose that $\Sigma_{jj}^2 \geq \underline{\sigma}^2 > 0$ for all $j = 1, \dots, m$, and let*

$$\Delta = \max_{1 \leq j, k \leq m} |\Sigma_{jk}^1 - \Sigma_{jk}^2|.$$

Then

$$\sup_{y \in \mathbb{R}^m} |\mathbb{P}(Z_1 \leq y) - \mathbb{P}(Z_2 \leq y)| \leq C (\Delta \log^2 m)^{1/2},$$

where the inequalities inside the probabilities are componentwise and C depends only on $\underline{\sigma}$.

Lemma SA-36 (Gaussian maximum anti-concentration; Chernozhukov et al. [2017], Lemma A.1). *Let $Z = (Z_1, \dots, Z_m)^\top$ be a centered Gaussian vector with $\min_{1 \leq j \leq m} \mathbb{V}[Z_j] \geq \underline{\sigma}^2 > 0$. Then, for every $\epsilon > 0$,*

$$\sup_{t \in \mathbb{R}} \mathbb{P} \left(\left| \max_{1 \leq j \leq m} Z_j - t \right| \leq \epsilon \right) \leq C \epsilon \sqrt{\log m},$$

where C depends only on $\underline{\sigma}$.

Lemma SA-37 (Cone restricted maximization on finite Gaussian grids). *Let $\mathcal{S}$ be a finite index set and let*

$$Z_s = (G_s^{(0)}, G_s^{(1)})^\top, \quad s \in \mathcal{S},$$

where $(G_s^{(0)} : s \in \mathcal{S})$ and $(G_s^{(1)} : s \in \mathcal{S})$ are independent copies of the same centered Gaussian vector. Let $\hat{s}$ be any index valued rule, with deterministic tie breaking, that is measurable with respect to the collection of norms $(\|Z_s\| : s \in \mathcal{S})$. Let A be any event measurable with respect to the same collection of norms. Then for every nonzero vector $v \in \mathbb{R}^2$ and every $\gamma \in (0, 1)$,

$$\mathbb{P}(A, |v^\top Z_{\hat{s}}| \geq \gamma \|v\| \|Z_{\hat{s}}\|) \geq \pi_\gamma \mathbb{P}(A), \quad \pi_\gamma = \frac{2 \arccos(\gamma)}{\pi} > 0.$$

Proof. Let $\mathcal{Z} = (Z_s : s \in \mathcal{S})$. The joint law of $\mathcal{Z}$ is invariant under a common orthogonal rotation of all two dimensional vectors, because the two coordinate processes are independent copies. Let R be an independent random rotation, uniform on the unit circle. Since $(RZ_s : s \in \mathcal{S}) \stackrel{d}{=} \mathcal{Z}$ and both A and $\hat{s}$ depend only on the norms, which are unchanged by R ,

$$\mathbb{P}(A, |v^\top Z_{\hat{s}}| \geq \gamma \|v\| \|Z_{\hat{s}}\|) = \mathbb{E}[\mathbf{1}(A) \mathbb{P}(|v^\top RZ_{\hat{s}}| \geq \gamma \|v\| \|Z_{\hat{s}}\| \mid \mathcal{Z})].$$

On $\{\|Z_{\hat{s}}\| > 0\}$, the direction of $RZ_{\hat{s}}$ is uniform on the circle, and the conditional probability in the last display is π_γ . On $\{\|Z_{\hat{s}}\| = 0\}$, the cone inequality is automatically satisfied. Hence the last display is at least $\pi_\gamma \mathbb{P}(A)$. $\square$

Lemma SA-38 (Weighted bivariate O-U maxima). *Let U_0 and U_1 be independent standard O-U processes with covariance $\mathbb{E}[U_d(s)U_d(t)] = \exp(-|s - t|/2)$, and let*

$$Q_\Lambda(t) = \lambda_0 U_0(t)^2 + \lambda_1 U_1(t)^2, \quad \Lambda = (\lambda_0, \lambda_1),$$

where $\lambda_0, \lambda_1 > 0$. For each fixed $c > 0$, there exist thresholds $w_n(u; \Lambda)$ and a constant $\kappa_\Lambda \in (0, \infty)$ such that, uniformly for u in compact subsets of $\mathbb{R}$,

$$\mathbb{P} \left(\sup_{0 \leq t \leq c \log n} Q_\Lambda(t) \leq w_n(u; \Lambda) \right) = \exp\{-c\kappa_\Lambda e^{-u}\} + o(1),$$

where, with $\lambda_\star = \max\{\lambda_0, \lambda_1\}$,

$$w_n(u; \Lambda) = 2\lambda_\star \log \log n + O(\log \log \log n + |u|).$$

If $\lambda_0 = \lambda_1$, this reduces to Lemma SA-32 after rescaling. If $\lambda_1 > \lambda_0$, then for every $\rho > 0$,

$$\mathbb{P} \left(\sup_{\substack{0 \leq t \leq c \log n \\ |U_0(t)| > \rho |U_1(t)|}} Q_\Lambda(t) > w_n(u; \Lambda) \right) = o(1),$$

provided ρ is fixed. The analogous statement with the roles of the two coordinates reversed holds when $\lambda_0 > \lambda_1$.

Proof. The equal weight case is exactly Lemma SA-32 applied to $\lambda_0^{-1/2} Q_\Lambda^{1/2}$. Suppose next that $\lambda_1 > \lambda_0$. The fixed window tail expansion for stationary Gaussian quadratic forms in Zhdanov [2022, Theorem 1], applied with $d = 2$, $\alpha = 1$, $C_0 = I_2/2$, and $D = \text{diag}(\lambda_0, \lambda_1)$, gives a positive intensity for exceedances of Q_Λ over any fixed interval. The standard O-U exponential mixing/blocking argument then extends the fixed window expansion to intervals of length $c \log n$ and yields the displayed Gumbel limit with some $\kappa_\Lambda \in (0, \infty)$. This is the same blocking step underlying the long-window Darling–Erdos statement above; only the fixed-window tail constant changes.

It remains to justify the displayed cone assertion. On the cone $|z_0| > \rho |z_1|$,

$$\frac{\lambda_0 z_0^2 + \lambda_1 z_1^2}{z_0^2 + z_1^2} \leq \lambda_{\text{bad}}(\rho) = \frac{\lambda_1 + \lambda_0 \rho^2}{1 + \rho^2} < \lambda_1.$$

Thus, on this cone, $Q_\Lambda(t) > w_n(u; \Lambda)$ implies $U_0(t)^2 + U_1(t)^2 > w_n(u; \Lambda)/\lambda_{\text{bad}}(\rho)$. Since $w_n(u; \Lambda)/\lambda_{\text{bad}}(\rho)$ has leading term $2\{\lambda_1/\lambda_{\text{bad}}(\rho)\} \log \log n$ with coefficient strictly larger than 2, Lemma SA-32 for $d = 2$ implies that the probability of such an exceedance over $[0, c \log n]$ tends to zero. The case $\lambda_0 > \lambda_1$ is symmetric. $\square$

SA-5.2 Proof of Theorem SA-1

First, we introduce some notation. Recall that for $\ell \in [p]$, π_ℓ denotes the permutation such that $(x_{\pi_\ell(i)} : 1 \leq i \leq n)$ is nondecreasing. Define the sample means at the left and right leaves at index $k \in [n]$ based on coordinate $\ell \in [p]$ by

$$\hat{\mu}_L(k, \ell) = \frac{1}{k} \sum_{i=1}^k y_{\pi_\ell(i)}, \quad \hat{\mu}_R(k, \ell) = \frac{1}{n-k} \sum_{i=k+1}^n y_{\pi_\ell(i)}, \quad k \in [n], \quad \ell \in [p].$$

Minimizing the *sum of squares* criterion in Equation (SA-3) is equivalent to maximizing the split criterion

$$(\hat{i}, \hat{j}) = \arg \max_{(i, j) \in [n] \times [p]} \mathcal{J}(i, j).$$

where

$$\mathcal{J}(k, \ell) = \frac{k(n-k)}{n} \left(\hat{\mu}_L(k, \ell) - \hat{\mu}_R(k, \ell) \right)^2, \quad k \in [n], \quad \ell \in [p].$$

The endpoint $k = n$ is never a valid split; throughout this proof, maxima over k are understood to range over valid split indices $1 \leq k < n$, or over the corresponding truncated subset. Since multiplying all errors by a positive constant does not change the maximizing split, we normalize $\mathbb{V}[\varepsilon_i] = 1$ in this proof. Moreover, under the constant conditional mean assumption, Assumption SA-1 (2), we have that $\hat{\mu}_L(k, \ell) - \hat{\mu}_R(k, \ell) = \frac{1}{k} \sum_{i=1}^k \varepsilon_{\pi_\ell(i)} - \frac{1}{n-k} \sum_{i=k+1}^n \varepsilon_{\pi_\ell(i)}$. Hence we may w.l.o.g. replace y_i by ε_i in the definition of $\hat{\mu}_L$ and $\hat{\mu}_R$, that is,

$$\hat{\mu}_L(k, \ell) = \frac{1}{k} \sum_{i=1}^k \varepsilon_{\pi_\ell(i)}, \quad \hat{\mu}_R(k, \ell) = \frac{1}{n-k} \sum_{i=k+1}^n \varepsilon_{\pi_\ell(i)}, \quad k \in [n], \quad \ell \in [p].$$

The rest of the proof is organized as follows. In Section SA-5.2, we prove the results under $p = 1$, showing a strong approximation of the split criterion $(\mathcal{J}(k, 1) : k \in [n])$ by the square of a time-transformed Ornstein–Uhlenbeck (O-U) process, and studying the argmax of the split criterion through the argmax of the O-U process. In Section SA-5.2, we generalize to allow for $p \geq 1$. We show that the split criteria over different coordinates, that is, $(\mathcal{J}(k, \ell) : k \in [n])$ for different ℓ 's, are asymptotically independent. This reduces our problem to one-dimensional calculations, and the same technique of approximation by an O-U process from Section SA-5.2 can be used.

Univariate Case

This is the case when $p = 1$. For notational simplicity, define partial sums by

$$S_k = \sum_{i=1}^k \varepsilon_{\pi_1(i)}, \quad k \in [n].$$

By Csörgő and Horváth [1997, Equation A.4.37], there exists a sequence of Brownian bridges $\{B_n(t) : 0 \leq t \leq 1\}$ on a suitable probability space such that

$$\left| \max_{1 \leq k < n} \sqrt{\mathcal{J}(k, 1)} - \sup_{1/n \leq t \leq 1-1/n} \frac{|B_n(t)|}{\sqrt{t(1-t)}} \right| = \left| \max_{1 \leq k < n} \frac{\left| \frac{1}{\sqrt{n}} S_k - \frac{k}{n} \frac{1}{\sqrt{n}} S_n \right|}{\sqrt{(k/n)(1-k/n)}} - \sup_{1/n \leq t \leq 1-1/n} \frac{|B_n(t)|}{\sqrt{t(1-t)}} \right| = \epsilon_n, \quad (\text{SA-10})$$

where $\epsilon_n = o_{\mathbb{P}}((\log \log(n))^{-1/2})$. Although Csörgő and Horváth [1997, Equation A.4.37] bounds the approximation error of the maximum over the full range $1 \leq k < n$ as in (SA-10), the same coupling controls the supremum over any deterministic subset of the split indices. Applying that uniform coupling to the subset $1 \leq k < n^a$ or $n^b < k < n$ gives

$$\left| \max_{\substack{1 \leq k < n^a \\ \text{or } n^b < k < n}} \frac{\left| \frac{1}{\sqrt{n}} S_k - \frac{k}{n} \frac{1}{\sqrt{n}} S_n \right|}{\sqrt{(k/n)(1-k/n)}} - \sup_{\substack{1/n \leq t < n^{a-1} \\ \text{or } n^{b-1} < t \leq 1-1/n}} \frac{|B_n(t)|}{\sqrt{t(1-t)}} \right| = \epsilon_n.$$

The standardized Brownian bridge $\{B_n(t)/\sqrt{t(1-t)} : 0 < t < 1\}$ is distributionally equivalent to a time-transformed Ornstein–Uhlenbeck (O-U) process $\{U(\log(t/(1-t))) : 0 < t < 1\}$, where $\{U(t) : t \in \mathbb{R}\}$ is an O-U process with mean $\mathbb{E}[U(t)] = 0$ and covariance $\mathbb{E}[U(s)U(t)] = e^{-|s-t|/2}$ [Csörgő and Révész, 1981, Section 1.9]. Define

$$A_n = \log\left(\frac{n^{a-1}(n-1)}{1-n^{a-1}}\right), \quad B_n = \log\left(\frac{n^{b-1}(n-1)}{1-n^{b-1}}\right), \quad C_n = 2 \log(n-1).$$

Then $A_n = a \log n + o(\log n)$, $C_n - B_n = (2-b) \log n + o(\log n)$, and $C_n = 2 \log n + o(\log n)$. By stationarity of $|U(t)|$,

$$\begin{aligned} & \mathbb{P}\left(\sup_{1/n \leq t \leq 1-1/n} \frac{|B_n(t)|}{\sqrt{t(1-t)}} > \sup_{\substack{1/n \leq t < n^{a-1} \\ \text{or } n^{b-1} < t \leq 1-1/n}} \frac{|B_n(t)|}{\sqrt{t(1-t)}} + 2\epsilon_n\right) \\ &= \mathbb{P}\left(\sup_{0 \leq t \leq C_n} |U(t)| > \sup_{\substack{0 \leq t < A_n \\ \text{or } B_n < t \leq C_n}} |U(t)| + 2\epsilon_n\right). \end{aligned} \quad (\text{SA-11})$$

Continuing from (SA-11), for any sequence u_n , we have

$$\begin{aligned} & \mathbb{P}\left(\sup_{0 \leq t \leq C_n} |U(t)| > \sup_{\substack{0 \leq t < A_n \\ \text{or } B_n < t \leq C_n}} |U(t)| + 2\epsilon_n\right) \\ & \geq \mathbb{P}\left(\sup_{\substack{0 \leq t < A_n \\ \text{or } B_n < t \leq C_n}} |U(t)| < u_n - 2\epsilon_n\right) - \mathbb{P}\left(\sup_{0 \leq t \leq C_n} |U(t)| < u_n\right). \end{aligned} \quad (\text{SA-12})$$

Since $U(t)$ is a continuous, mean-zero Gaussian process, it induces a centered Gaussian measure on the space of all continuous functions on $[0, C_n]$ equipped with the supremum norm (a separable Banach space). Thus, by the Gaussian correlation inequality [Latała and Matlak, 2017, Remark 3 (i)], we have that

$$\begin{aligned} & \mathbb{P}\left(\sup_{\substack{0 \leq t < A_n \\ \text{or } B_n < t \leq C_n}} |U(t)| < u_n - 2\epsilon_n\right) \\ & \geq \mathbb{P}\left(\sup_{0 \leq t < A_n} |U(t)| < u_n - 2\epsilon_n\right) \cdot \mathbb{P}\left(\sup_{B_n < t \leq C_n} |U(t)| < u_n - 2\epsilon_n\right) \\ & = \mathbb{P}\left(\sup_{0 \leq t < A_n} |U(t)| < u_n - 2\epsilon_n\right) \cdot \mathbb{P}\left(\sup_{0 < t \leq C_n - B_n} |U(t)| < u_n - 2\epsilon_n\right), \end{aligned} \quad (\text{SA-13})$$

where the last equality follows from stationarity.

Remark 3. The next step of our proof relies on a precise characterization of weak convergence for the suprema of a standardized empirical process, as studied in [Eicker, 1979]. However, Eicker [1979, Theorem 5] is incorrectly stated: the $2 \log(c)$ term appearing in the limiting probability should be $\log(c)$. This correction has important implications in our proof. $\perp$

By the Darling–Erdős Limit Theorem for the O-U process [Csörgö and Révész, 1981, Theorem 1.9.1] and [Eicker, 1979, Theorem 2.2 and the correct version of Theorem 5], for all $c > 0$ and $z \in \mathbb{R}$, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq t \leq (c+o(1)) \log(n)} |U(t)| < \frac{2 \log \log(n) + (1/2) \log \log \log(n) + z - (1/2) \log(\pi)}{\sqrt{2 \log \log(n)}} \right) \\ = \exp \left(-e^{-(z - \log(c))} \right). \end{aligned} \quad (\text{SA-14})$$

For a detailed proof of a generalized result on a multidimensional O-U process, see Section SA-4.

Let z^* maximize $z \mapsto \exp(-e^{-(z - \log(2 - (b-a)))}) - \exp(-e^{-(z - \log(2))})$, and set

$$u_n = \frac{2 \log \log(n) + (1/2) \log \log \log(n) + z^* - (1/2) \log(\pi)}{\sqrt{2 \log \log(n)}}.$$

Because $\epsilon_n = o_{\mathbb{P}}((\log \log n)^{-1/2})$, there is a deterministic sequence $\delta_n \downarrow 0$ such that $\mathbb{P}(|\epsilon_n| \leq \delta_n / \sqrt{\log \log n}) \rightarrow 1$. Hence replacing u_n by $u_n \pm 2\epsilon_n$ only changes the Darling–Erdős centering by an $o(1)$ perturbation in the z scale. We combine (SA-11), (SA-12), and (SA-13), and employ (SA-14) three times with $c = 2$, $c = 2 - b$, and $c = a$, using the preceding perturbation observation. We have that

$$\begin{aligned} \liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{1/n \leq t \leq 1-1/n} \frac{|B_n(t)|}{\sqrt{t(1-t)}} > \sup_{\substack{1/n \leq t < n^{a-1} \\ \text{or } n^{b-1} < t \leq 1-1/n}} \frac{|B_n(t)|}{\sqrt{t(1-t)}} + 2\epsilon_n \right) \\ \geq \exp \left(-e^{-(z^* - \log(a))} \right) \cdot \exp \left(-e^{-(z^* - \log(2-b))} \right) - \exp \left(-e^{-(z^* - \log(2))} \right) \\ = \exp \left(-e^{-(z^* - \log(2 - (b-a)))} \right) - \exp \left(-e^{-(z^* - \log(2))} \right) \\ = \frac{b-a}{2} \left(1 - \frac{b-a}{2} \right)^{\frac{2}{b-a} - 1} \\ \geq \frac{b-a}{2e}. \end{aligned} \quad (\text{SA-15})$$

Multivariate Case

For the general case $p \geq 1$, we show that the split criteria over different coordinates, that is, $(\mathcal{J}(k, \ell) : k \in [n])$ for different ℓ 's, are asymptotically independent after truncating away endpoint candidates with negligible probability. This reduces the lower bound calculation to the one-dimensional O-U argument in Section SA-5.2.

To show the split criteria over different coordinates are asymptotically independent, we divide the argument into two steps. In the first step, we show the partial sum process for n indices and p coordinates can be approximated by another partial sum process with Gaussian increments and the same covariance structure. In the second step, we show the covariance between the split criteria over any two different coordinates and any indices is vanishing. Together with Gaussianity, this implies asymptotic independence over different coordinates on the truncated candidate range.

Step 1: Non Gaussian to Gaussian Coupling.

For $1 \leq \ell \leq p$, denote by $H_n^\ell(\frac{k}{n})$ the scaled partial sum for the ℓ -th coordinate evaluated at time $\frac{k}{n}$, that is,

$$\begin{aligned} H_n^\ell \left(\frac{k}{n} \right) &= \sqrt{\frac{n}{k(n-k)}} \left\{ \sum_{i=1}^k \varepsilon_{\pi^\ell(i)} - \frac{k}{n} \sum_{i=1}^n \varepsilon_{\pi^\ell(i)} \right\} \\ &= \sqrt{\frac{n}{k(n-k)}} \sum_{i=1}^n \left(\mathbf{1}(\# \pi^\ell(i) \leq k) - \frac{k}{n} \right) \varepsilon_i, \end{aligned}$$

where $\#\pi^\ell : [n] \rightarrow [n]$ is the inverse mapping of π^ℓ .

We use a truncation argument for the proof. Fix $\varepsilon \in (0, 1)$. Take $r_n = \exp((\log n)^\varepsilon)$. And consider

$$\mathbf{C}_i = \sqrt{n} \left(\left(\sqrt{\frac{n}{k(n-k)}} (\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n}) : r_n \leq k \leq n - r_n \right)^\top : 1 \leq \ell \leq p \right)^\top \varepsilon_i,$$

where $\#\pi^\ell$ denotes the inverse mapping of π^ℓ . The factor $\sqrt{n}$ standardizes the vector. Let P denote the number of displayed coordinates, writing $P = p(n - 2r_n)$ below with the harmless rounding convention for split index cutoffs. Conditional on $\mathcal{B}$, the σ -algebra generated by the p permutations $\pi^1, \dots, \pi^p$, the vectors $\mathbf{C}_i$ are independent, and for all $1 \leq j \leq P$, $1 \leq \ell \leq p$, we have

$$n^{-1} \sum_{i=1}^n \mathbb{E}[C_{ij}^2 | \mathcal{B}] = \frac{n}{k(n-k)} \left[k \left(\frac{n-k}{n} \right)^2 + (n-k) \left(\frac{k}{n} \right)^2 \right] = 1,$$

where we assume row j in $\mathbf{C}_i$ corresponds to $\sqrt{n} \sqrt{\frac{n}{k(n-k)}} (\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n})$. We apply Lemma SA-33 conditionally on $\mathcal{B}$ with the lemma notation

$$m = P, \quad X_i = \mathbf{C}_i, \quad Y_i = \mathbf{D}_i, \quad \mathcal{A} = \mathcal{A}_P^{\text{re}}.$$

Since $r_n = o(n)$ and $p \geq 1$, $P \geq 3$ for all large n ; the finitely many smaller values of n are irrelevant for the asymptotic bound. We bound the quantities in the lemma. Suppose K_1 and K_2 are the universal constants given in Lemma SA-33,

$$\begin{aligned} L_n &= \max_{1 \leq j \leq P} \sum_{i=1}^n \mathbb{E}[|C_{ij}|^3 | \mathcal{B}] / n \\ &= \max_{1 \leq \ell \leq p} \max_{r_n \leq k \leq n - r_n} \left[k \left(\frac{n-k}{n} \right)^{3/2} + (n-k) \left(\frac{k}{n} \right)^{3/2} \right] \mathbb{E}[|\varepsilon_i|^3] / n \\ &\lesssim \max_{1 \leq \ell \leq p} \max_{r_n \leq k \leq n - r_n} \frac{1}{n} \left(\frac{(n-k)^{3/2}}{k^{1/2}} + \frac{k^{3/2}}{(n-k)^{1/2}} \right) \\ &\lesssim \sqrt{n/r_n}. \end{aligned} \tag{SA-16}$$

Take $\bar{L}_n = L_n$, then

$$\phi_n = K_2 \left(\frac{\bar{L}_n^2 \log^4(P)}{n} \right)^{-1/6} = K_2 \left(\frac{r_n}{\log^4(P)} \right)^{1/6}.$$

The definition of $\mathbf{C}_i$ implies C_{ij} is $\sqrt{n/r_n}$ -exponential. Hence

$$\begin{aligned} M_{n,X}(\phi_n) &= n^{-1} \sum_{i=1}^n \mathbb{E} \left[\max_{1 \leq j \leq P} |C_{ij}|^3 \mathbf{1} \left(\max_{1 \leq j \leq P} |C_{ij}| > \sqrt{n}/(4\phi_n \log(P)) \right) \middle| \mathcal{B} \right] \\ &\leq n^{-1} \sum_{i=1}^n \mathbb{E} \left[\max_{1 \leq j \leq P} |C_{ij}|^6 \middle| \mathcal{B} \right]^{1/2} \mathbb{P} \left[\max_{1 \leq j \leq P} |C_{ij}| > \sqrt{n}/(4\phi_n \log(P)) \middle| \mathcal{B} \right]^{1/2} \\ &\leq n^{-1} \sum_{i=1}^n \left[\sum_{1 \leq j \leq P} \mathbb{E}[C_{ij}^6 | \mathcal{B}] \right]^{1/2} \left[\sum_{1 \leq j \leq P} \mathbb{P}(|C_{ij}| > \sqrt{n}/(4\phi_n \log(P)) | \mathcal{B}) \right]^{1/2} \\ &\lesssim n^{-1} \sum_{i=1}^n (P(n/r_n)^3)^{1/2} \left[P \exp \left(- \frac{\sqrt{n}/(4\phi_n \log(P))}{\sqrt{n/r_n}} \right) \right]^{1/2} \\ &\lesssim P(n/r_n^3)^{1/2} \exp \left(- \frac{1}{4} \left(\frac{r_n}{\log P} \right)^{1/3} \right) \\ &\lesssim n^{-2}, \end{aligned}$$

since $r_n = \exp((\log n)^\varepsilon)$ and ε, p are fixed. Conditional on $\mathcal{B}$, let $\mathbf{D}_i, 1 \leq i \leq n$ be independent mean-zero Gaussian random vectors such that

$$\mathbf{D}_i \sim N(\mathbf{0}, \mathbb{E}[\mathbf{C}_i \mathbf{C}_i^\top | \mathcal{B}]), \quad \text{conditional on } \mathcal{B}.$$

Then for each $1 \leq j \leq P$, $1 \leq i \leq n$, we have D_{ij} is $\sqrt{n/r_n}$ -subGaussian. Hence the same argument implies

$$M_{n,Y}(\phi_n) \lesssim n^{-2}.$$

Lemma SA-33 then implies

$$\begin{aligned} \sup_{A \in \mathcal{A}^{\text{re}}} \left| \mathbb{P} \left(n^{-1/2} \sum_{i=1}^n \mathbf{C}_i \in A \middle| \mathcal{B} \right) - \mathbb{P} \left(n^{-1/2} \sum_{i=1}^n \mathbf{D}_i \in A \middle| \mathcal{B} \right) \right| &\leq K_1 \left[\left(\frac{\bar{L}_n^2 \log^7(P)}{n} \right)^{1/6} + \frac{M_{n,X}(\phi_n) + M_{n,Y}(\phi_n)}{\bar{L}_n} \right] \\ &\lesssim \left(\frac{\log^7(P)}{r_n} \right)^{1/6} + \sqrt{\frac{r_n}{n}} \frac{1}{n^2} \\ &\lesssim \left(\frac{\log^7(n)}{r_n} \right)^{1/6}, \end{aligned} \quad (\text{SA-17})$$

where $\mathcal{A}^{\text{re}}$ is the class of all rectangles A of the form

$$A = \{\mathbf{u} \in \mathbb{R}^P : a_j \leq u_j \leq b_j, \forall j = 1, 2, \dots, P\},$$

for some $-\infty \leq a_j \leq b_j \leq \infty$, $j = 1, 2, \dots, P$. In particular, under the variance normalization above, suppose $u_i, 1 \leq i \leq n$ are i.i.d. $N(0, 1)$ random variables. Then $\mathbf{D}_i$ can be taken such that

$$\mathbf{D}_i = \sqrt{n} \left(\left(\sqrt{\frac{n}{k(n-k)}} (\mathbb{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n}) : r_n \leq k \leq n - r_n \right)^\top : 1 \leq \ell \leq p \right)^\top u_i.$$

The above result shows if we define

$$G_n^\ell \left(\frac{k}{n} \right) = \sqrt{\frac{n}{k(n-k)}} \left\{ \sum_{i=1}^k u_{\pi^\ell(i)} - \frac{k}{n} \sum_{i=1}^n u_{\pi^\ell(i)} \right\},$$

then Equation (SA-17) and unconditioning on $\mathcal{B}$, we get

$$\sup_{t_1, \dots, t_p \in \mathbb{R}} \left| \mathbb{P} \left(\max_{r_n \leq k \leq n - r_n} |H_n^\ell(k/n)| \leq t_\ell, 1 \leq \ell \leq p \right) - \mathbb{P} \left(\max_{r_n \leq k \leq n - r_n} |G_n^\ell(k/n)| \leq t_\ell, 1 \leq \ell \leq p \right) \right| \lesssim \left(\frac{\log^7(n)}{r_n} \right)^{1/6}.$$

Step 2: Gaussian to Gaussian Coupling.

For $1 \leq \ell \leq p$, denote by $G_n^\ell(\frac{k}{n})$ the partial sum for the ℓ -th coordinate evaluated at time $\frac{k}{n}$, that is,

$$G_n^\ell \left(\frac{k}{n} \right) = \sqrt{\frac{n}{k(n-k)}} \left\{ \sum_{i=1}^k u_{\pi^\ell(i)} - \frac{k}{n} \sum_{i=1}^n u_{\pi^\ell(i)} \right\}.$$

Let $\mathbf{G}_n = ((G_n^1(k/n) : 1 \leq k < n)^\top, \dots, (G_n^p(k/n) : 1 \leq k < n)^\top)^\top$. Then $\mathbf{G}_n$ is a $(n-1)p$ -dimensional Gaussian random vector; denote its covariance matrix by $\mathbf{\Sigma}_n$. It remains to show that $\mathbf{\Sigma}_n$ is close to one with covariance between different coordinates zero.

Consider two different coordinates, $\ell_1, \ell_2 \in [p]$. Assume w.l.o.g. that $\ell_1 = 1$ and $\ell_2 = 2$. Let $k, j \in [n]$. Denote by σ the sigma-algebra generated by $\pi_1, \dots, \pi_p$. Then

$$\begin{aligned} &\text{Cov} \left[G_n^1 \left(\frac{k}{n} \right), G_n^2 \left(\frac{j}{n} \right) \middle| \sigma \right] \\ &= \sqrt{\frac{n}{k(n-k)} \frac{n}{j(n-j)}} \left\{ \sum_{i=1}^k \sum_{i'=1}^j \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] - \frac{j}{n} \sum_{i=1}^k \sum_{i'=1}^n \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] \right. \\ &\quad \left. - \frac{k}{n} \sum_{i=1}^n \sum_{i'=1}^j \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] + \frac{kj}{n^2} \sum_{i=1}^n \sum_{i'=1}^n \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] \right\} \\ &= \sqrt{\frac{n}{k(n-k)} \frac{n}{j(n-j)}} \frac{jk}{n} \left\{ \frac{n}{jk} \sum_{i=1}^k \sum_{i'=1}^j \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] - 1 \right\}. \end{aligned}$$

To calculate $\sum_{i=1}^k \sum_{i'=1}^j \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma]$, first condition on π_1 , and let $\mathcal{J} = \{\pi_1(i) : 1 \leq i \leq k\}$. Then $\sum_{i=1}^k \sum_{i'=1}^j \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] = |\{i' \in [j] : \pi_2(i') \in \mathcal{J}\}|$. Consider

$$f(\pi) = \frac{n}{jk} |\{i \in [j] : \pi(i) \in \mathcal{J}\}|,$$

where π is a random permutation of $[n]$. Changing the order of the first j values of π does not change the value of $f(\pi)$, and $|f(\pi) - f(\pi^{s,t})| \leq \frac{n}{jk}$ for all $\pi, s \in \{1, \dots, j\}, t \in \{j+1, \dots, n\}$, where the permutation $\pi^{s,t}$ is obtained from π by transposition of its s th and t th coordinates. Below we reduce to $j, k \leq \lceil n/2 \rceil$. Then by Lemma 2 from [El-Yaniv and Pechyony \[2009\]](#), for any $t \geq 0$,

$$\begin{aligned} & \mathbb{P}\left(\left|\frac{n}{jk} \sum_{i=1}^k \sum_{i'=1}^j \mathbb{E}[u_{\pi_1(i)} u_{\pi_2(i')} | \sigma] - 1\right| \geq t \mid \pi_1\right) \\ &= \mathbb{P}(|f(\pi_2) - \mathbb{E}[f(\pi_2)]| \geq t \mid \pi_1) \\ &\leq 2 \exp\left(-\frac{2t^2}{j(\frac{n}{jk})^2} \frac{n-1/2}{n-j} \left(1 - \frac{1}{2 \max(j, n-j)}\right)\right). \end{aligned}$$

Since $\frac{n-1/2}{n-j} \left(1 - \frac{1}{2 \max(j, n-j)}\right) \geq 1 - \frac{1}{n}$, a union bound over all coordinate pairs and all $r_n \leq j, k \leq n - r_n$ gives, with probability tending to one, a positive constant C such that

$$\max_{\ell_1 \neq \ell_2} \max_{r_n \leq j, k \leq n - r_n} \left| \frac{n}{jk} \sum_{i=1}^k \sum_{i'=1}^j \mathbb{E}[u_{\pi_{\ell_1}(i)} u_{\pi_{\ell_2}(i')} | \sigma] - 1 \right| \leq C \sqrt{\log n} \frac{n}{\sqrt{jk}}.$$

which implies

$$\max_{\ell_1 \neq \ell_2} \max_{r_n \leq j, k \leq n - r_n} |\text{Cov}[G_n^{\ell_1}(\frac{k}{n}), G_n^{\ell_2}(\frac{j}{n}) | \sigma]| \leq C \sqrt{\frac{\log n}{r_n}} \quad (\text{SA-18})$$

The reduction to $j, k \leq \lceil n/2 \rceil$ is because

$$\begin{aligned} G_n^\ell\left(\frac{k}{n}\right) &= \sqrt{\frac{n}{k(n-k)}} \left\{ \sum_{i=1}^k u_{\pi^\ell(i)} - \frac{k}{n} \sum_{i=1}^n u_{\pi^\ell(i)} \right\} \\ &= -\sqrt{\frac{n}{k(n-k)}} \left\{ \sum_{i=k+1}^n u_{\pi^\ell(i)} - \frac{n-k}{n} \sum_{i=1}^n u_{\pi^\ell(i)} \right\}. \end{aligned}$$

Consider a $(n-1)p$ -dimensional mean-zero Gaussian random vector

$$\mathbf{Z}_n = ((Z_n^1(k/n) : 1 \leq k < n)^\top, \dots, (Z_n^p(k/n) : 1 \leq k < n)^\top)^\top,$$

where for each $1 \leq \ell \leq p$, $(Z_n^\ell(k/n) : 1 \leq k < n)^\top$ has the same joint distribution as the partial sum random vector $(G_n^\ell(k/n) : 1 \leq k < n)^\top$, and for any $\ell \neq \ell'$ and any valid split indices j, k ,

$$\text{Cov}[Z_n^\ell(j/n), Z_n^{\ell'}(k/n)] = 0.$$

Denote by $\mathbf{\Gamma}_n$ the covariance matrix of $\mathbf{Z}_n$. It remains to show $\mathbf{\Gamma}_n$ is close to $\mathbf{\Sigma}_n$. For a tight control on the rate of convergence, consider the truncated random vector,

$$\begin{aligned} T_{r_n}(\mathbf{G}_n) &= ((G_n^\ell(k/n) : r_n \leq k \leq n - r_n)^\top : 1 \leq \ell \leq p)^\top, \\ T_{r_n}(\mathbf{Z}_n) &= ((Z_n^\ell(k/n) : r_n \leq k \leq n - r_n)^\top : 1 \leq \ell \leq p)^\top. \end{aligned}$$

Also by an abuse of notations, denote by $T_{r_n}(\mathbf{\Sigma}_n)$ and $T_{r_n}(\mathbf{\Gamma}_n)$ the covariance matrix of $T_{r_n}(\mathbf{G}_n)$ and $T_{r_n}(\mathbf{Z}_n)$, respectively. Then Equation (SA-18) implies, with probability tending to one,

$$\|T_{r_n}(\mathbf{\Sigma}_n) - T_{r_n}(\mathbf{\Gamma}_n)\|_{\max} = O\left(\sqrt{\frac{\log n}{r_n}}\right). \quad (\text{SA-19})$$

Additionally, the variance of each item of $T_{r_n}(\mathbf{Z}_n)$ admits the following conditioning lower bound: condition on the permutations π_ℓ , $1 \leq \ell \leq p$, then

$$\begin{aligned} \mathbb{V}[\mathbf{Z}_n^\ell(k/n) | \pi_\ell, 1 \leq \ell \leq p] &= \mathbb{V}[\mathbf{G}_n^\ell(k/n) | \pi_\ell, 1 \leq \ell \leq p] \\ &= \mathbb{V}\left[\sqrt{\frac{n}{k(n-k)}} \left(\sum_{i=1}^k u_{\pi_\ell(i)} - \frac{k}{n} \sum_{i=1}^n u_{\pi_\ell(i)} \right) \middle| \pi_\ell, 1 \leq \ell \leq p\right] \\ &= \mathbb{V}\left[\sqrt{\frac{n}{k(n-k)}} \left(\sum_{i=1}^k u_i - \frac{k}{n} \sum_{i=1}^n u_i \right)\right] \\ &= 1, \quad 1 \leq k < n, 1 \leq \ell \leq p, \end{aligned}$$

where in the third line, we have used the fact that conditional on $\pi_\ell, 1 \leq \ell \leq p$, $(u_{\pi_\ell(i)})_{i \in [n]}$'s are i.i.d. $\mathcal{N}(0, 1)$. Let $\mathcal{G}_n$ be the permutation event on which (SA-19) holds. Since $\mathbb{P}(\mathcal{G}_n) \rightarrow 1$, the following Gaussian comparison is applied conditionally on $\mathcal{B}$ and $\mathcal{G}_n$, and then unconditioned; the complement $\mathcal{G}_n^c$ contributes only $o(1)$. In the notation of Lemma SA-35, take

$$Z_1 = T_{r_n}(\mathbf{G}_n), \quad Z_2 = T_{r_n}(\mathbf{Z}_n), \quad m = pT(n),$$

where $T(n) = \lceil n - r_n \rceil - \lfloor r_n \rfloor$. The covariance matrices are $T_{r_n}(\mathbf{\Sigma}_n)$ and $T_{r_n}(\mathbf{\Gamma}_n)$. The variance lower bound above gives $\underline{\sigma} = 1$, $m \geq 3$ for all large n , and

$$\Delta_n = \|T_{r_n}(\mathbf{\Sigma}_n) - T_{r_n}(\mathbf{\Gamma}_n)\|_{\max}.$$

Therefore Lemma SA-35 gives

$$\sup_{\mathbf{y} \in \mathbb{R}^{pT(n)}} |\mathbb{P}(T_{r_n}(\mathbf{G}_n) \leq \mathbf{y}) - \mathbb{P}(T_{r_n}(\mathbf{Z}_n) \leq \mathbf{y})| \lesssim (\Delta_n \log^2 m)^{1/2} + o(1),$$

Combining with Equation (SA-19) and applying the comparison to the rectangle with coordinatewise bounds $-t_\ell \leq G_n^\ell(k/n) \leq t_\ell$ for each ℓ , we get

$$\begin{aligned} &\sup_{t_1, \dots, t_p \in \mathbb{R}} \left| \mathbb{P}\left(\max_{r_n \leq k \leq n-r_n} |G_n^\ell(k/n)| \leq t_\ell, 1 \leq \ell \leq p\right) - \mathbb{P}\left(\max_{r_n \leq k \leq n-r_n} |Z_n^\ell(k/n)| \leq t_\ell, 1 \leq \ell \leq p\right) \right| \\ &= O\left(\log(n) \left(\frac{\log n}{r_n}\right)^{1/4}\right) + o(1) = o(1). \end{aligned} \tag{SA-20}$$

Step 3: Reduction of calculations for a one-dimensional O-U process

As in the previous two sections, fix $\varepsilon > 0$, and take $r_n = \exp((\log n)^\varepsilon)$. Let $\mathcal{G} = \{\exists \ell \in [p] : \arg \max_k \mathcal{J}(k, \ell) < r_n \text{ or } \arg \max_k \mathcal{J}(k, \ell) > n - r_n\}$. Then by [Csörgő and Horváth, 1997, proof of Theorem A.4.2], and a union bound argument, we have

$$\mathbb{P}(\mathcal{G}) \leq \sum_{\ell=1}^p \mathbb{P}(\arg \max_k \mathcal{J}(k, \ell) < r_n \text{ or } \arg \max_k \mathcal{J}(k, \ell) > n - r_n) = o(1).$$

Hence we may effectively restrict the candidates of $\arg \max$ to $[r_n, n - r_n]$. Because $\log r_n = o(\log n)$, replacing the full endpoint range by this truncated range changes the corresponding O-U time intervals only by $o(\log n)$ and does not alter the Darling–Erdős constants below. W.l.o.g., consider coordinate $\ell = 1$, and

$$\begin{aligned} &\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq i \leq n^b, j = \ell) \\ &= \liminf_{n \rightarrow \infty} \mathbb{P}\left(\max_{k \in [n]} \mathcal{J}(k, 1) > \max_{k, j \neq 1} \mathcal{J}(k, j), \max_{k \in [n]} \mathcal{J}(k, 1) > \max_{k \notin [n^a, n^b]} \mathcal{J}(k, 1)\right) \\ &\geq \liminf_{n \rightarrow \infty} \mathbb{P}\left(\max_{k \in [n]} \mathcal{J}(k, 1) > \max_{k, j \neq 1} \mathcal{J}(k, j), \max_{k \in [n]} \mathcal{J}(k, 1) > \max_{k \notin [n^a, n^b]} \mathcal{J}(k, 1), \mathcal{G}^c\right) - \limsup_{n \rightarrow \infty} \mathbb{P}(\mathcal{G}) \\ &\geq \liminf_{n \rightarrow \infty} \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} \mathcal{J}(k, 1) > \max_{k, j \neq 1} \mathcal{J}(k, j), \max_{k \in [r_n, n-r_n]} \mathcal{J}(k, 1) > \max_{k \notin [n^a, n^b]} \mathcal{J}(k, 1)\right). \end{aligned}$$

The *coupling result* developed previously applies. With our notation, $\mathcal{J}(k, \ell) = (H_n^\ell(k/n))^2$. Hence

$$\begin{aligned}
& \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} \mathcal{J}(k, 1) > \max_{j \neq 1, k \in [r_n, n-r_n]} \mathcal{J}(k, j), \max_{k \in [r_n, n-r_n]} \mathcal{J}(k, 1) > \max_{k \notin [n^a, n^b]} \mathcal{J}(k, 1)\right) \\
&= \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} |H_n^1\left(\frac{k}{n}\right)| > \max_{\ell \neq 1, k \in [r_n, n-r_n]} |H_n^\ell\left(\frac{k}{n}\right)|, \max_{k \in [r_n, n-r_n]} |H_n^1\left(\frac{k}{n}\right)| > \max_{k \notin [n^a, n^b]} |H_n^1\left(\frac{k}{n}\right)|\right) \\
&\geq \sup_{z \in \mathbb{R}} \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} |H_n^1\left(\frac{k}{n}\right)| > z > \max_{\ell \neq 1, k \in [r_n, n-r_n]} |H_n^\ell\left(\frac{k}{n}\right)|, \right. \\
&\quad \left. \max_{k \in [r_n, n-r_n]} |H_n^1\left(\frac{k}{n}\right)| > z > \max_{\substack{k \in [r_n, n-r_n] \\ k \notin [n^a, n^b]}} |H_n^1\left(\frac{k}{n}\right)|\right) - o(1) \\
&\geq \sup_{z \in \mathbb{R}} \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} |Z_n^1\left(\frac{k}{n}\right)| > z > \max_{\ell \neq 1, k \in [r_n, n-r_n]} |Z_n^\ell\left(\frac{k}{n}\right)|, \right. \\
&\quad \left. \max_{k \in [r_n, n-r_n]} |Z_n^1\left(\frac{k}{n}\right)| > z > \max_{\substack{k \in [r_n, n-r_n] \\ k \notin [n^a, n^b]}} |Z_n^1\left(\frac{k}{n}\right)|\right) - o(1),
\end{aligned}$$

where the first $o(1)$ accounts for removing endpoint candidates from the outside window maximum before the threshold comparison; the same O-U endpoint bound as above applies because $\log r_n = o(\log n)$. We have also used the rectangle approximation in (SA-17), the Gaussian comparison in (SA-20), and a deterministic buffer tending to zero; the strict inequalities are handled by replacing z with $z \pm \eta_n$ and then taking $\eta_n \downarrow 0$. For the extreme value limit, restrict the preceding supremum to deterministic thresholds

$$z_n(u) = \frac{2 \log \log(n) + 1/2 \log \log \log(n) + u - 1/2 \log(\pi)}{\sqrt{2 \log \log(n)}}, \quad u \in \mathbb{R}.$$

The comparison error is uniform over rectangles, so restricting to this threshold family gives a valid lower bound after taking the supremum over u . Since we choose $r_n = \exp((\log n)^\varepsilon)$, both $\log^{7/6}(n)r_n^{-1/6} = o(1)$ and $\log(n)(\log n/r_n)^{1/4} = o(1)$. It then follows from independence and symmetry between Z_n^ℓ 's across different ℓ 's that

$$\begin{aligned}
& \liminf_{n \rightarrow \infty} \sup_{u \in \mathbb{R}} \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} |Z_n^1\left(\frac{k}{n}\right)| > z_n(u) > \max_{\ell \neq 1, k \in [r_n, n-r_n]} |Z_n^\ell\left(\frac{k}{n}\right)|, \right. \\
&\quad \left. \max_{k \in [r_n, n-r_n]} |Z_n^1\left(\frac{k}{n}\right)| > z_n(u) > \max_{\substack{k \in [r_n, n-r_n] \\ k \notin [n^a, n^b]}} |Z_n^1\left(\frac{k}{n}\right)|\right) \\
&\geq \liminf_{n \rightarrow \infty} \sup_{u \in \mathbb{R}} \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} |Z_n^1\left(\frac{k}{n}\right)| < z_n(u)\right)^{p-1} \\
&\quad \cdot \mathbb{P}\left(\max_{k \in [r_n, n-r_n]} |Z_n^1\left(\frac{k}{n}\right)| > z_n(u) > \max_{\substack{k \in [r_n, n-r_n] \\ k \notin [n^a, n^b]}} |Z_n^1\left(\frac{k}{n}\right)|\right) \\
&\geq \sup_{u \in \mathbb{R}} \exp\left(-(p-1)e^{-(u-\log(2))}\right) \left(\exp\left(-e^{-(u-\log(2)-(b-a))}\right) - \exp\left(-e^{-(u-\log(2))}\right)\right) \\
&= \frac{b-a}{2p} \left(1 - \frac{b-a}{2p}\right)^{\frac{2p}{b-a}-1} \\
&\geq \frac{b-a}{2pe},
\end{aligned}$$

where the third line is by a similar calculation as in Section SA-5.2. Putting these bounds together, we have

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i} \leq n^b, \hat{j} = \ell) \geq \frac{b-a}{2pe},$$

and by symmetry, we have

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i} \leq n - n^a, \hat{j} = \ell) \geq \frac{b-a}{2pe}.$$

SA-5.3 Proof of Theorem SA-2

For simplicity, we denote $\hat{\mu}^{\text{stump}}(\mathbf{x})$ by $\hat{\mu}(\mathbf{x})$. The split index arguments use the variance normalization from Theorem SA-1; for a general error variance σ^2 , apply the same proof to ε_i/σ and multiply the final estimation error threshold by σ . We divide the proofs into two parts, one for uniform estimation and one for pointwise results near the boundary.

Part 1: Inconsistency for Uniform Estimation Rates

For notational simplicity, introduce the *partial sum based on ordering for the ℓ 's coordinate*,

$$S(k, \ell) = \sum_{i=1}^k \varepsilon_{\pi_\ell(i)}, \quad k \in [n], \quad \ell \in [p],$$

and define the optimal index for splitting based on the ℓ 's coordinate by

$$\iota_\ell = \arg \max_{k \in [n]} \mathcal{J}(k, \ell), \quad \ell \in [p].$$

Consider the event

$$\begin{aligned} \text{Imbalance}_\ell &= \{\hat{j} = \ell, \hat{i} < n^b \text{ or } \hat{i} > n - n^b\} \\ &= \{\max_k \mathcal{J}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}(k, j), \max_k \mathcal{J}(k, \ell) > \max_{k \in [n^b, n-n^b]} \mathcal{J}(k, \ell)\}, \quad \ell \in [p]. \end{aligned}$$

Consider the case $\hat{i} < n^b$ on Imbalance_ℓ . The other case where $\hat{i} > n - n^b$ can be dealt with by symmetry. Then

$$\begin{aligned} &\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\mu}(\mathbf{x}) - \mu|^2 \\ &\geq \frac{S^2(\iota_\ell, \ell)}{\iota_\ell^2} \\ &\geq \frac{1}{\iota_\ell} \left[\frac{S^2(\iota_\ell, \ell)}{\iota_\ell} + \frac{(S(n, \ell) - S(\iota_\ell, \ell))^2}{n - \iota_\ell} - \frac{(S(n, \ell) - S(\iota_\ell, \ell))^2}{n - \iota_\ell} \right] \\ &\geq \frac{1}{\min\{\iota_\ell, n - \iota_\ell\}} \left(\max_{1 \leq k < n} \left(\frac{S^2(k, \ell)}{k} + \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} \right) \right. \\ &\quad \left. - \max_{\lfloor n/2 \rfloor \leq k \leq n} \frac{S^2(k, \ell)}{k} - \max_{1 \leq k \leq \lceil n/2 \rceil} \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} \right). \end{aligned}$$

where the last line is because ι_ℓ is the index that maximizes the split criterion based on the ℓ 's coordinate, i.e.,

$$\begin{aligned} \iota_\ell &= \arg \max_{1 \leq k < n} \sum_{i=1}^k (y_i - \bar{y})^2 - \sum_{i=1}^k (y_{\pi_\ell(i)} - S(k, \ell)/k)^2 - \sum_{i=k+1}^n (y_i - (S(n, \ell) - S(k, \ell))/(n - k))^2 \\ &= \arg \max_{1 \leq k < n} \frac{S^2(k, \ell)}{k} + \frac{(S(n, \ell) - S(k, \ell))^2}{n - k}. \end{aligned}$$

Fix $\epsilon > 0$. Consider the events

$$\begin{aligned} A_\ell^\epsilon &= \left\{ \max_{1 \leq k < n} \left(\frac{S^2(k, \ell)}{k} + \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} \right) \geq (2 - \epsilon) \log \log(n) \right\}, \\ B_\ell^\epsilon &= \left\{ \max_{\lfloor n/2 \rfloor \leq k \leq n} \frac{S^2(k, \ell)}{k} + \max_{1 \leq k \leq \lceil n/2 \rceil} \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} \leq 2\epsilon \log \log(n) \right\}. \end{aligned}$$

By [Csörgö and Horváth, 1997, Theorem A.4.1] $\liminf_{n \rightarrow \infty} \mathbb{P}(A_\ell^\epsilon) = \liminf_{n \rightarrow \infty} \mathbb{P}(B_\ell^\epsilon) = 1$. Hence for any $\epsilon > 0$,

$$\mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\mu}(\mathbf{x}) - \mu|^2 \geq \frac{(2 - 3\epsilon) \log \log(n)}{n^b} \right) \geq \sum_{\ell=1}^p \mathbb{P}(\text{Imbalance}_\ell \cap A_\ell^\epsilon \cap B_\ell^\epsilon) \geq \frac{b}{e} + o(1),$$

where we have used the fact that Imbalance_ℓ 's are disjoint for different ℓ 's and Theorem SA-1. Equation (SA-5) then follows.

Part 2: Inconsistency for Points Near the Boundary

Consider the event

$$\begin{aligned} \text{Off}_\ell &= \{\hat{j} = \ell, \hat{i} \in [n^a, n^b]\} \\ &= \{\max_k \mathcal{J}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}(k, j), \max_k \mathcal{J}(k, \ell) > \max_{k \notin [n^a, n^b]} \mathcal{J}(k, \ell)\}, \quad \ell \in [p]. \end{aligned}$$

On Off_ℓ , $\iota_\ell \geq n^a$, so $x_{\pi_\ell(\iota_\ell), \ell} \geq x_{\pi_\ell(\lceil n^a \rceil), \ell}$. For any deterministic sequence $\eta_n \downarrow 0$, the marginal probability integral transform gives

$$\mathbb{P}(x_{\pi_\ell(\lceil n^a \rceil), \ell} \geq \eta_n n^{a-1}) \rightarrow 1.$$

Together with Theorem SA-1,

$$\mathbb{P}(\text{Off}_\ell, x_{\pi_\ell(\iota_\ell), \ell} \geq \eta_n n^{a-1}) \geq \frac{b-a}{2pe} + o(1).$$

Then on the event Off_ℓ and $x_{\pi_\ell(\iota_\ell), \ell} \geq \eta_n n^{a-1}$, for any $\mathbf{x} \in [0, 1]^p$ such that $x_\ell \leq \eta_n n^{a-1}$, we have $x_\ell \leq x_{\pi_\ell(\iota_\ell), \ell}$, and

$$\begin{aligned} |\hat{\mu}(\mathbf{x}) - \mu|^2 &= \frac{S^2(\iota_\ell, \ell)}{\iota_\ell^2} \\ &= \frac{1}{\iota_\ell} \left(\frac{S^2(\iota_\ell, \ell)}{\iota_\ell} + \frac{(S(n, \ell) - S(\iota_\ell, \ell))^2}{n - \iota_\ell} - \frac{(S(n, \ell) - S(\iota_\ell, \ell))^2}{n - \iota_\ell} \right) \\ &\geq \frac{1}{\iota_\ell} \left(\max_{1 \leq k < n} \left(\frac{S^2(k, \ell)}{k} + \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} \right) - \max_{1 \leq k \leq n^b} \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} \right). \end{aligned}$$

The last step uses the same boundary transfer as Part 1, but we record the two ingredients explicitly. On the event where coordinate ℓ is selected with $n^a \leq \iota_\ell \leq n^b$, the selected split attains the full CART split maximum along that coordinate. The first part of this proof and the preceding comparison imply that this full maximum is $(2 + o_{\mathbb{P}}(1)) \log \log(n)$ on the selected imbalanced window event. In contrast,

$$\max_{1 \leq k \leq n^b} \frac{(S(n, \ell) - S(k, \ell))^2}{n - k} = O_{\mathbb{P}}(1) + O_{\mathbb{P}}\left(\frac{n^b \log \log(n)}{n}\right) = o_{\mathbb{P}}(\log \log(n)),$$

because $n - k \asymp n$ over $k \leq n^b$, $S^2(n, \ell)/n = O_{\mathbb{P}}(1)$, and the maximal partial sum bound gives $\max_{k \leq n^b} S^2(k, \ell) = O_{\mathbb{P}}(n^b \log \log(n))$. Therefore

$$\liminf_{n \rightarrow \infty} \inf_{\mathbf{x} \in \mathcal{X}_n(a, \eta_n)} \mathbb{P}\left(|\hat{\mu}(\mathbf{x}) - \mu|^2 \geq \frac{(2 + o(1)) \log \log(n)}{n^b}\right) \geq \frac{b-a}{2pe},$$

which is Equation (SA-6).

SA-5.4 Proof of Theorem SA-3

Let

$$\bar{\varepsilon}(\mathbf{t}) = \frac{1}{n(\mathbf{t})} \sum_{\mathbf{x}_i \in \mathbf{t}} \varepsilon_i$$

denote the centered average over a node. By Part 1 in the proof of Theorem SA-2, with probability at least $b/e + o(1)$ the root split creates a child node $\mathbf{t}$ such that

$$|\bar{\varepsilon}(\mathbf{t})| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log n}.$$

Any later recursive splitting only refines this root child. If $\mathcal{P}(\mathbf{t})$ denotes the collection of terminal descendants of $\mathbf{t}$ in the final tree, with $\mathcal{P}(\mathbf{t}) = \{\mathbf{t}\}$ if the child is never split again, then

$$\bar{\varepsilon}(\mathbf{t}) = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t})} \frac{n(\mathbf{t}')}{n(\mathbf{t})} \bar{\varepsilon}(\mathbf{t}').$$

Thus at least one terminal descendant $\mathbf{t}' \in \mathcal{P}(\mathbf{t})$ satisfies $|\bar{\varepsilon}(\mathbf{t}')| \geq |\bar{\varepsilon}(\mathbf{t})|$. Since $y_i = \mu + \varepsilon_i$, the terminal node estimator equals $\mu + \bar{\varepsilon}(\mathbf{t}')$ on that descendant. Taking the supremum over $\mathbf{x} \in \mathcal{X}$ and using the root split event proves the theorem.

SA-5.5 Proof of Theorem SA-4

This follows directly from [Klusowski and Tian \[2024, Theorem 4.3\]](#), choosing $g^* \equiv \mu$ and $g \equiv \mu$, and changing the subGaussian rate to the subexponential rate by choosing $U \asymp \log(n)$ instead of $U \asymp \sqrt{\log(n)}$ in the truncation argument step. The last statement follows from the proof of [Klusowski and Tian \[2024, Theorem 4.3\]](#).

SA-5.6 Proof of Theorem SA-5

Throughout the proof, abbreviate the honest tree $\hat{\mu}^{\text{HON}}(\mathbf{x})$ by $\check{\mu}(\mathbf{x})$. Use $(y_i, \mathbf{x}_i^\top)_{i=1}^M$ to denote the construction sample $\mathcal{D}_{\text{HON},1}$ and $(\tilde{y}_i, \tilde{\mathbf{x}}_i^\top)_{i=1}^N$ to denote the estimation sample $\mathcal{D}_{\text{HON},2}$, with $n \lesssim M, N \lesssim n$. Let $(\hat{i}, \hat{j})$ be the root splitting index and coordinate selected from the construction sample, and let

$$\mathbf{t}_L = \{\mathbf{x} \in \mathcal{X} : x_{\hat{j}} \leq x_{\pi_{\hat{j}}(\hat{i}, \hat{j})}\}$$

be the corresponding left root child. If the target tree has depth larger than one, later splits only refine $\mathbf{t}_L$. Let $\mathcal{P}(\mathbf{t}_L)$ denote the terminal descendants of this child, with $\mathcal{P}(\mathbf{t}_L) = \{\mathbf{t}_L\}$ for a stump. Define the number of estimation fold observations falling in the root child by

$$\tilde{i} = \sum_{i=1}^N \mathbf{1}(\tilde{x}_{i,\hat{j}} \leq x_{\pi_{\hat{j}}(\hat{i}, \hat{j})}), \quad \mathcal{A}_{\tilde{i}} = \{\tilde{i} > 0\}.$$

On $\mathcal{A}_{\tilde{i}}$, choose a terminal descendant $\mathbf{t}_\star \in \mathcal{P}(\mathbf{t}_L)$ with positive estimation fold count, using any fixed deterministic rule, and let $\tilde{i}_\star = n_\mu(\mathbf{t}_\star)$. This rule depends only on the construction tree and the estimation fold covariates, not on the estimation fold outcomes. Hence, conditional on the selected descendant and its estimation fold count, the selected estimation errors remain independent draws from the original error law. Then $1 \leq \tilde{i}_\star \leq \tilde{i}$, and for any $\mathbf{x}_\star \in \mathbf{t}_\star$,

$$\sup_{\mathbf{x} \in \mathcal{X}} |\check{\mu}(\mathbf{x}) - \mu| \geq |\check{\mu}(\mathbf{x}_\star) - \mu| = \left| \frac{1}{\tilde{i}_\star} \sum_{i: \tilde{\mathbf{x}}_i \in \mathbf{t}_\star} (\tilde{y}_i - \mu) \right|.$$

Conditional on the tree, the estimation fold covariates, and $\tilde{i}_\star$, the errors in $\mathbf{t}_\star$ are i.i.d. and independent of the selected partition. Let $\sigma_\mu^2 = \mathbb{V}[y_i] > 0$. The exponential moment assumption implies a finite fourth moment, so Rosenthal's inequality and Paley-Zygmund applied to the square of the displayed average give

$$\mathbb{P} \left(|\check{\mu}(\mathbf{x}_\star) - \mu| \geq c_\mu \frac{\mathbb{E}[|y_i - \mu|]}{\sqrt{\tilde{i}_\star}} \mid \mathbf{T}, \tilde{\mathbf{X}}, \tilde{i}_\star \right) \geq c_\mu \frac{\mathbb{E}[|y_i - \mu|]^2}{\mathbb{V}[y_i]},$$

for a positive constant c_μ depending only on the distribution of y_i . It remains to obtain a high probability upper bound on $\tilde{i}$ given ν . Work on the left boundary event and fix a selected coordinate j and split index k . Let F_j be the marginal distribution function of x_{ij} , and set $U_{(k),j} = F_j(x_{\pi_j(k),j})$. Because the CART criterion in this constant regression model depends on the rank ordering and errors, but not on the order statistic spacings, conditional on $(\hat{j}, \hat{i}) = (j, k)$ the variable $U_{(k),j}$ has the $\text{Beta}(k, M - k + 1)$ distribution. Suppose $1 \leq k \leq M/2$. By a Bernstein bound for Beta variables [[Skorski, 2023, Theorem 1](#)], we have for all $\epsilon > 0$,

$$\mathbb{P}(U_{(k),j} > k/M + \epsilon) \leq \exp \left(- \frac{\epsilon^2}{2v + \frac{c\epsilon}{3}} \right),$$

where for large enough n ,

$$\begin{aligned} v &= \frac{k(M - k + 1)}{(M + 1)^2(M + 2)} \leq 2 \frac{k}{M^2}, \\ c &= \frac{2(M - 2k + 1)}{M(M + 2)} \leq \frac{2}{M}. \end{aligned}$$

Hence with probability at least $1 - M^{-1}$,

$$U_{(k),j} \leq k/M + 2 \frac{\sqrt{\log(M)k}}{M} + 3 \frac{\log(M)}{M}.$$

Conditional on the construction sample and $(\hat{j}, \hat{i}) = (j, k)$, the indicators $\mathbf{1}(\tilde{x}_{i,j} \leq x_{\pi_j(k),j})$ are i.i.d. $\text{Bernoulli}(U_{(k),j})$. Hence, conditional on the construction sample, with probability at least $1 - N^{-1}$,

$$\tilde{i}/N \leq U_{(i),\hat{j}} + 2\sqrt{\frac{\log(N)U_{(i),\hat{j}}}{N}}.$$

Hence, conditional on the event $\hat{i} \leq M^b$, we have with probability at least $1 - 2N^{-1}$,

$$\tilde{i} \leq Cn^b,$$

where C is a constant depending only on the lower and upper limiting construction to estimation sample size ratios. In addition, the beta binomial identity for the honest fold count gives a constant $c_0 > 0$, depending only on the limiting ratio of M and N , such that uniformly over $1 \leq k \leq M^b$ and $j \in [p]$,

$$\mathbb{P}(\mathcal{A}_{\tilde{i}} \mid \hat{i} = k, \hat{j} = j) \geq c_0 + o(1).$$

Therefore the event $\mathcal{B}_n = \{\mathcal{A}_{\tilde{i}}, \tilde{i} \leq Cn^b\}$ has conditional probability bounded away from zero on $\{\hat{i} \leq M^b\}$, up to the same $O(n^{-1})$ error; the constant is absorbed below.

Theorem SA-1 gives the corresponding root split lower bound. Using the left boundary event,

$$\liminf_{M \rightarrow \infty} \mathbb{P}(\hat{i} \leq M^b) \geq \frac{b}{2e}.$$

The right boundary event only improves the constant. On $\mathcal{B}_n$, $\tilde{i}_* \leq \tilde{i} \leq Cn^b$, so the conditional lower bound above implies

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\tilde{\mu}(\mathbf{x}) - \mu| \geq \frac{C_1 \mathbb{E}[|y_i - \mu|]}{n^{b/2}}\right) \geq C_2 \frac{\mathbb{E}[|y_i - \mu|^2]}{\mathbb{V}[y_i]} b,$$

after absorbing fixed constants. This proves the conclusion.

SA-5.7 Proof of Theorem SA-6

For notational simplicity, we use $\mathbf{T}$ to denote the data-driven decision tree. We follow the proof strategy from Klusowski and Tian [2024, Theorem 4.3], conditioning on $\mathcal{D}_{\mathbf{T}}$. Denote by $\mathcal{G}_0$ the class of constant functions. Decompose $\|\hat{\mu}(\mathbf{T}) - \mu\|^2 = E_1 + E_2$, where

$$E_1 = \|\hat{\mu}(\mathbf{T}) - \mu\|^2 - 2(\|y - \hat{\mu}(\mathbf{T})\|_{\mathcal{D}_{\mu}}^2 - \|y - \mu\|_{\mathcal{D}_{\mu}}^2) - \alpha - \beta,$$

and

$$E_2 = 2(\|y - \hat{\mu}(\mathbf{T})\|_{\mathcal{D}_{\mu}}^2 - \|y - \mu\|_{\mathcal{D}_{\mu}}^2) + \alpha + \beta.$$

Denote the partition for $\mathbf{T}$ by $\mathcal{P}$. Since $\mathcal{P}$ is independent of $\mathcal{D}_{\mu}$, the bound (E.27) from Klusowski and Tian [2024] does not apply automatically. Instead, we consider $\mathcal{G}_0$ as the reference class. Given the partitions of $\mathbf{T}$, the values of leaf nodes are obtained by least squares projection using $\mathcal{D}_{\mu}$. This immediately implies

$$\|y - \hat{\mu}(\mathbf{T})\|_{\mathcal{D}_{\mu}}^2 \leq \|y - \bar{y}\|_{\mathcal{D}_{\mu}}^2 \leq \|y - g\|_{\mathcal{D}_{\mu}}^2,$$

for any constant function $g \in \mathcal{G}_0$. Hence for all $g \in \mathcal{G}_0$,

$$\begin{aligned} \mathbb{E}_{\mathcal{D}_{\mu}}[E_2 | \mathcal{D}_{\mathbf{T}}] &\leq 2\mathbb{E}_{\mathcal{D}_{\mu}}[\|y - g\|_{\mathcal{D}_{\mu}}^2 - \|y - \mu\|_{\mathcal{D}_{\mu}}^2 | \mathcal{D}_{\mathbf{T}}] + \alpha + \beta \\ &= 2\|g - \mu\|^2 + \alpha + \beta. \end{aligned}$$

For the term E_1 , first assume $|y_i| \leq U$. Conditional on $\mathcal{D}_{\mathbf{T}}$, $\hat{\mu}(\mathbf{T})$ is still a member of the class $\mathcal{G}_{n_{\mathbf{T}}}[\mathcal{P}]$, which is the collection of all piecewise constant functions (bounded by U) on the partition $\mathcal{P}$. Since for any $\varepsilon \in (0, 1)$,

$$N(\varepsilon U, \mathcal{G}_{n_{\mathbf{T}}}[\mathcal{P}], \|\cdot\|_{P_{X^{n_{\mu}}}}) \leq N(\varepsilon U, \mathcal{G}_{n_{\mathbf{T}}}[\mathcal{P}], \|\cdot\|_{\infty}) \leq \left(\frac{2}{\varepsilon}\right)^{2^{\kappa}},$$

Györfi et al. [2002, Theorem 11.4] and the same argument from Equation (B.30) to (B.33) in Klusowski and Tian [2024] still give

$$\mathbb{P}_{\mathcal{D}_\mu}(E_1 \geq 0 \mid \mathcal{D}_\mathsf{T}) \leq 14 \left(\frac{2U^2}{\beta} \right)^{2^\kappa} \exp \left(- \frac{\alpha n_\mu}{2568U^4} \right).$$

Choosing $\alpha \asymp U^4 2^K \log(n_\mu)/n_\mu$ and $\beta \asymp U^2/n_\mu$ makes the preceding conditional probability bound polynomially small and gives the corresponding conditional expectation bound on the bounded outcome event. Taking $U = C_0 \log(n_\mu)$ and using the same truncation argument over the subexponential ε_i 's yields both the expectation and high probability conclusions after unconditioning on $\mathcal{D}_\mathsf{T}$. The sample size balance condition then replaces n_μ by n up to constants depending only on ρ .

SA-5.8 Proof of Lemma SA-7

For the honest estimator, condition on the selected tree partition and on the covariates in the final estimation sample. If the terminal node containing $\mathbf{x}$ has positive final estimation count, the conditional expectation of the terminal average is μ ; if the count is zero, the estimator is defined to be zero. Taking expectations gives the displayed empty cell expression.

For the estimator without sample splitting, condition on the covariates and write $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)$. Assumption SA-1 makes ε independent of the covariates, and the additional residual symmetry implies $\varepsilon \stackrel{d}{=} -\varepsilon$ conditional on the covariates. Under the constant mean model, every CART split criterion is a function of squared differences of residual averages, and is therefore invariant under the global sign flip $\varepsilon \mapsto -\varepsilon$. The valid candidate set depends only on the covariates and the deterministic tie breaking rule is fixed, so the selected recursive partition is also invariant under this sign flip. The terminal node containing $\mathbf{x}$ is nonempty because the NSS construction and final estimation use the same sample, and the centered terminal average $\hat{\mu}^{\text{NSS}}(\mathbf{x}) - \mu$ is an odd function of ε . Conditional central symmetry of ε therefore gives conditional expectation zero, and the result follows by iterated expectations.

SA-5.9 Proof of Corollary SA-8

Let $\tilde{y}_i = y_i(d_i - \xi)/\{\xi(1 - \xi)\}$ and $\tilde{\varepsilon}_i = \tilde{y}_i - \tau$. Under Assumption SA-2, $\mathbb{E}[\tilde{y}_i \mid \mathbf{x}_i] = \tau$, $\tilde{\varepsilon}_i$ is independent of $\mathbf{x}_i$, and $\tilde{\varepsilon}_i$ satisfies the same subexponential and nondegeneracy conditions needed in Assumption SA-1. The NSS-IPW split criterion is exactly the CART split criterion applied to $\tilde{y}_i$. Therefore Theorem SA-1 applied to the transformed outcome regression problem gives the stated split index result.

SA-5.10 Proof of Corollary SA-9

With the same transformed outcome $\tilde{y}_i = y_i(d_i - \xi)/\{\xi(1 - \xi)\}$, the IPW terminal node estimate in a stump is the corresponding left or right node average of $\tilde{y}_i$. Hence Theorem SA-2 applied to $\tilde{y}_i$ gives both the uniform stump lower bound and the boundary lower bound, with $\sigma^2 = \mathbb{V}(\tilde{\varepsilon}_i)$. The boundary statement follows after the same marginal probability integral transform normalization used in Theorem SA-2.

SA-5.11 Proof of Corollary SA-10

Let

$$\tilde{y}_i = y_i \frac{d_i - \xi}{\xi(1 - \xi)}, \quad \tilde{\varepsilon}_i = \tilde{y}_i - \tau.$$

Under Assumption SA-2, $\mathbb{E}[\tilde{y}_i \mid \mathbf{x}_i] = \tau$ and $\tilde{\varepsilon}_i$ is independent of $\mathbf{x}_i$. The NSS-IPW tree is therefore the CART tree constructed from the transformed outcomes $\tilde{y}_i$, and each terminal node IPW estimate is exactly the corresponding terminal node average of $\tilde{y}_i$.

By Part 1 in the proof of Theorem SA-2 applied to $\tilde{y}_i$, with probability at least $b/e + o(1)$ the root split creates a child node $\mathbf{t}$ with $n(\mathbf{t}) \leq n^b$ and

$$\left| \frac{1}{n(\mathbf{t})} \sum_{\mathbf{x}_i \in \mathbf{t}} (\tilde{y}_i - \tau) \right| \geq c n^{-b/2} \sqrt{\log \log n}$$

for a positive constant c depending only on the distribution of $\tilde{\varepsilon}_i$. Any later recursive splitting only refines $\mathbf{t}$. If $\mathcal{P}(\mathbf{t})$ denotes the terminal descendants of $\mathbf{t}$, then the root child average is the convex combination

$$\frac{1}{n(\mathbf{t})} \sum_{\mathbf{x}_i \in \mathbf{t}} (\tilde{y}_i - \tau) = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t})} \frac{n(\mathbf{t}')}{n(\mathbf{t})} \left(\frac{1}{n(\mathbf{t}')} \sum_{\mathbf{x}_i \in \mathbf{t}'} (\tilde{y}_i - \tau) \right).$$

Hence at least one terminal descendant has absolute IPW estimation error at least as large as the root child transformed outcome average. Taking the supremum over terminal nodes proves the result, after renaming the positive constant.

SA-5.12 Proof of Corollary SA-11

The NSS-IPW estimator is the CART regression estimator based on the transformed outcome $\tilde{y}_i$. Applying Theorem SA-4 to this transformed outcome regression problem gives the expectation and probability bounds, with constants depending on the distribution of $\tilde{\varepsilon}_i$.

SA-5.13 Proof of Corollary SA-12

Conditional on the construction fold, the honest IPW estimator averages $\tilde{y}_i$ over the independent estimation fold in the selected terminal node. Thus Theorem SA-5 applies to the transformed outcome regression problem. The constants C_1 and C_2 in Corollary SA-12 are the transformed outcome versions of the constants in Theorem SA-5, with fixed factors absorbed into their definitions.

SA-5.14 Proof of Corollary SA-13

Apply Theorem SA-6 to the transformed outcome regression problem. The sample splitting ratio condition is the same, with n_μ in the regression statement replaced by the causal estimation fold size n_τ .

SA-5.15 Proof of Lemma SA-14

Since the number of coordinates p is fixed, a union bound over the approximation error for the p coordinates lets us assume w.l.o.g. that $p = 1$ and drop the second index on the coordinate ℓ from $\mathcal{J}^{\text{DIM}}(k, \ell)$ and $\bar{\mathcal{J}}^{\text{IPW}}(k, \ell)$ everywhere. Throughout, we assume the data is already sorted so that

$$x_1 \leq x_2 \leq \dots \leq x_n.$$

All ratios below are evaluated only for valid candidates in $\mathcal{V}_{\text{DIM}}$; equivalently, every maximum over an index range is taken over the intersection of that range with the valid candidate set. Displays containing denominators such as $b_i = \sum_{\ell \leq i} d_\ell$ are read only on indices where the denominator is positive; invalid zero denominator indices are omitted by the valid candidate convention. On the balanced ranges considered in this lemma, the omitted invalid candidate event has probability tending to zero.

Expand the square. Define

$$\begin{aligned} R_1(k) &= \hat{\tau}_{\mathbf{t}_L}^{\text{DIM}}(k) - \hat{\tau}_{\mathbf{t}_R}^{\text{DIM}}(k) + \bar{\tau}_{\mathbf{t}_L}^{\text{IPW}}(k) - \bar{\tau}_{\mathbf{t}_R}^{\text{IPW}}(k), \\ R_2(k) &= \hat{\tau}_{\mathbf{t}_L}^{\text{DIM}}(k) - \hat{\tau}_{\mathbf{t}_R}^{\text{DIM}}(k) - \bar{\tau}_{\mathbf{t}_L}^{\text{IPW}}(k) + \bar{\tau}_{\mathbf{t}_R}^{\text{IPW}}(k). \end{aligned}$$

Then, for any $k = 1, 2, \dots, n$,

$$\mathcal{J}^{\text{DIM}}(k) - \bar{\mathcal{J}}^{\text{IPW}}(k) = \frac{k(n-k)}{n} R_1(k) R_2(k). \quad (\text{SA-21})$$

We focus on the case where $1 \leq k \leq \frac{n}{2}$, the other case where $\frac{n}{2} < k \leq n$ follow from symmetry. Consider the term $R_2(k)$. First, consider the term corresponding to i from 1 to k . The other term corresponding to i from

$k + 1$ to n can be handled similarly. Breaking down $y_i(1) = \mu_1(x_i) + \varepsilon_i(1)$ and $y_i(0) = \mu_0(x_i) + \varepsilon_i(0)$, we have

$$\begin{aligned}
|R_2(k)| &= \left| \frac{\sum_{i=1}^k d_i y_i(1)}{\sum_{i=1}^k d_i} - \frac{1}{k} \sum_{i=1}^k \frac{d_i}{\xi} \varepsilon_i(1) - \frac{\sum_{i=1}^k (1-d_i) y_i(0)}{\sum_{i=1}^k (1-d_i)} + \frac{1}{k} \sum_{i=1}^k \frac{1-d_i}{1-\xi} \varepsilon_i(0) + \text{counterpart for } t_R \right| \\
&\leq \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| \cdot \left| \frac{1}{k} \sum_{i=1}^k \left(\frac{d_i}{\xi} - 1 \right) \right| + \left| \frac{\sum_{i=1}^k (1-d_i) \varepsilon_i(0)}{\sum_{i=1}^k (1-d_i)} \right| \cdot \left| \frac{1}{k} \sum_{i=1}^k \left(\frac{1-d_i}{1-\xi} - 1 \right) \right| \\
&\quad + \left| \frac{\sum_{i=k+1}^n d_i \varepsilon_i(1)}{\sum_{i=k+1}^n d_i} \right| \cdot \left| \frac{1}{n-k} \sum_{i=k+1}^n \left(\frac{d_i}{\xi} - 1 \right) \right| + \left| \frac{\sum_{i=k+1}^n (1-d_i) \varepsilon_i(0)}{\sum_{i=k+1}^n (1-d_i)} \right| \cdot \left| \frac{1}{n-k} \sum_{i=k+1}^n \left(\frac{1-d_i}{1-\xi} - 1 \right) \right| \\
&\quad + \left| \frac{\sum_{i=1}^k d_i \mu_1(x_i)}{\sum_{i=1}^k d_i} - \frac{\sum_{i=1}^k (1-d_i) \mu_0(x_i)}{\sum_{i=1}^k (1-d_i)} - \frac{\sum_{i=k+1}^n d_i \mu_1(x_i)}{\sum_{i=k+1}^n d_i} + \frac{\sum_{i=k+1}^n (1-d_i) \mu_0(x_i)}{\sum_{i=k+1}^n (1-d_i)} \right|. \quad (\text{SA-22})
\end{aligned}$$

Assumption SA-2 (iii) implies that the last term is zero. Since $x_i \perp d_i$, even though the data is ordered according to x_i , $\{d_i/\xi - 1 : 1 \leq i \leq n\}$ are i.i.d. mean-zero with bounded second moment. By Theorem A.4.1 in Csörgő and Horváth [1997],

$$\max_{r_n \leq k < n-r_n} \sqrt{k} \cdot \left| \frac{1}{k} \sum_{i=1}^k \left(\frac{d_i}{\xi} - 1 \right) \right| = O_{\mathbb{P}}(\sqrt{\log \log(n)}).$$

Take $b_i = \sum_{1 \leq \ell \leq i} d_\ell$. By Equation (8) from Shorack and Smythe [1976], for any $\lambda > 0$,

$$\begin{aligned}
\mathbb{P} \left(\max_{r_n \leq k \leq n-r_n} \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| \geq \lambda \middle| (d_i)_{1 \leq i \leq n} \right) &\leq 16 \sum_{r_n \leq i \leq n-r_n} \frac{d_i \mathbb{V}[\varepsilon_i(1)]}{b_i^2} \lambda^{-2} \\
&\leq 16 \sum_{i \geq b_{r_n}} \frac{1}{i^2} \lambda^{-2} \mathbb{V}[\varepsilon_i(1)] \\
&\leq \frac{8}{3} \pi^2 \lambda^{-2} \mathbb{V}[\varepsilon_i(1)] \frac{1}{b_{r_n}},
\end{aligned}$$

Since $r_n \rightarrow \infty$ and $b_{r_n}/r_n \rightarrow \xi$ in probability,

$$(b_{r_n})^{-1} = r_n^{-1} \left(\xi + \frac{1}{r_n} \sum_{i=1}^{r_n} (d_i - \xi) \right)^{-1} = O_{\mathbb{P}}(r_n^{-1}).$$

Unconditioning on $(d_i)_{1 \leq i \leq n}$ gives

$$\max_{r_n \leq k \leq n-r_n} \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| = O_{\mathbb{P}}(r_n^{-1/2}). \quad (\text{SA-23})$$

Hence

$$\max_{r_n \leq k < n-r_n} \sqrt{k} \cdot \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| \cdot \left| \frac{1}{k} \sum_{i=1}^k \left(\frac{d_i}{\xi} - 1 \right) \right| = O_{\mathbb{P}} \left(\sqrt{\frac{\log \log(n)}{r_n}} \right).$$

The same calculation applies to the control left ratio, the treated right ratio, and the control right ratio, namely to

$$\frac{\sum_{i=1}^k (1-d_i) \varepsilon_i(0)}{\sum_{i=1}^k (1-d_i)}, \quad \frac{\sum_{i=k+1}^n d_i \varepsilon_i(1)}{\sum_{i=k+1}^n d_i}, \quad \frac{\sum_{i=k+1}^n (1-d_i) \varepsilon_i(0)}{\sum_{i=k+1}^n (1-d_i)},$$

with the corresponding empirical treatment share multiplier on the same left or right child. The valid candidate convention ensures that the displayed denominators are positive, and the delayed partial sum and treatment count bounds are unchanged after replacing $(d_i, \xi, \varepsilon_i(1))$ by $(1-d_i, 1-\xi, \varepsilon_i(0))$ or by the complementary index set $\{k+1, \dots, n\}$. Hence

$$\max_{r_n \leq k < n-r_n} \sqrt{k} |R_2(k)| = O_{\mathbb{P}} \left(\sqrt{\frac{\log \log(n)}{r_n}} \right).$$

Under the assumption that $\mu_0 \equiv c_0$ and $\mu_1 \equiv c_1$, we have

$$\begin{aligned} R_1(k) &= \left| \frac{\sum_{i=1}^k d_i y_i}{\sum_{i=1}^k d_i} + \frac{1}{k} \sum_{i=1}^k \frac{d_i}{\xi} \varepsilon_i(1) - \frac{\sum_{i=1}^k (1-d_i) y_i}{\sum_{i=1}^k (1-d_i)} - \frac{1}{k} \sum_{i=1}^k \frac{1-d_i}{1-\xi} \varepsilon_i(0) + \text{counterpart for } t_R \right| \\ &= \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} + \frac{1}{k} \sum_{i=1}^k \frac{d_i}{\xi} \varepsilon_i(1) - \frac{\sum_{i=1}^k (1-d_i) \varepsilon_i(0)}{\sum_{i=1}^k (1-d_i)} - \frac{1}{k} \sum_{i=1}^k \frac{1-d_i}{1-\xi} \varepsilon_i(0) + \text{counterpart for } t_R \right|. \end{aligned}$$

By Equation (SA-23) and Theorem A.4.1 in Csörgö and Horváth [1997] for the terms $k^{-1} \sum_{i=1}^k \xi^{-1} d_i \varepsilon_i(1)$, $k^{-1} \sum_{i=1}^k (1-\xi)^{-1} (1-d_i) \varepsilon_i(0)$ and the counterparts for t_R , we have

$$\max_{r_n \leq k < n-r_n} \sqrt{k} |R_1(k)| = O_{\mathbb{P}} \left(\sqrt{\log \log(n)} \right).$$

Putting together the parts for R_1 and R_2 , we have

$$\max_{r_n \leq k < n-r_n} |\mathcal{J}^{\text{DIM}}(k) - \bar{\mathcal{J}}^{\text{IPW}}(k)| = O_{\mathbb{P}} \left(\frac{\log \log(n)}{r_n^{1/2}} \right).$$

SA-5.16 Proof of Lemma SA-15

Since the number of coordinates p is fixed, a union bound over the approximation error for the p coordinates lets us assume w.l.o.g. that $p = 1$ and drop the second index on the coordinate ℓ from $\mathcal{J}^{\text{DIM}}(k, \ell)$ and $\bar{\mathcal{J}}^{\text{IPW}}(k, \ell)$ everywhere. Throughout, we assume the data is already sorted so that

$$x_1 \leq x_2 \leq \dots \leq x_n.$$

All ratios and maxima below are understood over valid candidates in $\mathcal{V}_{\text{DIM}}$. This convention is the same as the one used by the tree construction rule, which omits candidates with zero treated or control denominators in either child. In particular, displays involving b_i^{-1} or the corresponding control denominator are interpreted after omitting zero denominator indices.

For $1 \leq k \leq s_n$ and $n - s_n \leq k \leq n$, Equations (SA-21) and (SA-22) still hold. W.l.o.g., assume $1 \leq k \leq s_n$. First, we upper bound the IPW terms. The definition of s_n and Equation (A.4.3) in Csörgö and Horváth [1997] imply

$$\begin{aligned} \max_{1 \leq k \leq s_n} \left| \frac{1}{\sqrt{k}} \sum_{i=1}^k \frac{d_i}{\xi} \varepsilon_i(1) \right| &= O_{\mathbb{P}}(u_n), \\ \max_{1 \leq k \leq s_n} \left| \frac{1}{\sqrt{k}} \sum_{i=1}^k \frac{1-d_i}{1-\xi} \varepsilon_i(0) \right| &= O_{\mathbb{P}}(u_n). \end{aligned} \tag{SA-24}$$

with $u_n = \sqrt{\rho_n \log \log(n)}$. Also, Equation (A.4.2) in Csörgö and Horváth [1997] implies

$$\begin{aligned} \max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{1}{n-k} \sum_{i=k+1}^n \frac{d_i}{\xi} \varepsilon_i(1) \right| &= O_{\mathbb{P}}(v_n), \\ \max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{1}{n-k} \sum_{i=k+1}^n \frac{1-d_i}{1-\xi} \varepsilon_i(0) \right| &= O_{\mathbb{P}}(v_n). \end{aligned} \tag{SA-25}$$

where $v_n = \sqrt{\frac{s_n}{n-s_n} \log \log(n)}$. Again, Equation (A.4.3) from Csörgö and Horváth [1997] implies that

$$\max_{1 \leq k \leq s_n} \left| \frac{1}{\sqrt{k}} \sum_{i=1}^k \left(\frac{d_i}{\xi} - 1 \right) \right| = O_{\mathbb{P}}(u_n).$$

Take $b_i = \sum_{1 \leq \ell \leq i} d_\ell$. By Equation (8) from [Shorack and Smythe \[1976\]](#), for any $\lambda > 0$,

$$\begin{aligned} \mathbb{P}\left(\max_{1 \leq k \leq s_n} \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| \geq \lambda \mid (d_i)_{1 \leq i \leq n}\right) &\leq 16 \sum_{1 \leq i \leq s_n} \frac{d_i \mathbb{V}[\varepsilon_i(1)]}{b_i^2} \lambda^{-2} \\ &\leq 16 \sum_{1 \leq i \leq s_n} \frac{1}{i^2} \lambda^{-2} \mathbb{V}[\varepsilon_i(1)] \\ &\leq \frac{8}{3} \pi^2 \lambda^{-2} \mathbb{V}[\varepsilon_i(1)], \end{aligned}$$

Unconditioning on $(d_i)_{1 \leq i \leq n}$ gives

$$\max_{1 \leq k \leq s_n} \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| = O_{\mathbb{P}}(1).$$

Therefore

$$\max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} - \frac{1}{k} \sum_{i=1}^k \frac{d_i}{\xi} \varepsilon_i(1) \right| = \max_{1 \leq k \leq s_n} \left| \frac{1}{\sqrt{k}} \sum_{i=1}^k \left(\frac{d_i}{\xi} - 1 \right) \cdot \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| = O_{\mathbb{P}}(u_n). \quad (\text{SA-26})$$

The control left term is identical after replacing $(d_i, \xi, \varepsilon_i(1), b_i)$ by $(1 - d_i, 1 - \xi, \varepsilon_i(0), \sum_{\ell \leq i} (1 - d_\ell))$. Thus Equation (SA-26), together with Equation (SA-24), gives both treatment arms explicitly:

$$\begin{aligned} \max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| &= O_{\mathbb{P}}(u_n), \\ \max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{\sum_{i=1}^k (1 - d_i) \varepsilon_i(0)}{\sum_{i=1}^k (1 - d_i)} \right| &= O_{\mathbb{P}}(u_n). \end{aligned} \quad (\text{SA-27})$$

Apply Equation (A.4.2) in [Csörgö and Horváth \[1997\]](#) for the partial sum with at least $n - s_n$ terms and using $\max_{1 \leq k \leq s_n} \left| \frac{1}{n-k} \sum_{i=k+1}^n (d_i - \xi) \right| = o_{\mathbb{P}}(1)$, we have

$$\begin{aligned} \max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{\sum_{i=k+1}^n d_i \varepsilon_i(1)}{\sum_{i=k+1}^n d_i} \right| &= \max_{1 \leq k \leq s_n} \sqrt{k} \cdot \left| \frac{n-k}{\sum_{i=k+1}^n d_i} \right| \cdot \left| \frac{1}{n-k} \sum_{i=k+1}^n d_i \varepsilon_i(1) \right| \\ &\leq \sqrt{\frac{s_n}{n-s_n}} \left(\xi + \min_{1 \leq k \leq s_n} \frac{1}{n-k} \sum_{i=k+1}^n (d_i - \xi) \right)^{-1} \\ &\quad \times \max_{1 \leq k \leq s_n} \left| \frac{1}{\sqrt{n-k}} \sum_{i=k+1}^n d_i \varepsilon_i(1) \right| \\ &= O_{\mathbb{P}}(v_n). \end{aligned} \quad (\text{SA-28})$$

The control right term follows from the same display after replacing d_i by $1 - d_i$ and ξ by $1 - \xi$; the denominator is bounded away from zero uniformly because $n - k \geq n - s_n$ and the treatment assignments are i.i.d. Bernoulli. Putting together Equations (SA-24), (SA-25), (SA-27), (SA-28), we have

$$\max_{r=1,2} \max_{1 \leq k \leq s_n} \sqrt{k} |R_r(k)| = O_{\mathbb{P}}(u_n + v_n).$$

From the decomposition in Equation (SA-22) and the symmetry for $k \in [1, s_n]$ and $k \in [n - s_n, n]$, the conclusion follows.

SA-5.17 Proof of Theorem SA-16

We break down the proofs into two steps.

Step 1: Approximation of reg score by IPW score

Let $0 < a < b < 1$. Let ρ_n be a sequence of real numbers taking values in $(0, 1)$ to be determined, and take $s_n = \exp((\log n)^{\rho_n})$. Then for large enough n , we have $s_n \leq n^a \leq n^b \leq n - s_n$. Consider the event

$$A_n = \left\{ \exists \ell \in [p] : \max_{\substack{k \in [n] \\ k \notin [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, \ell) \geq \max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{DIM}}(k, \ell) \right\}.$$

By Equation (A.4.18) from Csörgö and Horváth [1997],

$$\max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)} = O_{\mathbb{P}}(\sqrt{\rho_n \log \log(n)}).$$

Then controlling the difference between $\mathcal{J}^{\text{IPW}}(k, \ell)$ and $\mathcal{J}^{\text{DIM}}(k, \ell)$ by Lemma SA-15,

$$\max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \mathcal{J}^{\text{DIM}}(k, \ell) = O_{\mathbb{P}}\left(\rho_n \log \log(n) + \frac{s_n}{n-s_n} \log \log(n)\right) \quad (\text{SA-29})$$

By Lemma SA-14 with the choice $r_n = s_n$,

$$\begin{aligned} \max_{s_n < k < n-s_n} \sqrt{\mathcal{J}^{\text{DIM}}(k, \ell)} &= \max_{s_n < k < n-s_n} \sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)} + O_{\mathbb{P}}\left(\frac{\sqrt{\log \log(n)}}{s_n^{1/4}}\right) \\ &\geq \max_{1 \leq k \leq n} \sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)} - \max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)} - O_{\mathbb{P}}\left(\frac{\sqrt{\log \log(n)}}{s_n^{1/4}}\right). \end{aligned}$$

Equation (A.4.20) in Csörgö and Horváth [1997] implies that $(2 \log \log(n))^{-1/2} \max_{1 \leq k \leq n} \sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)} = 1 + o_{\mathbb{P}}(1)$ and $(2 \log \log(n))^{-1/2} \max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)} = \sqrt{\rho_n}(1 + o_{\mathbb{P}}(1))$. Hence

$$\max_{s_n < k < n-s_n} \sqrt{\mathcal{J}^{\text{DIM}}(k, \ell)} \geq \sqrt{2 \log \log(n)}(1 + o_{\mathbb{P}}(1)) \quad (\text{SA-30})$$

Choose $\log \log \log \log(n) / \log \log(n) \ll \rho_n \ll 1$, then by Equation (SA-29) and (SA-30),

$$\max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \mathcal{J}^{\text{DIM}}(k, \ell) = o_{\mathbb{P}}(\log \log(n)), \text{ and } \max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{DIM}}(k, \ell) \geq 2 \log \log(n)(1 + o_{\mathbb{P}}(1)).$$

Hence

$$\max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \mathcal{J}^{\text{DIM}}(k, \ell) = o_{\mathbb{P}}\left(\max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{DIM}}(k, \ell)\right), \quad \ell \in [p],$$

which by a union bound implies

$$\limsup_{n \rightarrow \infty} \mathbb{P}(A_n) = 0.$$

On the event A_n^c , the argmax for $\mathcal{J}^{\text{DIM}}$ lies inside $[s_n, n - s_n]$. Hence

$$\begin{aligned} &\mathbb{P}\left(\exists \ell \in [p] : \max_k \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}^{\text{DIM}}(k, j), \max_k \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k \notin [n^a, n^b]} \mathcal{J}^{\text{DIM}}(k, \ell)\right) \\ &\geq \mathbb{P}\left(\exists \ell \in [p] : \max_k \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}^{\text{DIM}}(k, j), \max_k \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k \notin [n^a, n^b]} \mathcal{J}^{\text{DIM}}(k, \ell) \text{ and } A_n^c\right) - \mathbb{P}(A_n) \\ &\geq \mathbb{P}\left(\exists \ell \in [p] : \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{\substack{j \neq \ell \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, j), \right. \\ &\quad \left. \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k \notin [n^a, n^b], k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell)\right) - 2\mathbb{P}(A_n). \end{aligned}$$

Focus on the first term. By symmetry in the p coordinates,

$$\begin{aligned}
& \mathbb{P}\left(\exists \ell \in [p] : \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{\substack{j \neq \ell \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, j), \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, \ell)\right) \\
&= p \mathbb{P}\left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, 1) > \max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, j), \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, 1) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, 1)\right) \\
&\geq p \sup_{z \in \mathbb{R}} \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{DIM}}(k, j)} < z, \max_{k \in [s_n, n-s_n]} \sqrt{\mathcal{J}^{\text{DIM}}(k, 1)} > z > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{DIM}}(k, 1)}\right) \\
&\geq p \sup_{z \in \mathbb{R}} \left\{ \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{DIM}}(k, j)} < z, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{DIM}}(k, 1)} < z\right) \right. \\
&\quad \left. - \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{DIM}}(k, j)} < z, \max_{k \in [s_n, n-s_n]} \sqrt{\mathcal{J}^{\text{DIM}}(k, 1)} < z\right) \right\}.
\end{aligned}$$

Then using the fact that $\sqrt{\mathcal{J}^{\text{IPW}}(k, \ell)}$ approximates $\sqrt{\mathcal{J}^{\text{DIM}}(k, \ell)}$ from Lemma SA-14, we have

$$\begin{aligned}
& \mathbb{P}\left(\exists \ell \in [p] : \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{\substack{j \neq \ell \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, j), \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{DIM}}(k, \ell)\right) \\
&\geq p \sup_{z \in \mathbb{R}} \left\{ \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, j)} < z - v_n, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, 1)} < z - v_n\right) \right. \\
&\quad \left. - \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, j)} < z + v_n, \max_{k \in [s_n, n-s_n]} \sqrt{\mathcal{J}^{\text{IPW}}(k, 1)} < z + v_n\right) \right\},
\end{aligned}$$

where $v_n = O_{\mathbb{P}}((\log \log(n))^{1/2} s_n^{-1/4})$. Here v_n denotes a nonnegative random envelope for the square root approximation error. Since the chosen s_n makes $v_n = o_{\mathbb{P}}((\log \log(n))^{-1/2})$, there is a deterministic envelope $\bar{v}_n = o((\log \log(n))^{-1/2})$ with $\mathbb{P}(v_n \leq \bar{v}_n) \rightarrow 1$. Replacing v_n by $\bar{v}_n$ in the threshold displays loses only $o(1)$ probability; the next display records this replacement through an arbitrary fixed $\epsilon > 0$.

Step 2: IPW score approximation by Gaussian approximation

The choice $s_n = \exp(\log^{\rho_n}(n))$ for $\log \log \log \log(n) / \log \log(n) \ll \rho_n \ll 1$ implies $v_n = o_{\mathbb{P}}((\log \log(n))^{-1/2})$. Let $\epsilon > 0$. Then

$$\begin{aligned}
& \sup_{z \in \mathbb{R}} \left\{ \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, j)} < z - v_n, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, 1)} < z - v_n\right) \right. \\
&\quad \left. - \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, j)} < z + v_n, \max_{k \in [s_n, n-s_n]} \sqrt{\mathcal{J}^{\text{IPW}}(k, 1)} < z + v_n\right) \right\} \\
&\geq \sup_{z \in \mathbb{R}} \left\{ \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, j)} < z - \frac{\epsilon}{\sqrt{2 \log \log(n)}}, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, 1)} < z - \frac{\epsilon}{\sqrt{2 \log \log(n)}}\right) \right. \\
&\quad \left. - \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\mathcal{J}^{\text{IPW}}(k, j)} < z + \frac{\epsilon}{\sqrt{2 \log \log(n)}}, \max_{k \in [s_n, n-s_n]} \sqrt{\mathcal{J}^{\text{IPW}}(k, 1)} < z + \frac{\epsilon}{\sqrt{2 \log \log(n)}}\right) \right\} \\
&\quad - \mathbb{P}(|v_n| > \frac{\epsilon}{\sqrt{2 \log \log(n)}}).
\end{aligned}$$

The last probability term is $o(1)$ and is omitted in the following liminf calculation. Choosing $z_n(u) = \frac{2 \log \log(n) + 1/2 \log \log \log(n) + u - 1/2 \log(\pi)}{\sqrt{2 \log \log(n)}}$, and using the proof of Theorem SA-1, we have

$$\begin{aligned}
& \liminf_{n \rightarrow \infty} \sup_{z \in \mathbb{R}} \left\{ \mathbb{P} \left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, j)} < z - \frac{\epsilon}{\sqrt{2 \log \log n}}, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, 1)} < z - \frac{\epsilon}{\sqrt{2 \log \log n}} \right) \right. \\
& \quad - \mathbb{P} \left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, j)} < z + \frac{\epsilon}{\sqrt{2 \log \log n}}, \max_{k \in [s_n, n-s_n]} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, 1)} < z + \frac{\epsilon}{\sqrt{2 \log \log n}} \right) \Big\} \\
& \quad - \mathbb{P}(|v_n| > \frac{\epsilon}{\sqrt{2 \log \log n}}) \\
& \geq \liminf_{n \rightarrow \infty} \sup_{u \in \mathbb{R}} \left\{ \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, 1)} < z_n(u) - \frac{\epsilon}{\sqrt{2 \log \log n}} \right)^{p-1} \right. \\
& \quad \times \mathbb{P} \left(\max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, 1)} < z_n(u) - \frac{\epsilon}{\sqrt{2 \log \log n}} \right) \\
& \quad - \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, 1)} < z_n(u) + \frac{\epsilon}{\sqrt{2 \log \log n}} \right)^{p-1} \\
& \quad \times \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \sqrt{\tilde{\mathcal{J}}^{\text{IPW}}(k, 1)} < z_n(u) + \frac{\epsilon}{\sqrt{2 \log \log n}} \right) \Big\} \\
& \geq \sup_{u \in \mathbb{R}} \left\{ \exp \left(- (p-1) e^{-(u-\epsilon-\log(2))} \right) \exp \left(- e^{-(u-\epsilon-\log(2-(b-a)))} \right) \right. \\
& \quad \left. - \exp \left(- (p-1) e^{-(u+\epsilon-\log(2))} \right) \exp \left(- e^{-(u+\epsilon-\log(2))} \right) \right\}.
\end{aligned}$$

Let $\epsilon \downarrow 0$. Combining the previous steps gives

$$\begin{aligned}
& \liminf_{n \rightarrow \infty} \mathbb{P} \left(\exists \ell \in [p] : \max_k \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}^{\text{DIM}}(k, j), \max_k \mathcal{J}^{\text{DIM}}(k, \ell) > \max_{k \notin [n^a, n^b]} \mathcal{J}^{\text{DIM}}(k, \ell) \right) \\
& \geq p \sup_{u \in \mathbb{R}} \exp \left(- (p-1) e^{-(u-\log(2))} \right) \left(\exp \left(- e^{-(u-\log(2-(b-a)))} \right) - \exp \left(- e^{-(u-\log(2))} \right) \right) \\
& \geq \frac{b-a}{2e}.
\end{aligned}$$

The last display is an existence statement over coordinates. The corresponding events are disjoint across split coordinates once the deterministic tie breaking rule is applied, and coordinate symmetry makes their probabilities equal asymptotically. Therefore, for each fixed $\ell \in [p]$,

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(n^a \leq \hat{i}_{\text{DIM}} \leq n^b, \hat{j}_{\text{DIM}} = \ell \right) \geq \frac{b-a}{2pe}.$$

Applying the same argument after reversing the ordering along coordinate ℓ gives the analogous lower bound for $n - n^b \leq \hat{i}_{\text{DIM}} \leq n - n^a$.

SA-5.18 Proof of Theorem SA-17

The proofs follow essentially the same logic as the proof for Theorem SA-2, with some tricks for the random numerator in $\frac{\sum_{1 \leq i \leq k} d_i \varepsilon_i(1)}{\sum_{1 \leq i \leq k} d_i}$. The split-index comparison is invariant to multiplying all transformed residuals by a positive constant. Thus, in the displays below, we use the standardized transformed residual with variance one; multiplying back by $\sigma^2 = \mathbb{V}[\tilde{\varepsilon}_i]$ restores the σ factor in Theorem SA-17.

Part 1: Inconsistency for Uniform Estimation Rates

Denote the optimal index for splitting based on the ℓ 's coordinate by

$$\hat{i}_{\text{reg},\ell} = \arg \max_{k \in [n]} \mathcal{J}^{\text{DIM}}(k, \ell), \quad \ell \in [p].$$

For notational simplicity, denote

$$\begin{aligned} \bar{\tau}_L^{\text{DIM}}(k, \ell) &= \tau_L^{\text{DIM}}(k, \ell) - \tau = \frac{\sum_{1 \leq i \leq k} d_{\pi_\ell(i)} \varepsilon_{\pi_\ell(i)}(1)}{\sum_{1 \leq i \leq k} d_{\pi_\ell(i)}} - \frac{\sum_{1 \leq i \leq k} (1 - d_{\pi_\ell(i)}) \varepsilon_{\pi_\ell(i)}(0)}{\sum_{1 \leq i \leq k} (1 - d_{\pi_\ell(i)})}, \\ \bar{\tau}_L^{\text{DIM}}(\ell) &= \bar{\tau}_L^{\text{DIM}}(\hat{i}_{\text{reg},\ell}, \ell), \\ \bar{\tau}_R^{\text{DIM}}(k, \ell) &= \tau_R^{\text{DIM}}(k, \ell) - \tau = \frac{\sum_{k < i \leq n} d_{\pi_\ell(i)} \varepsilon_{\pi_\ell(i)}(1)}{\sum_{k < i \leq n} d_{\pi_\ell(i)}} - \frac{\sum_{k < i \leq n} (1 - d_{\pi_\ell(i)}) \varepsilon_{\pi_\ell(i)}(0)}{\sum_{k < i \leq n} (1 - d_{\pi_\ell(i)})}, \\ \bar{\tau}_R^{\text{DIM}}(\ell) &= \bar{\tau}_R^{\text{DIM}}(\hat{i}_{\text{reg},\ell}, \ell), \end{aligned}$$

and consider the event

$$\text{Imbalance}_\ell^{\text{DIM}} = \{\hat{j}_{\text{reg}} = \ell, \hat{i}_{\text{reg},\ell} < n^b \text{ or } \hat{i}_{\text{reg},\ell} > n - n^b\}, \quad \ell \in [p].$$

Since we assume $\mu_0 \equiv c_0$ and $\mu_1 \equiv c_1$ with $c_1 - c_0 = \tau$, we have on $\text{Imbalance}_\ell^{\text{DIM}} \cap \{\hat{i}_{\text{reg},\ell} \leq n/2\}$,

$$\begin{aligned} \sup_{x \in \mathcal{X}} |\hat{\tau}(x) - \tau|^2 &\geq \bar{\tau}_L^{\text{DIM}}(\ell)^2 \\ &\geq \frac{1}{\min\{\hat{i}_{\text{reg},\ell}, n - \hat{i}_{\text{reg},\ell}\}} \left(\hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 - (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} \leq n/2) \right). \end{aligned}$$

Take $\bar{\tau}^{\text{DIM}} = \frac{\hat{i}_{\text{reg},\ell}}{n} \bar{\tau}_L^{\text{DIM}}(\ell) + \frac{n - \hat{i}_{\text{reg},\ell}}{n} \bar{\tau}_R^{\text{DIM}}(\ell)$. Then

$$\begin{aligned} \hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 &\geq \hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 - n(\bar{\tau}^{\text{DIM}})^2 \\ &= \frac{\hat{i}_{\text{reg},\ell}(n - \hat{i}_{\text{reg},\ell})}{n} \left(\bar{\tau}_L^{\text{DIM}}(\ell) - \bar{\tau}_R^{\text{DIM}}(\ell) \right)^2. \end{aligned}$$

By Lemma SA-14 and Lemma SA-15 with $r_n = s_n = \exp((\log n)^{\rho_n})$ for $\log \log \log \log(n) / \log \log(n) \ll \rho_n \ll 1$,

$$\begin{aligned} \frac{\hat{i}_{\text{reg},\ell}(n - \hat{i}_{\text{reg},\ell})}{n} \left(\bar{\tau}_L^{\text{DIM}}(\ell) - \bar{\tau}_R^{\text{DIM}}(\ell) \right)^2 &= \mathcal{J}^{\text{DIM}}(\hat{i}_{\text{reg},\ell}, \ell) \\ &= \max_{1 \leq k < n} \mathcal{J}^{\text{DIM}}(k, \ell) \\ &\geq \max_{1 \leq k < n} \mathcal{J}^{\text{IPW}}(k, \ell) + o_{\mathbb{P}}(\log \log n). \end{aligned}$$

By Theorem A.4.1 in Csörgő and Horváth [1997], under this standardization, $\max_{1 \leq k < n} \mathcal{J}^{\text{IPW}}(k, \ell) = 2 \log \log(n)(1 + o_{\mathbb{P}}(1))$. Moreover,

$$\begin{aligned} \hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} > n/2) &\leq \max_{k > n/2} k \cdot \left(\frac{\sum_{1 \leq i \leq k} d_{\pi_\ell(i)} \varepsilon_{\pi_\ell(i)}(1)}{\sum_{1 \leq i \leq k} d_{\pi_\ell(i)}} - \frac{\sum_{1 \leq i \leq k} (1 - d_{\pi_\ell(i)}) \varepsilon_{\pi_\ell(i)}(0)}{\sum_{1 \leq i \leq k} (1 - d_{\pi_\ell(i)})} \right)^2 \\ &\leq 2 \max_{k > n/2} k \cdot \left(\frac{\sum_{1 \leq i \leq k} d_{\pi_\ell(i)} \varepsilon_{\pi_\ell(i)}(1)}{\sum_{1 \leq i \leq k} d_{\pi_\ell(i)}} \right)^2 \\ &\quad + 2 \max_{k > n/2} k \cdot \left(\frac{\sum_{1 \leq i \leq k} (1 - d_{\pi_\ell(i)}) \varepsilon_{\pi_\ell(i)}(0)}{\sum_{1 \leq i \leq k} (1 - d_{\pi_\ell(i)})} \right)^2. \end{aligned}$$

For simplicity in showing the upper bound for $\hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} > n/2)$, we assume π is the identity permutation. Take $b_i = \sum_{1 \leq j \leq i} d_j$. By Equation (8) from [Shorack and Smythe \[1976\]](#), for any $\lambda > 0$,

$$\begin{aligned} \mathbb{P}\left(\max_{k > n/2} \left| \frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right| \geq \lambda \mid (d_i)_{1 \leq i \leq n}\right) &\leq 16 \sum_{i > n/2} \frac{d_i \mathbb{V}[\varepsilon_i(1)]}{b_i^2} \lambda^{-2} \\ &\leq 16 \sum_{i > b_{n/2}} \frac{1}{i^2} \lambda^{-2} \mathbb{V}[\varepsilon_i(1)] \\ &\leq \frac{8}{3} \pi^2 \lambda^{-2} \mathbb{V}[\varepsilon_i(1)] \frac{1}{b_{n/2}}, \end{aligned}$$

And since d_i 's are i.i.d with $\mathbb{E}[d_i] = \xi > 0$, we have

$$(b_{n/2})^{-1} = (n/2)^{-1} \left(\xi + \frac{2}{n} \sum_{i=1}^{n/2} (d_i - \xi) \right)^{-1} = O_{\mathbb{P}}(n^{-1}).$$

Unconditioning on $(d_i)_{1 \leq i \leq n}$ gives

$$\max_{k \geq n/2} k \cdot \left(\frac{\sum_{i=1}^k d_i \varepsilon_i(1)}{\sum_{i=1}^k d_i} \right)^2 = O_{\mathbb{P}}(1) = o_{\mathbb{P}}(\log \log(n)).$$

By a similar term for control, and a symmetric argument for the right node,

$$\hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} > n/2) + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} \leq n/2) = o_{\mathbb{P}}(\log \log(n)).$$

Fix $\epsilon > 0$. Consider the events

$$\begin{aligned} A_{\ell}^{\epsilon} &= \left\{ \hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 \geq (2 - \epsilon) \log \log(n) \right\}, \\ B_{\ell}^{\epsilon} &= \left\{ \hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} > n/2) + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 \mathbf{1}(\hat{i}_{\text{reg},\ell} \leq n/2) \leq 2\epsilon \log \log(n) \right\}. \end{aligned}$$

The above arguments show that $\liminf_{n \rightarrow \infty} \mathbb{P}(A_{\ell}^{\epsilon}) = \liminf_{n \rightarrow \infty} \mathbb{P}(B_{\ell}^{\epsilon}) = 1$. From Theorem [SA-16](#), after summing the left and right boundary events and letting the lower boundary exponent tend to zero, $\liminf_{n \rightarrow \infty} \mathbb{P}(\text{Imbalance}_{\ell}^{\text{DIM}}) \geq b/(pe)$. It then follows from a union bound argument that

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)}\right) \geq \frac{b}{e}.$$

Part 2: Inconsistency for Points near the Boundary

Define the left boundary split event

$$\text{Off}_{\ell}^{\text{DIM}} = \{n^a \leq \hat{i}_{\text{reg},\ell} \leq n^b, \hat{j}_{\text{reg}} = \ell\}.$$

By Theorem [SA-16](#), $\mathbb{P}(\text{Off}_{\ell}^{\text{DIM}}) \geq (b - a)/(2pe) + o(1)$. On this event,

$$x_{\pi_{\ell}(\hat{i}_{\text{reg},\ell}),\ell} \geq x_{\pi_{\ell}(\lceil n^a \rceil),\ell}.$$

For any deterministic sequence $\eta_n \downarrow 0$, the marginal probability integral transform gives

$$\mathbb{P}(x_{\pi_{\ell}(\lceil n^a \rceil),\ell} \geq \eta_n n^{a-1}) \rightarrow 1.$$

Thus, with probability at least $(b - a)/(2pe) + o(1)$, $\text{Off}_{\ell}^{\text{DIM}}$ occurs and the selected left child contains every $\mathbf{z} \in \mathcal{X}_n(a, \eta_n)$ such that $z_{\ell} \leq \eta_n n^{a-1}$. On this event,

$$\begin{aligned} |\hat{\tau}_{\text{DIM}}(\mathbf{z}) - \tau|^2 &= \bar{\tau}_L^{\text{DIM}}(\ell)^2 \geq \frac{1}{\hat{i}_{\text{reg},\ell}} \left(\hat{i}_{\text{reg},\ell} \bar{\tau}_L^{\text{DIM}}(\ell)^2 + (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 - (n - \hat{i}_{\text{reg},\ell}) \bar{\tau}_R^{\text{DIM}}(\ell)^2 \right) \\ &\geq \frac{1}{\hat{i}_{\text{reg},\ell}} \left(\max_{1 \leq k < n} \left(k \bar{\tau}_L^{\text{DIM}}(k, \ell)^2 + (n - k) \bar{\tau}_R^{\text{DIM}}(k, \ell)^2 \right) - \max_{k \leq n^b} (n - k) \bar{\tau}_R^{\text{DIM}}(k, \ell)^2 \right) \\ &\geq \frac{(2 + o_{\mathbb{P}}(1)) \log \log(n)}{\hat{i}_{\text{reg},\ell}} \\ &\geq \frac{(2 + o_{\mathbb{P}}(1)) \log \log(n)}{n^b}, \end{aligned}$$

The penultimate inequality follows from the same explicit boundary transfer used in Part 1. The full DIM quadratic split maximum along the selected coordinate is $(2 + o_{\mathbb{P}}(1)) \log \log(n)$ on the imbalanced window event, while the long child remainder is negligible:

$$\max_{k \leq n^b} (n - k) \bar{\tau}_R^{\text{DIM}}(k, \ell)^2 = o_{\mathbb{P}}(\log \log(n)).$$

Indeed, after writing the DIM contrast in terms of the transformed residual partial sums, the condition $k \leq n^b$ implies $n - k \asymp n$; the terminal long child average is therefore bounded by the full sample transformed residual average plus the first n^b partial sum maximum, giving the same $O_{\mathbb{P}}(1) + O_{\mathbb{P}}(n^b \log \log(n)/n)$ bound as above. By a symmetry argument for the event $\{n - n^b \leq \hat{t}_{\text{reg}, \ell} \leq n - n^a\}$, we have

$$\liminf_{n \rightarrow \infty} \inf_{\mathbf{x} \in \mathcal{X}_n(a, \eta_n)} \mathbb{P}\left(|\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq \sigma n^{-b/2} \sqrt{(2 + o(1)) \log \log(n)}\right) \geq \frac{b - a}{2pe},$$

for any deterministic sequence $\eta_n \downarrow 0$, where $\sigma^2 = \mathbb{V}\left[\frac{d_i \varepsilon_i(1)}{\xi} - \frac{(1 - d_i) \varepsilon_i(0)}{1 - \xi}\right]$. This completes the proof of Theorem SA-17.

SA-5.19 Proof of Theorem SA-18

Recall the centered DIM contrast $\Delta(\mathbf{t})$ defined above. Under Assumption SA-2, $\mu_1(\mathbf{x}) \equiv c_1$, $\mu_0(\mathbf{x}) \equiv c_0$, and $\tau = c_1 - c_0$. Hence, for every valid terminal node $\mathbf{t}$,

$$\hat{\tau}_{\text{DIM}}(\mathbf{t}; \mathbf{T}, \mathcal{D}) - \tau = \Delta(\mathbf{t}).$$

Lemma SA-39 (Uniform rectangle control). *Suppose Assumption SA-2 holds and p is fixed. Let $\mathcal{R}_n$ be the finite collection of all axis-aligned sample rectangles whose coordinate endpoints are chosen from the observed coordinate values and the support boundaries. Then*

$$|\mathcal{R}_n| \leq (n + 2)^{2p} \leq 3^{2p} n^{2p}.$$

Let $\underline{\xi} = \xi \wedge (1 - \xi)$. There exist positive constants $C_0, C_{\max}, C_{\xi}$, and $C_{\bar{\varepsilon}}$, depending only on p, ξ and the exponential moment constants in Assumption SA-2, such that the following holds. Set $h_n = C_0 \log n$. Then there is an event $\mathcal{E}_n$ satisfying $\mathbb{P}(\mathcal{E}_n) \rightarrow 1$ such that, on $\mathcal{E}_n$, the following bounds hold simultaneously for all $\mathbf{t} \in \mathcal{R}_n$:

$$\max_{\substack{1 \leq i \leq n \\ d \in \{0, 1\}}} |\varepsilon_i(d)| \leq C_{\max} \log n,$$

and, whenever $n(\mathbf{t}) \geq h_n$,

$$\left| \frac{n_1(\mathbf{t})}{n(\mathbf{t})} - \xi \right| \leq C_{\xi} \sqrt{\frac{\log n}{n(\mathbf{t})}},$$

and $n_0(\mathbf{t}) \wedge n_1(\mathbf{t}) > 0$. Moreover,

$$|\bar{\varepsilon}_{d, \mathbf{t}}| \leq C_{\bar{\varepsilon}} \sqrt{\frac{\log n}{n(\mathbf{t})}}, \quad d \in \{0, 1\}.$$

In particular, on $\mathcal{E}_n$, whenever $n(\mathbf{t}) \geq h_n$,

$$\frac{\xi}{2} \leq \frac{n_1(\mathbf{t})}{n(\mathbf{t})} \leq 1 - \frac{1 - \xi}{2}$$

for all sufficiently large n .

Proof. The cardinality bound follows because each coordinate interval is determined by an ordered pair of endpoints chosen from at most $n + 2$ possibilities. Thus there are at most $(n + 2)^2$ choices in each coordinate and at most $(n + 2)^{2p}$ rectangles. Since $n + 2 \leq 3n$ for $n \geq 1$, this gives the displayed bound.

Let $\mathcal{E}_{n, \max}$ be the maximum error event. The exponential moment condition in Assumption SA-2 implies that there are positive constants A_{ε} and a_{ε} such that

$$\mathbb{P}(|\varepsilon_i(d)| > t) \leq A_{\varepsilon} \exp(-a_{\varepsilon} t), \quad t \geq 0, \quad d \in \{0, 1\}.$$

Choose $C_{\max}$ so that $a_\varepsilon C_{\max} > 2$. Then

$$\mathbb{P} \left(\max_{\substack{1 \leq i \leq n \\ d \in \{0,1\}}} |\varepsilon_i(d)| > C_{\max} \log n \right) \leq 2A_\varepsilon n^{1-a_\varepsilon C_{\max}} = o(1).$$

Thus $\mathbb{P}(\mathcal{G}_{n,\max}) \rightarrow 1$.

Next condition on $\mathbf{x}_1, \dots, \mathbf{x}_n$. For any fixed $\mathbf{t} \in \mathcal{R}_n$,

$$n_1(\mathbf{t}) \mid \mathbf{x}_1, \dots, \mathbf{x}_n \sim \text{Binomial}(n(\mathbf{t}), \xi).$$

Hoeffding's inequality gives, for any $A > 0$,

$$\mathbb{P} \left(|n_1(\mathbf{t}) - \xi n(\mathbf{t})| > A \sqrt{n(\mathbf{t}) \log n} \mid \mathbf{x}_1, \dots, \mathbf{x}_n \right) \leq 2n^{-2A^2}.$$

Choose C_ξ so that $2C_\xi^2 > 2p + 2$. Since $|\mathcal{R}_n| \leq 3^{2p} n^{2p}$, a union bound gives an event $\mathcal{G}_{n,\xi}$ satisfying $\mathbb{P}(\mathcal{G}_{n,\xi}) \rightarrow 1$ on which

$$\left| \frac{n_1(\mathbf{t})}{n(\mathbf{t})} - \xi \right| \leq C_\xi \sqrt{\frac{\log n}{n(\mathbf{t})}}$$

for every $\mathbf{t} \in \mathcal{R}_n$ with $n(\mathbf{t}) > 0$. We shall choose C_0 large enough that

$$C_0 \geq \frac{4C_\xi^2}{\underline{\xi}}.$$

Then, on $\mathcal{G}_{n,\xi}$, if $n(\mathbf{t}) \geq h_n = C_0 \log n$, the treatment share error is at most $\underline{\xi}/2$. Hence

$$\frac{\underline{\xi}}{2} \leq \frac{n_1(\mathbf{t})}{n(\mathbf{t})} \leq 1 - \frac{1 - \underline{\xi}}{2},$$

and $n_0(\mathbf{t}) \wedge n_1(\mathbf{t}) > 0$.

Finally fix $d \in \{0, 1\}$. Conditional on the covariates and treatment assignments, for any fixed $\mathbf{t} \in \mathcal{R}_n$,

$$S_{d,\mathbf{t}} = \sum_{i=1}^n \mathbf{1}(\mathbf{x}_i \in \mathbf{t}) \mathbf{1}(d_i = d) \varepsilon_i(d)$$

is a sum of independent mean-zero subexponential random variables. We apply Bernstein's inequality only to rectangles satisfying $n(\mathbf{t}) \geq h_n$. For $u = A \sqrt{n(\mathbf{t}) \log n}$,

$$\mathbb{P}(|S_{d,\mathbf{t}}| > u \mid \mathbf{x}_1, \dots, \mathbf{x}_n, d_1, \dots, d_n) \leq 2 \exp \left[-c \min \left\{ \frac{A^2 n(\mathbf{t}) \log n}{n_d(\mathbf{t}) \vee 1}, A \sqrt{n(\mathbf{t}) \log n} \right\} \right].$$

Since $n_d(\mathbf{t}) \leq n(\mathbf{t})$ and $n(\mathbf{t}) \geq h_n = C_0 \log n$, the right hand side is bounded by

$$2n^{-c \min\{A^2, A\sqrt{C_0}\}}.$$

Choose $A = A_\varepsilon$ and then increase C_0 , if necessary, so that $c \min\{A_\varepsilon^2, A_\varepsilon \sqrt{C_0}\} > 2p + 2$. A union bound over $d \in \{0, 1\}$ and over all $\mathbf{t} \in \mathcal{R}_n$ with $n(\mathbf{t}) \geq h_n$ gives an event $\mathcal{G}_{n,\varepsilon}$ satisfying $\mathbb{P}(\mathcal{G}_{n,\varepsilon}) \rightarrow 1$ on which

$$|S_{d,\mathbf{t}}| \leq A_\varepsilon \sqrt{n(\mathbf{t}) \log n}$$

simultaneously for all such d and $\mathbf{t}$. On $\mathcal{G}_{n,\xi} \cap \mathcal{G}_{n,\varepsilon}$, $n_d(\mathbf{t}) \geq \underline{\xi} n(\mathbf{t})/2$. Therefore

$$|\bar{\varepsilon}_{d,\mathbf{t}}| = \frac{|S_{d,\mathbf{t}}|}{n_d(\mathbf{t})} \leq \frac{2A_\varepsilon}{\underline{\xi}} \sqrt{\frac{\log n}{n(\mathbf{t})}},$$

so the displayed residual average bound holds with $C_\varepsilon = 2A_\varepsilon/\underline{\xi}$. The event

$$\mathcal{G}_n = \mathcal{G}_{n,\max} \cap \mathcal{G}_{n,\xi} \cap \mathcal{G}_{n,\varepsilon}$$

has probability tending to one and satisfies all claimed conclusions. $\square$

Lemma SA-40 (Refinement transfer with random arm weights). *Suppose Assumption SA-2 holds. Fix $0 < a < b < 1$. Let $\mathbf{t}_n$ be a possibly random sample rectangle satisfying*

$$M_n = n(\mathbf{t}_n) \in [n^a, n^b], \quad n_0(\mathbf{t}_n) \wedge n_1(\mathbf{t}_n) > 0.$$

Let $\mathcal{P}(\mathbf{t}_n)$ be a possibly random valid recursive refinement of $\mathbf{t}_n$ into terminal descendants. Assume that every $\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)$ belongs to $\mathcal{R}_n$, that the descendants partition $\mathbf{t}_n$, that $n_0(\mathbf{t}') \wedge n_1(\mathbf{t}') > 0$ for every $\mathbf{t}'$, and that $|\mathcal{P}(\mathbf{t}_n)| \leq J_n$ for a deterministic sequence J_n . If

$$J_n \log^2 n = o\left(n^{a/2} \sqrt{\log \log n}\right),$$

then

$$\Delta(\mathbf{t}_n) = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} \frac{n_1(\mathbf{t}')}{n_1(\mathbf{t}_n)} \Delta(\mathbf{t}') + o_{\mathbb{P}}\left(M_n^{-1/2} \sqrt{\log \log n}\right).$$

Proof. Work on the event $\mathcal{G}_n$ from Lemma SA-39. Since $M_n \geq n^a$, we have $M_n \geq h_n$ for all sufficiently large n . Therefore, on $\mathcal{G}_n$,

$$\hat{\xi}_{\mathbf{t}_n} = \frac{n_1(\mathbf{t}_n)}{M_n}$$

is bounded away from zero and one. More precisely, with $\underline{\xi} = \xi \wedge (1 - \xi)$,

$$n_1(\mathbf{t}_n) \geq \frac{\underline{\xi}}{2} M_n, \quad n_0(\mathbf{t}_n) \geq \frac{1 - \underline{\xi}}{2} M_n, \quad \hat{\xi}_{\mathbf{t}_n} (1 - \hat{\xi}_{\mathbf{t}_n}) \geq \frac{\underline{\xi}(1 - \underline{\xi})}{4}.$$

Let $C_\star = \max\{C_{\max}, C_\xi, C_{\bar{\varepsilon}}\}$, where $C_{\max}$, C_ξ , and $C_{\bar{\varepsilon}}$ are the constants from Lemma SA-39. For any descendant $\mathbf{t}'$ with $r_{\mathbf{t}'} = n(\mathbf{t}') > 0$, write

$$\hat{\xi}_{\mathbf{t}'} = \frac{n_1(\mathbf{t}')}{r_{\mathbf{t}'}}.$$

For each $\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)$, define

$$\alpha_{\mathbf{t}'} = \frac{n_1(\mathbf{t}')}{n_1(\mathbf{t}_n)}, \quad \beta_{\mathbf{t}'} = \frac{n_0(\mathbf{t}')}{n_0(\mathbf{t}_n)}.$$

These are the treated and control weights with which the descendant averages aggregate back to the parent. Because the terminal descendants partition $\mathbf{t}_n$,

$$\bar{\varepsilon}_{1, \mathbf{t}_n} = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} \alpha_{\mathbf{t}'} \bar{\varepsilon}_{1, \mathbf{t}'}, \quad \bar{\varepsilon}_{0, \mathbf{t}_n} = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} \beta_{\mathbf{t}'} \bar{\varepsilon}_{0, \mathbf{t}'}.$$

Subtracting gives the exact identity

$$\Delta(\mathbf{t}_n) = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} \alpha_{\mathbf{t}'} \Delta(\mathbf{t}') + R_n,$$

where

$$R_n = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} (\alpha_{\mathbf{t}'} - \beta_{\mathbf{t}'}) \bar{\varepsilon}_{0, \mathbf{t}'}.$$

It remains to show that R_n is of smaller order than the root child fluctuation.

For each $\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)$,

$$\alpha_{\mathbf{t}'} - \beta_{\mathbf{t}'} = \frac{r_{\mathbf{t}'}}{M_n} \frac{\hat{\xi}_{\mathbf{t}'} - \hat{\xi}_{\mathbf{t}_n}}{\hat{\xi}_{\mathbf{t}_n} (1 - \hat{\xi}_{\mathbf{t}_n})}.$$

Thus, on $\mathcal{G}_n$,

$$|\alpha_{\mathbf{t}'} - \beta_{\mathbf{t}'}| \leq \frac{4}{\underline{\xi}(1 - \underline{\xi})} \frac{r_{\mathbf{t}'}}{M_n} |\hat{\xi}_{\mathbf{t}'} - \hat{\xi}_{\mathbf{t}_n}|.$$

We split the sum in R_n according to whether the descendant is large enough for the uniform rectangle bounds to apply. First suppose $r_{\mathbf{t}'} \geq h_n$. On $\mathcal{G}_n$,

$$|\hat{\xi}_{\mathbf{t}'} - \hat{\xi}_{\mathbf{t}_n}| \leq C_\star \sqrt{\frac{\log n}{r_{\mathbf{t}'}}} + C_\star \sqrt{\frac{\log n}{M_n}}, \quad |\bar{\varepsilon}_{0, \mathbf{t}'}| \leq C_\star \sqrt{\frac{\log n}{r_{\mathbf{t}'}}}.$$

Thus

$$\begin{aligned} |(\alpha_{\mathbf{t}'} - \beta_{\mathbf{t}'})\bar{\varepsilon}_{0,\mathbf{t}'}| &\leq \frac{4C_\star^2}{\xi(1-\xi)} \frac{r_{\mathbf{t}'}}{M_n} \left(\sqrt{\frac{\log n}{r_{\mathbf{t}'}}} + \sqrt{\frac{\log n}{M_n}} \right) \sqrt{\frac{\log n}{r_{\mathbf{t}'}}} \\ &\leq \frac{8C_\star^2}{\xi(1-\xi)} \frac{\log n}{M_n} \leq C_{\text{large}} \frac{\log n}{M_n}, \end{aligned}$$

where $C_{\text{large}} = 8C_\star^2/\{\xi(1-\xi)\}$ and the last inequality uses $r_{\mathbf{t}'} \leq M_n$. Summing over at most J_n such descendants gives a contribution bounded by $C_{\text{large}} J_n \log n / M_n$.

Now suppose $r_{\mathbf{t}'} < h_n$. For such small descendants, the treatment share bound need not be useful, so we use only the validity of the parent and the maximum error bound. First,

$$|\alpha_{\mathbf{t}'} - \beta_{\mathbf{t}'}| \leq \alpha_{\mathbf{t}'} + \beta_{\mathbf{t}'} \leq \left(\frac{2}{\xi} + \frac{2}{1-\xi} \right) \frac{r_{\mathbf{t}'}}{M_n},$$

because $n_1(\mathbf{t}_n) \geq \xi M_n/2$ and $n_0(\mathbf{t}_n) \geq (1-\xi)M_n/2$ on $\mathcal{E}_n$. Also, on $\mathcal{E}_n$,

$$|\bar{\varepsilon}_{0,\mathbf{t}'}| \leq \max_{\substack{1 \leq i \leq n \\ d \in \{0,1\}}} |\varepsilon_i(d)| \leq C_\star \log n.$$

Hence

$$|(\alpha_{\mathbf{t}'} - \beta_{\mathbf{t}'})\bar{\varepsilon}_{0,\mathbf{t}'}| \leq C_{\text{small},0} \frac{r_{\mathbf{t}'} \log n}{M_n} \leq C_{\text{small}} \frac{\log^2 n}{M_n},$$

where $C_{\text{small},0} = C_\star\{2/\xi + 2/(1-\xi)\}$ and $C_{\text{small}} = C_{\text{small},0}C_0$. Summing over at most J_n small descendants gives a contribution bounded by $C_{\text{small}} J_n \log^2 n / M_n$. Combining the two parts, with $C_{\text{rem}} = C_{\text{large}} + C_{\text{small}}$,

$$|R_n| \leq C_{\text{rem}} \frac{J_n \log^2 n}{M_n}$$

on $\mathcal{E}_n$. Since $\mathbb{P}(\mathcal{E}_n) \rightarrow 1$,

$$R_n = O_{\mathbb{P}}\left(\frac{J_n \log^2 n}{M_n}\right) = o_{\mathbb{P}}\left(M_n^{-1/2} \sqrt{\log \log n}\right),$$

because

$$\frac{J_n \log^2 n / M_n}{M_n^{-1/2} \sqrt{\log \log n}} = \frac{J_n \log^2 n}{\sqrt{M_n} \log \log n} \leq \frac{J_n \log^2 n}{n^{a/2} \sqrt{\log \log n}} \rightarrow 0.$$

Substituting this bound into the exact identity above proves the lemma. $\square$

Corollary SA-41 (Terminal inheritance from a root child with random size). *Under the conditions of Lemma SA-40, suppose that $\mathcal{E}_n$ is any event on which*

$$M_n = n(\mathbf{t}_n) \in [n^a, n^b], \quad |\Delta(\mathbf{t}_n)| \geq c_0 M_n^{-1/2} \sqrt{\log \log n}$$

for some constant $c_0 > 0$. Then

$$\mathbb{P}\left(\mathcal{E}_n, \max_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} |\Delta(\mathbf{t}')| \geq \frac{c_0}{2} M_n^{-1/2} \sqrt{\log \log n}\right) \geq \mathbb{P}(\mathcal{E}_n) - o(1).$$

Consequently,

$$\mathbb{P}\left(\mathcal{E}_n, \max_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} |\Delta(\mathbf{t}')| \geq \frac{c_0}{2} n^{-b/2} \sqrt{\log \log n}\right) \geq \mathbb{P}(\mathcal{E}_n) - o(1).$$

Proof. By Lemma SA-40,

$$\Delta(\mathbf{t}_n) = \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} \alpha_{\mathbf{t}'} \Delta(\mathbf{t}') + \rho_n, \quad \alpha_{\mathbf{t}'} = \frac{n_1(\mathbf{t}')}{n_1(\mathbf{t}_n)}$$

where $\rho_n = o_{\mathbb{P}}(M_n^{-1/2} \sqrt{\log \log n})$. Since $\alpha_{\mathbf{t}'} \geq 0$ and $\sum_{\mathbf{t}'} \alpha_{\mathbf{t}'} = 1$,

$$\left| \sum_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} \alpha_{\mathbf{t}'} \Delta(\mathbf{t}') \right| \leq \max_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} |\Delta(\mathbf{t}')|.$$

Set

$$\lambda_n = M_n^{-1/2} \sqrt{\log \log n}.$$

Then $\rho_n/\lambda_n = o_{\mathbb{P}}(1)$, and therefore

$$\mathbb{P}\left(\mathcal{E}_n, |\rho_n| > \frac{c_0}{2} \lambda_n\right) \rightarrow 0.$$

On $\mathcal{E}_n$, outside the event in the preceding display,

$$\max_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} |\Delta(\mathbf{t}')| \geq |\Delta(\mathbf{t}_n)| - |\rho_n| \geq \frac{c_0}{2} M_n^{-1/2} \sqrt{\log \log n}.$$

Since $M_n \leq n^b$, we also have $M_n^{-1/2} \sqrt{\log \log n} \geq n^{-b/2} \sqrt{\log \log n}$. This gives the second display. $\square$

We can now complete the proof of Theorem SA-18. Set $a = b/2$. The proof of Theorem SA-17 gives more than the final depth one supremum bound. Before replacing the selected child size by the deterministic upper bound n^b , it gives a root child event. Combining the left boundary event with its symmetric right boundary counterpart, there exist constants $c_{\text{root}}, q_{\text{root}} > 0$, depending only on the distributional constants in Assumption SA-2 and on the fixed value of p , and an event $\mathcal{E}_{\text{DIM},n}$ with

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\mathcal{E}_{\text{DIM},n}) \geq q_{\text{root}}(b - a),$$

such that, on $\mathcal{E}_{\text{DIM},n}$, the DIM root split creates a child $\mathbf{t}_n$ satisfying

$$M_n = n(\mathbf{t}_n) \in [n^a, n^b], \quad n_0(\mathbf{t}_n) \wedge n_1(\mathbf{t}_n) > 0, \quad |\Delta(\mathbf{t}_n)| \geq c_{\text{root}} M_n^{-1/2} \sqrt{\log \log n}.$$

Here $\mathbf{t}_n$ is the left root child on the left boundary event and the right root child on the reflected event. The validity condition in the tree construction ensures that both treatment-arm denominators in $\mathbf{t}_n$ are positive.

Let $\mathcal{P}(\mathbf{t}_n)$ be the terminal descendants of this root child in the final NSS-DIM tree. Later splits only refine $\mathbf{t}_n$ into sample rectangles, and valid recursive splitting preserves positive treated and control counts in every terminal descendant. Since the tree has depth at most K , $|\mathcal{P}(\mathbf{t}_n)| \leq 2^K$. The assumed depth condition implies the leaf count condition in Lemma SA-40 with $a = b/2$. Applying Corollary SA-41 with $c_0 = c_{\text{root}}$ gives

$$\mathbb{P}\left(\mathcal{E}_{\text{DIM},n}, \max_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} |\Delta(\mathbf{t}')| \geq \frac{c_{\text{root}}}{2} n^{-b/2} \sqrt{\log \log n}\right) \geq \mathbb{P}(\mathcal{E}_{\text{DIM},n}) - o(1).$$

For every terminal descendant $\mathbf{t}'$ and every $\mathbf{x} \in \mathbf{t}'$,

$$\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau = \Delta(\mathbf{t}').$$

Thus the supremum over $\mathbf{x} \in \mathcal{X}$ is at least the maximum over these terminal descendants. Consequently,

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{DIM}}(\mathbf{x}) - \tau| \geq \frac{c_{\text{root}}}{2} n^{-b/2} \sqrt{\log \log n}\right) \geq q_{\text{root}}(b - a) = \frac{q_{\text{root}}}{2} b.$$

Thus the theorem holds with $c_{\text{DIM}} = c_{\text{root}}/2$ and $q_{\text{DIM}} = q_{\text{root}}/2$.

SA-5.20 Proof of Theorem SA-19

For notational simplicity, denote $\hat{\tau}_{\text{DIM}}$ by $\hat{\tau}$ and the data-driven partition by $\mathcal{P}$.

Reduction to Least Squares Prediction Error

The leaf node values coincide with the least squares projection given $\mathcal{P}$: for $\mathbf{t} \in \mathcal{P}$, we have $\hat{\tau}(\mathbf{t}) = \hat{b}_{\mathbf{t}}$, where

$$\hat{a}_{\mathbf{t}}, \hat{b}_{\mathbf{t}} = \begin{cases} \arg \min_{a,b} \sum_{i=1}^n \mathbf{1}(\mathbf{x}_i \in \mathbf{t}) (y_i - a - b d_i)^2 & \text{if } \sum_{i=1}^n \mathbf{1}(\mathbf{x}_i \in \mathbf{t}) > 0, \\ 0, 0 & \text{otherwise.} \end{cases}$$

If a nonempty terminal node contains only one treatment arm, the least squares minimizer is not unique; throughout this proof we choose the minimizer with $\hat{b}_{\mathbf{t}} = 0$. This agrees with the convention in Definition SA-4,

which sets the DIM estimate to zero when one arm is absent, and it preserves the empirical risk minimization inequality below. Consider the outcome prediction model based on partition $\mathcal{P}$:

$$\begin{aligned}\hat{g}(\mathbf{x}, d) &= \sum_{\mathbf{t} \in \mathcal{P}} \mathbf{1}(\mathbf{x} \in \mathbf{t})(\hat{a}_{\mathbf{t}} + \hat{b}_{\mathbf{t}}d) \\ &= \hat{A}(\mathbf{x}) + \hat{B}(\mathbf{x})d,\end{aligned}\tag{SA-31}$$

where

$$\hat{A}(\mathbf{x}) = \sum_{\mathbf{t} \in \mathcal{P}} \mathbf{1}(\mathbf{x} \in \mathbf{t})\hat{a}_{\mathbf{t}}, \quad \hat{B}(\mathbf{x}) = \sum_{\mathbf{t} \in \mathcal{P}} \mathbf{1}(\mathbf{x} \in \mathbf{t})\hat{b}_{\mathbf{t}}.$$

First, we show that for integrated L_2 convergence rates for treatment effect estimation, it is enough to look at the L_2 loss for outcome prediction. Denote by $P_{X,d}$ the joint distribution of $(\mathbf{x}_i, d_i)$. Since we assumed $\mathbf{x}_i$ and d_i are independent, we have $P_{X,d} = P_X \times P_d$, where P_X and P_d are the marginal distributions of $\mathbf{x}_i$ and d_i . Given Assumption SA-2, the target outcome prediction model is

$$g^*(\mathbf{x}_i, d_i) = \mathbb{E}[y_i | \mathbf{x}_i, d_i] = \mu + \tau d_i, \quad \mu = \mathbb{E}[y_i(0)], \quad \tau = \mathbb{E}[y_i(1) - y_i(0)].$$

Hence

$$\begin{aligned}\mathbb{E}[\|\hat{g} - g^*\|^2] &= \mathbb{E}\left[\int_{\mathcal{X} \times \{0,1\}} (\hat{g}(\mathbf{x}, d) - \mu - \tau d)^2 dP_{X,d}(\mathbf{x}, d)\right] \\ &= \mathbb{E}\left[\int_{\mathcal{X} \times \{0,1\}} (\hat{A}(\mathbf{x}) + \hat{B}(\mathbf{x})d - \mu - \tau d)^2 dP_X(\mathbf{x}) \times P_d(d)\right] \\ &= \mathbb{E}\left[\int_{\mathcal{X} \times \{0,1\}} (d(\hat{A}(\mathbf{x}) + \hat{B}(\mathbf{x}) - \mu - \tau) + (1-d)(\hat{A}(\mathbf{x}) - \mu))^2 dP_X(\mathbf{x}) \times P_d(d)\right] \\ &= \mathbb{E}\left[\int_{\mathcal{X} \times \{0,1\}} d(\hat{A}(\mathbf{x}) + \hat{B}(\mathbf{x}) - \mu - \tau)^2 + (1-d)(\hat{A}(\mathbf{x}) - \mu)^2 dP_X(\mathbf{x}) \times P_d(d)\right] \\ &= \mathbb{E}\left[\xi \int_{\mathcal{X}} (\hat{A}(\mathbf{x}) + \hat{B}(\mathbf{x}) - \mu - \tau)^2 dP_X(\mathbf{x}) + (1-\xi) \int_{\mathcal{X}} (\hat{A}(\mathbf{x}) - \mu)^2 dP_X(\mathbf{x})\right] \\ &= \xi \mathbb{E}[\|\hat{A} + \hat{B} - \mu - \tau\|^2] + (1-\xi) \mathbb{E}[\|\hat{A} - \mu\|^2].\end{aligned}\tag{SA-32}$$

For each realization of $(\hat{A}, \hat{B})$,

$$\|\hat{B} - \tau\|^2 \leq 2\|\hat{A} + \hat{B} - \mu - \tau\|^2 + 2\|\hat{A} - \mu\|^2.$$

Taking expectations and using (SA-32),

$$\mathbb{E}[\|\hat{\tau} - \tau\|^2] = \mathbb{E}[\|\hat{B} - \tau\|^2] \leq \frac{4}{\min\{\xi, 1-\xi\}} \mathbb{E}[\|\hat{g} - g^*\|^2].$$

Error Bound for Least Squares Prediction

We bound the least squares error $\mathbb{E}[\|\hat{g} - g^*\|^2]$ following the strategy for [Klusowski and Tian, 2024, Theorem 4.3]. First, assume $|y_i(t)| \leq U$, $i = 1, 2, \dots, n$, $t = 0, 1$, for some $U \geq 0$. Decompose by

$$\|\hat{g} - g^*\|^2 = E_1 + E_2,$$

where

$$\begin{aligned}E_1 &= \|\hat{g} - g^*\|^2 - 2(\|y - \hat{g}\|_{\mathcal{D}}^2 - \|y - g^*\|_{\mathcal{D}}^2) - \alpha - \beta, \\ E_2 &= 2(\|y - \hat{g}\|_{\mathcal{D}}^2 - \|y - g^*\|_{\mathcal{D}}^2) + \alpha + \beta.\end{aligned}$$

The least squares representation (SA-31) implies that

$$\|y - \hat{g}\|_{\mathcal{D}}^2 \leq \frac{1}{n} \min_{a \in \mathbb{R}, b \in \mathbb{R}} \sum_{i=1}^n (y_i - a - b d_i)^2 \leq \frac{1}{n} \sum_{i=1}^n (y_i - \mu - \tau d_i)^2 = \|y - g^*\|_{\mathcal{D}}^2, \quad (\text{SA-33})$$

which implies

$$E_2 \leq \alpha + \beta.$$

We control E_1 using uniform law of large number arguments. On the bounded event, $\hat{g}$ is one member of the class

$$\mathcal{G}_n = \{(1-d)G_0(\mathbf{x}) + dG_1(\mathbf{x}) : G_0, G_1 \in \mathcal{H}_n\},$$

where $\mathcal{H}_n$ is the class of piecewise constant functions, bounded by U , on partitions $\mathbb{P} \in \Pi_n$. Here

$$\Pi_n = \{\mathcal{P}(\{(\mathbf{x}_1, d_1, y_1), \dots, (\mathbf{x}_n, d_n, y_n)\}) : (\mathbf{x}_i, d_i, y_i) \in \mathbb{R}^p \times \mathbb{R} \times \mathbb{R}\},$$

is the family of all achievable partitions $\mathcal{P}$ by growing a depth K binary tree on n points by iteratively splitting in $\mathbf{x}$ -space based on any criterion. By [Klusowski and Tian, 2024, Equation B.33],

$$N\left(\frac{\beta}{40U}, \mathcal{H}_n, \|\cdot\|_{P_{X^n, 1}}\right) \leq (np)^{2^K} \left(\frac{417eU^2}{\beta}\right)^{2^{K+1}}.$$

A product covering argument then gives, for a universal constant C ,

$$N\left(\frac{\beta}{80U}, \mathcal{G}_n, \|\cdot\|_{P_{(X, d)^n, 1}}\right) \leq (np)^{2^{K+1}} \left(\frac{CU^2}{\beta}\right)^{2^{K+2}},$$

Here P_{X^n} is the empirical measure of $X_1, \dots, X_n$, and $P_{(X, d)^n}$ is the empirical measure of $(X_i, d_i)_{i=1}^n$. Since $\hat{g} \in \mathcal{G}_n$, [Györfi et al., 2002, Theorem 11.4] gives

$$\begin{aligned} \mathbb{P}(E_1 \geq 0) &\leq \mathbb{P}(\exists g \in \mathcal{G}_n : \|\hat{g} - g^*\|^2 \geq 2(\|y - \hat{g}\|_{\mathcal{D}}^2 - \|y - g^*\|_{\mathcal{D}}^2) + \alpha + \beta) \\ &\leq 14 \sup_{(X, d)^n} N\left(\frac{\beta}{80U}, \mathcal{G}_n, \|\cdot\|_{P_{(X, d)^n, 1}}\right) \exp\left(-\frac{\alpha n}{2568U^4}\right) \\ &\leq 14(np)^{2^{K+1}} \left(\frac{CU^2}{\beta}\right)^{2^{K+2}} \exp\left(-\frac{\alpha n}{2568U^4}\right). \end{aligned}$$

Choosing $\alpha \propto \frac{U^4 2^K \log(np)}{n}$ and $\beta \propto \frac{U^2}{n}$ gives

$$\mathbb{E}[\|\hat{g} - g^*\|^2] \leq C \left(\frac{U^4 2^K \log(np)}{n} + \frac{U^2}{n} \right),$$

where C is a positive universal constant.

Relax the condition that $|y_i(t)| \leq U$. Take $A = \{|y_i(t)| \leq U, \forall i = 1, \dots, n, t = 0, 1\}$. Then

$$\begin{aligned} \mathbb{E}[\|\hat{g} - g^*\|^2] &= \mathbb{E}[\|\hat{g} - g^*\|^2 \mathbf{1}(A)] + \mathbb{E}[\|\hat{g} - g^*\|^2 \mathbf{1}(A^c)] \\ &\leq C \left(\frac{U^4 2^K \log(np)}{n} + \frac{U^2}{n} \right) + \mathbb{E}[\|\hat{g} - g^*\|^2 \mathbf{1}(A^c)]. \end{aligned} \quad (\text{SA-34})$$

A union bound and subexponentiality give constants $C, c > 0$ such that

$$\begin{aligned} \mathbb{P}(A^c) &\leq n\mathbb{P}(|y_i(0)| \geq U) + n\mathbb{P}(|y_i(1)| \geq U) \\ &\leq Cn \exp(-cU). \end{aligned}$$

For the truncation remainder, use the deterministic bound that every fitted leaf value is either zero or an average of observed outcomes in one treatment arm. Hence, for all samples,

$$\|\hat{g} - g^*\|^2 \leq C \left(1 + \max_{1 \leq i \leq n, t \in \{0, 1\}} |y_i(t)|^2 \right),$$

where C depends only on the fixed constants c_0, c_1 . The exponential moment assumption then gives

$$\mathbb{E}[\|\hat{g} - g^*\|^4] \leq C \mathbb{E} \left[1 + \max_{i,t} |y_i(t)|^4 \right] \leq C \log^4(n) \leq Cn.$$

Using this bound and Cauchy–Schwarz,

$$\begin{aligned} \mathbb{E}[\|\hat{g} - g^*\|^2 \mathbf{1}(A^c)] &\leq \sqrt{\mathbb{E}[\|\hat{g} - g^*\|^4] \mathbb{P}(A^c)} \\ &\leq Cn \exp(-cU/2). \end{aligned}$$

Choosing $U = C_0 \log(n)$ with C_0 sufficiently large, we have

$$\mathbb{E}[\|\hat{g} - g^*\|^2 \mathbf{1}(A^c)] \leq \frac{C}{n},$$

for some absolute constant C . Putting this bound into Equation (SA-34) gives the desired expectation conclusion.

For the high probability bound, the deterministic version of the comparison in Equation (SA-32) gives

$$\|\hat{\tau} - \tau\|^2 = \|\hat{B} - \tau\|^2 \leq \frac{4}{\min\{\xi, 1 - \xi\}} \|\hat{g} - g^*\|^2.$$

On the bounded event A , the preceding empirical process bound, with the same choices of α and β , implies

$$\mathbb{P} \left(\|\hat{g} - g^*\|^2 > C_1 \left(\frac{U^4 2^K \log(np)}{n} + \frac{U^2}{n} \right), A \right) \leq n^{-C_2},$$

for positive constants C_1 and C_2 . Taking again $U = C_0 \log(n)$ with C_0 sufficiently large gives $\mathbb{P}(A^c) \leq n^{-C_3}$ for some $C_3 > 0$. Combining the two probability bounds with the deterministic comparison proves the stated high probability conclusion. Indeed, for a sufficiently large constant C ,

$$\mathbb{P} \left(\|\hat{\tau} - \tau\|^2 > C \frac{2^K \log^4(n) \log(np)}{n} \right) \leq n^{-C_2} + \mathbb{P}(A^c) \leq n^{-c},$$

after increasing C and decreasing $c > 0$ if necessary; the smaller U^2/n term is absorbed by the displayed rate.

SA-5.21 Proof of Theorem SA-20

The honest proof uses sample splitting, so terminal node estimation can be handled conditionally on the constructed tree. The construction fold first selects, with probability proportional to b , an imbalanced root child of size at most order n^b . Conditional on the resulting tree, the independent estimation fold places a comparable number of observations in this root child, and an occupancy argument shows that at least one terminal descendant receives two or more estimation fold observations with probability tending to one. That descendant is chosen using only the construction tree and estimation fold covariates, so its treatment assignments and outcomes remain independent of the selection step. Lemma SA-42 then supplies the fixed conditional probability that the honest terminal contrast has size at least a constant multiple of $N_\star^{-1/2}$, which is at least order $n^{-b/2}$ on the event constructed below.

Lemma SA-42 (Independent finite-arm anti-concentration). *Suppose Assumption SA-2 holds. Let $\mathbf{t}_\star$ be a terminal node selected using only a construction fold tree and estimation fold covariates, not estimation fold treatment assignments or outcomes. Conditional on $\mathbf{t}_\star$ and on treatment counts $N_{1,\star}, N_{0,\star}$ satisfying $N_{1,\star} > 0$ and $N_{0,\star} > 0$, the honest DIM terminal contrast obeys*

$$\mathbb{P} \left(|\hat{\tau}_{\text{DIM}}(\mathbf{t}_\star) - \tau| \geq c N_\star^{-1/2} \mid \mathbf{t}_\star, N_{1,\star}, N_{0,\star} \right) \geq q, \quad N_\star = N_{1,\star} + N_{0,\star},$$

for constants $c, q > 0$ depending only on the distribution of $(y_i(0), y_i(1), d_i)$.

Proof. Conditional on $\mathbf{t}_\star$ and the treatment counts, the estimation fold errors in $\mathbf{t}_\star$ are independent draws from their original treatment-arm error laws. The centered terminal error is

$$\frac{1}{N_{1,\star}} \sum_{i=1}^{N_{1,\star}} \varepsilon_i(1) - \frac{1}{N_{0,\star}} \sum_{i=1}^{N_{0,\star}} \varepsilon_i(0).$$

Its conditional variance is bounded below by $c_\varepsilon(N_{1,\star}^{-1} + N_{0,\star}^{-1}) \geq c_\varepsilon N_\star^{-1}$, while the exponential moment assumption gives a fourth moment bounded above by $C_\varepsilon(N_{1,\star}^{-1} + N_{0,\star}^{-1})^2$. Paley–Zygmund applied to the square of the display gives the stated anti-concentration bound after decreasing c if necessary. The constants are uniform over all positive treatment counts. $\square$

Let $M = n_\tau$ and $N = n_\tau$ denote the construction and estimation fold sample sizes, and write $\tilde{\tau}(\mathbf{x}) = \tilde{\tau}_{\text{DIM}}(\mathbf{x})$. Fix $b \in (0, 1)$ and set $a = b/2$. Let $(\hat{i}, \hat{j})$ be the root splitting index and coordinate selected by the construction fold, and consider the root event

$$\mathcal{E}_n = \{\exists \ell \in [p] : M^a \leq \hat{i} \leq M^b, \hat{j} = \ell\}.$$

By Theorem SA-16 and the fact that p is fixed, $\liminf_{n \rightarrow \infty} \mathbb{P}(\mathcal{E}_n) \geq cb$ for a positive constant c . On $\mathcal{E}_n$, let $\mathbf{t}_L$ be the smaller root child selected by the event, and put $m = n_\tau(\mathbf{t}_L) = \hat{i}$.

Let $\mathcal{P}(\mathbf{t}_L) = \{\mathbf{t}_1, \dots, \mathbf{t}_J\}$ be the terminal descendants of $\mathbf{t}_L$ in the final tree. Because each valid causal split requires positive treated and control construction-sample denominators in both child nodes, every terminal descendant of $\mathbf{t}_L$ contains at least one treated and one control construction observation. Hence

$$J \leq m/2.$$

Let R be the number of estimation fold observations that fall in $\mathbf{t}_L$. Conditional on the construction fold, R is binomial with success probability equal to the population mass of $\mathbf{t}_L$. Since this mass is the order statistic $F_{\hat{j}}(x_{\pi_{\hat{j}}(m), \hat{j}})$, the beta concentration bound for order statistics used above, the honest splitting condition $n \lesssim n_\tau, n_\tau \lesssim n$, and a binomial Chernoff bound give constants $0 < c_R < C_R < \infty$ such that, conditional on $\mathcal{E}_n$,

$$\mathbb{P}(c_R m \leq R \leq C_R m \mid \mathcal{E}_n) \rightarrow 1.$$

In particular, on this event $R \leq C_R M^b \leq C n^b$.

Conditional on the construction fold and on R , the R estimation fold observations in $\mathbf{t}_L$ are allocated among the J terminal descendants according to their conditional probabilities $q_1, \dots, q_J$. If no terminal descendant receives two estimation fold observations, then the R draws are all assigned to distinct descendants. For $R \leq J$, this probability is $R!e_R(q_1, \dots, q_J)$, where e_R is the R th elementary symmetric polynomial; for $R > J$ it is zero. By Maclaurin's inequality, $e_R(q_1, \dots, q_J) \leq \binom{J}{R} J^{-R}$. Therefore

$$\mathbb{P}\left(\max_{1 \leq r \leq J} n_\tau(\mathbf{t}_r) \leq 1 \mid R, J\right) \leq \prod_{s=0}^{R-1} \left(1 - \frac{s}{J}\right) \leq \exp\left(-\frac{R(R-1)}{2J}\right)$$

with the product interpreted as zero when $R > J$. Since $J \leq m/2$ and $R \geq c_R m$ with probability tending to one conditional on $\mathcal{E}_n$, the last display implies that, conditional on $\mathcal{E}_n$, with probability tending to one at least one terminal descendant receives at least two estimation fold observations. Choose one such terminal descendant by a fixed deterministic rule and denote it by $\mathbf{t}_\star$. This selection uses only the construction fold tree and the estimation fold covariates through the descendant occupancy counts; it does not use estimation fold treatment assignments or outcomes. Consequently, conditional on $\mathbf{t}_\star$ and its occupancy count, the treatment indicators and potential outcome errors inside $\mathbf{t}_\star$ retain their original independence and Bernoulli/error laws. Let

$$N_\star = n_\tau(\mathbf{t}_\star), \quad N_{1,\star} = n_{1,\tau}(\mathbf{t}_\star), \quad N_{0,\star} = n_{0,\tau}(\mathbf{t}_\star).$$

Then $2 \leq N_\star \leq R \leq C n^b$ on the event just described.

Because the estimation fold treatment assignments are independent of the tree and of the estimation fold covariates,

$$\mathbb{P}(N_{1,\star} > 0, N_{0,\star} > 0 \mid N_\star) \geq 2\xi(1 - \xi).$$

This is the common treatment-arm positivity factor used in the honest causal lower bounds. It is the positive denominator event for the terminal node DIM estimator; on its complement the estimator is defined to be zero by Definition SA-4. We therefore restrict the lower bound calculation to this event. On this event, for any $\mathbf{x}_\star \in \mathbf{t}_\star$,

$$|\tilde{\tau}(\mathbf{x}_\star) - \tau| = \left| \frac{1}{N_{1,\star}} \sum_{\tilde{\mathbf{x}}_i \in \mathbf{t}_\star} \tilde{d}_i \tilde{\varepsilon}_i(1) - \frac{1}{N_{0,\star}} \sum_{\tilde{\mathbf{x}}_i \in \mathbf{t}_\star} (1 - \tilde{d}_i) \tilde{\varepsilon}_i(0) \right|.$$

Lemma SA-42, applied conditionally on the independent estimation fold, implies

$$\mathbb{P}\left(|\tilde{\tau}(\mathbf{x}_*) - \tau| \geq cN_*^{-1/2} \mid \mathbf{t}_*, N_{1,*}, N_{0,*}\right) \geq c$$

whenever $N_{1,*} > 0$ and $N_{0,*} > 0$. Combining the previous displays and using $N_* \leq Cn^b$ gives, after renaming constants,

$$\liminf_{n \rightarrow \infty} \mathbb{P}\left(\sup_{\mathbf{x} \in \mathcal{X}} |\tilde{\tau}(\mathbf{x}) - \tau| \geq C_1 n^{-b/2}\right) \geq C_2 b.$$

The fixed factor $2\xi(1 - \xi)$ and the constants from the root event probability, allocation argument, and Paley–Zygmund bound are absorbed into C_2 , which is allowed because C_2 depends on the distribution of $(y_i(0), y_i(1), d_i)$ and on the sample splitting ratios.

SA-5.22 Proof of Theorem SA-21

For simplicity, denote $\tilde{\tau}_{\text{DIM}}$ by $\tilde{\tau}$. Conditional on the construction fold $\mathcal{D}_{\text{T}}$, the partition $\mathcal{P}$ is fixed and is independent of the estimation fold $\mathcal{D}_{\tau}$. The least squares representation in Equation (SA-31) continues to hold with empirical risk computed only over $\mathcal{D}_{\tau}$. Therefore the deterministic comparison in Equation (SA-32) reduces the problem to bounding the conditional prediction risk of $\hat{g}$.

Assume first that $|y_i(t)| \leq U$ for $t = 0, 1$. Conditional on $\mathcal{D}_{\text{T}}$, define the fixed partition prediction class

$$\mathcal{G}(\mathcal{P}) = \{(\mathbf{x}, d) \mapsto A(\mathbf{x}) + dB(\mathbf{x}) : A, B \text{ are piecewise constant on } \mathcal{P}, |A|, |B| \leq CU\},$$

where the constant C is universal. Since $\mathcal{P}$ has at most 2^K leaves,

$$N(\varepsilon U, \mathcal{G}(\mathcal{P}), \|\cdot\|_{P_{X^{n_{\tau}}, 1}}) \leq \left(\frac{C}{\varepsilon}\right)^{2^{K+1}}, \quad \varepsilon \in (0, 1).$$

Thus the empirical process argument used in the proof of Theorem SA-19, now conditional on $\mathcal{D}_{\text{T}}$ and with sample size n_{τ} , gives

$$\begin{aligned} \mathbb{E}_{\mathcal{D}_{\tau}} [\|\hat{g} - g^*\|^2 \mid \mathcal{D}_{\text{T}}] &\leq C \left(\frac{U^4 2^K \log(n_{\tau})}{n_{\tau}} + \frac{U^2}{n_{\tau}} \right), \\ \mathbb{P}_{\mathcal{D}_{\tau}} \left(\|\hat{g} - g^*\|^2 > C \left(\frac{U^4 2^K \log(n_{\tau})}{n_{\tau}} + \frac{U^2}{n_{\tau}} \right) \mid \mathcal{D}_{\text{T}} \right) &\leq n_{\tau}^{-c}, \end{aligned}$$

for constants $C, c > 0$. The factor 2^{K+1} in the covering exponent reflects that the causal prediction class has two leafwise components, A and B ; it only changes constants and hence keeps the same $2^K \log(n_{\tau})$ complexity term.

Taking $U = C_0 \log(n_{\tau})$ and repeating the truncation argument from the proof of Theorem SA-19 gives the conditional expectation and high probability bounds with rate $2^K \log^5(n_{\tau})/n_{\tau}$. Applying the deterministic comparison between $\tilde{\tau}$ and $\hat{g}$, unconditioning over $\mathcal{D}_{\text{T}}$, and using $\rho \leq n_{\text{T}}/n_{\tau} \leq \rho^{-1}$ completes the proof.

SA-5.23 Proof of Lemma SA-22

All maxima below involving $\mathcal{S}^{\text{SSE}}$ are taken over the intersection of the displayed index range with the valid candidate set $\mathcal{V}_{\text{SSE}}$. The corresponding proxy process maxima may be taken over the larger displayed range, which only weakens upper bounds.

Write $\omega_k = k(n - k)/n$. From the proof of Lemma SA-14,

$$\max_{1 \leq \ell \leq p} \sup_{r_n \leq k < n - r_n} \omega_k \left| (\hat{\mu}_{L,0}(k, \ell) - \hat{\mu}_{R,0}(k, \ell))^2 - (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 \right| = O_{\mathbb{P}} \left(\frac{\log \log n}{\sqrt{r_n}} \right).$$

Moreover, the proof of the term R_1 in Lemma SA-14 implies the weighted contrast bound

$$\max_{1 \leq \ell \leq p} \sup_{r_n \leq k < n - r_n} \omega_k \left\{ (\hat{\mu}_{L,0}(k, \ell) - \hat{\mu}_{R,0}(k, \ell))^2 + (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 \right\} = O_{\mathbb{P}}(\log \log n).$$

Consider the randomness induced by $n_0, n_{L,0}, n_{R,0}$. By Theorem A.4.1 in Csörgő and Horváth [1997], applied to the left and reversed right partial sums and then unioned over the fixed set of coordinates,

$$\max_{1 \leq \ell \leq p} \max_{r_n \leq k < n - r_n} \left[\sqrt{k} \left| \frac{n_{L,0}(k)}{(1-\xi)k} - 1 \right| + \sqrt{n-k} \left| \frac{n_{R,0}(k)}{(1-\xi)(n-k)} - 1 \right| \right] = O_{\mathbb{P}}(\sqrt{\log \log n}),$$

and $n_0/((1-\xi)n) - 1 = O_{\mathbb{P}}(\sqrt{\log \log n/n})$. Hence

$$\max_{1 \leq \ell \leq p} \sup_{r_n \leq k < n - r_n} \frac{1}{\sqrt{\omega_k}} \left| \frac{n_{L,0}(k)n_{R,0}(k)}{n_0} - (1-\xi)\omega_k \right| = O_{\mathbb{P}}(\sqrt{\log \log n}).$$

Combining this weighted count bound with the preceding weighted contrast bound and the fact that $\omega_k \gtrsim r_n$ on the displayed range gives

$$\max_{1 \leq \ell \leq p} \sup_{r_n \leq k < n - r_n} \left| \frac{n_{L,0}(k)n_{R,0}(k)}{n_0} - (1-\xi)\omega_k \right| (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 = O_{\mathbb{P}}\left(\frac{(\log \log n)^{3/2}}{\sqrt{r_n}}\right).$$

The same bounds hold for the treated arm. Also, $n_{L,d}(k)n_{R,d}(k)/n_d \leq \min\{k, n-k\} \leq 2\omega_k$ for $d = 0, 1$, so the empirical to proxy squared contrast error is controlled by the first display. Putting these pieces together, the triangle inequality implies

$$\max_{1 \leq \ell \leq p} \max_{r_n \leq k < n - r_n} \left| \mathcal{J}^{\text{SSE}}(k, \ell) - \mathcal{J}^{\text{prox}}(k, \ell) \right| = O_{\mathbb{P}}\left(\frac{(\log \log n)^{3/2}}{r_n^{1/2}}\right).$$

SA-5.24 Proof of Lemma SA-23

All maxima involving $\mathcal{J}^{\text{SSE}}$ are again over valid candidates in $\mathcal{V}_{\text{SSE}}$.

The proof of Lemma SA-15 implies that

$$\max_{1 \leq \ell \leq p} \max_{1 \leq k \leq s_n} k \left| (\hat{\mu}_{L,0}(k, \ell) - \hat{\mu}_{R,0}(k, \ell))^2 - (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 \right| = O_{\mathbb{P}}(\alpha_n),$$

where $\alpha_n = \rho_n \log \log n + \frac{s_n}{n-s_n} \log \log n$, and

$$\max_{1 \leq \ell \leq p} \max_{1 \leq k \leq s_n} k (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 = O_{\mathbb{P}}(\rho_n \log \log n).$$

Hence it also follows that

$$\max_{1 \leq \ell \leq p} \max_{1 \leq k \leq s_n} k (\hat{\mu}_{L,0}(k, \ell) - \hat{\mu}_{R,0}(k, \ell))^2 = O_{\mathbb{P}}\left(\frac{s_n}{n-s_n} \log \log n + \alpha_n\right) = O_{\mathbb{P}}(\alpha_n).$$

When $1 \leq k \leq s_n$, we have $\frac{n_{L,0}(k)n_{R,0}(k)}{n_0} \leq n_{L,0}(k) \leq k$. The right edge range follows by applying the same argument to the reversed ordering. The conclusion then follows.

SA-5.25 Proof of Lemma SA-25

The proof of Theorem SA-24 gives a joint Gaussian approximation to the treated and control components of the SSE criterion. After the normalization used in that proof, the imbalanced window criterion is approximated on a finite grid by

$$Q_{\Lambda}(Z_{k,\ell}) = \sigma_0^2 Z_{0,k,\ell}^2 + \sigma_1^2 Z_{1,k,\ell}^2, \quad Z_{k,\ell} = (Z_{0,k,\ell}, Z_{1,k,\ell})^{\top},$$

where $\sigma_d^2 = \mathbb{V}[\varepsilon_i(d)]$, each local vector $Z_{k,\ell}$ is centered Gaussian with standardized marginal variances, and the collection has the covariance structure described in the Gaussian comparison step of Theorem SA-24. For the corresponding small child $S_{k,\ell}$, write $m = m_{k,\ell} = n(S_{k,\ell})$ and $m_d = n_d(S_{k,\ell})$. The centered terminal CATE contrast satisfies

$$\hat{\tau}(S_{k,\ell}) - \tau = \frac{1}{m_1} \sum_{i \in S_{k,\ell}} d_i \varepsilon_i(1) - \frac{1}{m_0} \sum_{i \in S_{k,\ell}} (1 - d_i) \varepsilon_i(0).$$

On the balanced treatment count event, $m_1/m = \xi + o_{\mathbb{P}}(1)$ and $m_0/m = 1 - \xi + o_{\mathbb{P}}(1)$ uniformly over the selected valid candidates. Hence, with $v = (\xi^{-1/2}, -(1 - \xi)^{-1/2})^\top$ and with $Z_{k,\ell}$ denoting the corresponding standardized treated/control Gaussian fluctuation,

$$\hat{\tau}(S_{k,\ell}) - \tau = m_{k,\ell}^{-1/2} v_\Lambda^\top Z_{k,\ell} + o_{\mathbb{P}}\left(n^{-b/2} \sqrt{\log \log n}\right),$$

where v_Λ has both coordinates nonzero; the fixed variance scale constants are absorbed into v_Λ . Because the selected imbalanced child has $m_{k,\ell} \leq n^b$, a projection of order $\sqrt{\log \log n}$ in the v_Λ direction yields a terminal CATE error of order at least $n^{-b/2} \sqrt{\log \log n}$. Thus the only additional point, beyond the split index theorem, is to ensure that the selected bivariate fluctuation is not asymptotically concentrated in the line orthogonal to v_Λ .

If $\sigma_0^2 = \sigma_1^2$, we insert a fixed cone restriction into the same finite-grid Gaussian threshold comparison used in Theorem SA-24. On that finite grid, the event that the selected norm maximizer lies in the imbalanced window for coordinate ℓ and beats the competing coordinates and locations is measurable only through the Gaussian norms $\|Z_{k,j}\|$. Lemma SA-37 therefore applies to the selected Gaussian vector itself, not merely to a fixed vector: after imposing

$$|v_\Lambda^\top Z_{\hat{k},j}| \geq \gamma \|v_\Lambda\| \|Z_{\hat{k},j}\|$$

the finite-grid Gaussian probability is reduced by at most a fixed positive factor π_γ .

If the variances are unequal, suppose for concreteness that $\sigma_1^2 > \sigma_0^2$; the other case is symmetric. Choose $\rho > 0$ so small that $|v_{\Lambda,0}| \rho \leq |v_{\Lambda,1}|/2$. By the dominance part of Lemma SA-38, the probability that the selected weighted extreme lies in the cone $|Z_0| > \rho |Z_1|$ is $o(1)$ in the threshold comparison used to prove Theorem SA-24. On the complementary cone,

$$|v_\Lambda^\top Z| \geq (|v_{\Lambda,1}| - |v_{\Lambda,0}| \rho) |Z_1| \geq c_\Lambda \|Z\|,$$

for a positive constant c_Λ . Since $Q_\Lambda(Z)$ is of order $\log \log n$ on the selected threshold event, this gives $|v_\Lambda^\top Z|$ of order $\sqrt{\log \log n}$.

The cone events just described are finite unions of angular sectors; approximating those sectors by polygonal cones lets the same simple convex set CLT and Gaussian comparison bounds used in Theorem SA-24 apply with the same $o(1)$ error. The Gaussian correlation and O-U stationarity steps then give the same imbalanced window probability lower bound, multiplied only by a fixed cone factor in the equal variance case and with only an $o(1)$ loss in the unequal variance case.

On this restricted event, the selected small child has $m \in [n^a, n^b]$ observations, the selected weighted criterion is of order $\log \log n$, and the cone or dominance restriction gives a CATE direction projection of order $\sqrt{\log \log n}$. Returning to the terminal average normalization yields

$$m \Delta(S_{\hat{k},\ell})^2 \geq c^2 \log \log n$$

after decreasing $c > 0$ if necessary. Validity of the selected SSE split gives positive treated and control denominators in both children. Absorbing the cone probability, treatment probability constants, and approximation constants into c and q proves the left boundary statement. The right boundary statement follows by reversing the ordering along coordinate ℓ .

SA-5.26 Proof of Theorem SA-24

The proof is similar to the proof of Theorem SA-16. The difference is that Theorem SA-16 approximates the split criterion by a time-transformed O-U process; here, we approximate the split criterion by a weighted quadratic form of a bivariate time-transformed O-U process. We divide the proof into four steps.

Throughout this proof, every maximum involving $\mathcal{J}^{\text{SSE}}$ is taken over the relevant valid candidates retained by Definition SA-5; if the displayed index range contains no valid candidate, the maximum is interpreted as $-\infty$. On the balanced ranges used below, all treatment-arm denominators are positive with probability tending to one, so this convention does not affect the Gaussian comparison step. Let $\sigma_d^2 = \mathbb{V}[\varepsilon_i(d)]$ for $d \in \{0, 1\}$ and $\lambda_\star = \max\{\sigma_0^2, \sigma_1^2\}$.

Step 1: Approximation of fit-based processes by IPW-based processes

Let $0 < a < b < 1$. Let ρ_n be a sequence of real numbers taking values in $(0, 1)$ to be determined, and take $s_n = \exp((\log n)^{\rho_n})$. Then for large enough n , we have $s_n \leq n^a \leq n^b \leq n - s_n$. Consider the event

$$A_n = \left\{ \exists \ell \in [p] : \max_{\substack{k \in [n] \\ k \notin [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, \ell) \geq \max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{SSE}}(k, \ell) \right\}.$$

Equation (A.4.18) and (A.4.20) imply that for each $\ell \in [p]$,

$$\begin{aligned} \max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \mathcal{J}^{\text{prox}}(k, \ell) &= O_{\mathbb{P}}(\rho_n \log \log(n)), \\ \max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{prox}}(k, \ell) &= 2\lambda_{\star} \log \log(n)(1 + o_{\mathbb{P}}(1)). \end{aligned}$$

Hence

$$\max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \mathcal{J}^{\text{prox}}(k, \ell) = o_{\mathbb{P}}\left(\max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{prox}}(k, \ell)\right), \quad \ell \in [p],$$

The approximation errors from Lemma SA-22 (taking $r_n = s_n$) and Lemma SA-23 are both $o_{\mathbb{P}}(\log \log n)$ under $\log \log \log \log(n) / \log \log(n) \ll \rho_n \ll 1$. Therefore the imbalanced edge maximum of $\mathcal{J}^{\text{SSE}}$ remains $o_{\mathbb{P}}(\log \log n)$ while the balanced maximum remains $2 \log \log(n)(1 + o_{\mathbb{P}}(1))$, exactly as in the separation step for Theorem SA-16. Hence

$$\max_{\substack{1 \leq k \leq s_n \\ \text{or } n-s_n \leq k \leq n}} \mathcal{J}^{\text{SSE}}(k, \ell) = o_{\mathbb{P}}\left(\max_{s_n \leq k \leq n-s_n} \mathcal{J}^{\text{SSE}}(k, \ell)\right), \quad \ell \in [p].$$

Using a union bound, $\mathbb{P}(A_n) \rightarrow 0$ as $n \rightarrow \infty$. On the event A_n^c , the argmax for $\mathcal{J}^{\text{SSE}}$ lies inside $[s_n, n - s_n]$. Hence

$$\begin{aligned} &\mathbb{P}\left(\exists \ell \in [p] : \max_k \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}^{\text{SSE}}(k, j), \max_k \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{k \notin [n^a, n^b]} \mathcal{J}^{\text{SSE}}(k, \ell)\right) \\ &\geq \mathbb{P}\left(\exists \ell \in [p] : \max_k \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{k, j \neq \ell} \mathcal{J}^{\text{SSE}}(k, j), \max_k \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{k \notin [n^a, n^b]} \mathcal{J}^{\text{SSE}}(k, \ell) \text{ and } A_n^c\right) - \mathbb{P}(A_n) \\ &\geq \mathbb{P}\left(\exists \ell \in [p] : \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{\substack{j \neq \ell \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j), \right. \\ &\quad \left. \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, \ell)\right) - 2\mathbb{P}(A_n). \end{aligned}$$

Focus on the first term. By symmetry in the p coordinates,

$$\begin{aligned} &\mathbb{P}\left(\exists \ell \in [p] : \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{\substack{j \neq \ell \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j), \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, \ell)\right) \\ &= p\mathbb{P}\left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, 1) > \max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j), \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, 1) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, 1)\right) \\ &\geq p \sup_{z \in \mathbb{R}} \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j) < z, \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, 1) > z > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, 1)\right) \\ &\geq p \sup_{z \in \mathbb{R}} \left\{ \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j) < z, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, 1) < z\right) \right. \\ &\quad \left. - \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j) < z, \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, 1) < z\right) \right\}. \end{aligned}$$

Then using the fact that $\mathcal{J}^{\text{prox}}(k, \ell)$ approximates $\mathcal{J}^{\text{SSE}}(k, \ell)$ from Lemma SA-22, we have

$$\begin{aligned} & \mathbb{P}\left(\exists \ell \in [p] : \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{\substack{j \neq \ell \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, j), \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{SSE}}(k, \ell) > \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{SSE}}(k, \ell)\right) \\ & \geq p \sup_{z \in \mathbb{R}} \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{prox}}(k, j) < z - v_n, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{prox}}(k, 1) < z - v_n\right) \\ & \quad - p \mathbb{P}\left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{prox}}(k, j) < z + v_n, \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{prox}}(k, 1) < z + v_n\right), \end{aligned} \quad (\text{SA-35})$$

where $v_n = O_{\mathbb{P}}((\log \log(n))^{3/2} s_n^{-1/2})$. Here v_n denotes a nonnegative random envelope for the criterion approximation error. Under the chosen s_n , $v_n = o_{\mathbb{P}}(1)$, so there is a deterministic envelope $\bar{v}_n \downarrow 0$ with $\mathbb{P}(v_n \leq \bar{v}_n) \rightarrow 1$. In the following threshold comparisons we use this deterministic envelope, still denoted by v_n , losing only an $o(1)$ probability term.

Step 2: Gaussian approximation of IPW partial sums

Recall that

$$\begin{aligned} \mathcal{J}^{\text{prox}}(k, \ell) &= (1 - \xi) \frac{k(n-k)}{n} (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell))^2 \\ & \quad + \xi \frac{k(n-k)}{n} (\bar{\mu}_{L,1}(k, \ell) - \bar{\mu}_{R,1}(k, \ell))^2. \end{aligned} \quad (\text{SA-36})$$

We will show that the high-dimensional random vector Ξ formed by concatenating

$$\left(\sqrt{(1 - \xi) \frac{k(n-k)}{n}} (\bar{\mu}_{L,0}(k, \ell) - \bar{\mu}_{R,0}(k, \ell)) : k \in [n], \ell \in [p] \right)$$

and

$$\left(\sqrt{\xi \frac{k(n-k)}{n}} (\bar{\mu}_{L,1}(k, \ell) - \bar{\mu}_{R,1}(k, \ell)) : k \in [n], \ell \in [p] \right)$$

can be approximated by a Gaussian random vector with the same covariance structure. The proof will still be based on writing Ξ as $\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{C}_i$. Let

$$\mathbf{a}_i = \sqrt{n} \left(\left(\sqrt{\frac{n}{k(n-k)}} \left(\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n} \right) : r_n \leq k \leq n - r_n \right)^\top : 1 \leq \ell \leq p \right)^\top.$$

Then

$$\mathbf{C}_i = \begin{pmatrix} \mathbf{a}_i \sqrt{1 - \xi \frac{1-d_i}{1-\xi}} \varepsilon_i(0) \\ \mathbf{a}_i \sqrt{\xi \frac{d_i}{\xi}} \varepsilon_i(1) \end{pmatrix},$$

where $\#\pi^\ell$ denotes the inverse mapping of π^ℓ , as in the proof of Theorem SA-1.

The random vectors are $2np$ dimensional. For notational simplicity, denote by $\mathbf{e}_{t,k,\ell}$ the indicator of the position corresponding to $\sqrt{w_t \frac{k(n-k)}{n}} (\bar{\mu}_{L,t}(k, \ell) - \bar{\mu}_{R,t}(k, \ell))$, where $w_0 = 1 - \xi$ and $w_1 = \xi$, $t = 0, 1$, $k \in [n]$, $\ell \in [p]$.

However, the format of Equation (SA-36) induces a different geometry when approximating probabilities in Equation (SA-35). Instead of a high-dimensional CLT for hyperrectangles, we consider the class of simple convex sets [Chernozhukov et al., 2017, Section 3.1].

Let $\mathcal{J}$ be a subset of $[n] \times [p]$. Consider the class of closed convex sets $\mathcal{A}$ containing sets of the form

$$A = \{\mathbf{u} \in \mathbb{R}^{2np} : (\mathbf{e}_{0,k,\ell}^\top \mathbf{u}, \mathbf{e}_{1,k,\ell}^\top \mathbf{u}) \in B_2(s_{k,\ell}), s_{k,\ell} \in (0, n], (k, \ell) \in \mathcal{J}\}, \quad (\text{SA-37})$$

where $B_2(r)$ denotes the Euclidean ball centered at $\mathbf{0}$ with radius r in $\mathbb{R}^2$. That is, the class $\mathcal{A}$ contains intersections of cylinders $\{\mathbf{u} \in \mathbb{R}^{2np} : \|(\mathbf{e}_{j_1}^\top \mathbf{u}, \mathbf{e}_{j_2}^\top \mathbf{u})\|_2 \leq s\}$. For $z \in (0, n]$, the event in Equation (SA-36) (inside $\sup z$) can be characterized as the high-dimensional vector Ξ lying in a set in $\mathcal{A}$.

For each $A \in \mathcal{A}$, we consider its approximation by simple convex sets. For each $B_2(r)$, denote by $B_2^{\text{in},n}(r)$ and $B_2^{\text{out},n}(r)$ its inscribed and circumscribed regular n^2 -gon. Take $M = n^2|\mathcal{J}|$. Then for each $A \in \mathcal{A}$ of the form (SA-37), take

$$A^M = \{\mathbf{u} \in \mathbb{R}^{2np} : (\mathbf{e}_{0,k,\ell}^\top \mathbf{u}, \mathbf{e}_{1,k,\ell}^\top \mathbf{u}) \in B_2^{\text{in},n}(s_{k,\ell}), s_{k,\ell} \in (0, n], (k, \ell) \in \mathcal{J}\},$$

and

$$A^{M,\epsilon} = \{\mathbf{u} \in \mathbb{R}^{2np} : (\mathbf{e}_{0,k,\ell}^\top \mathbf{u}, \mathbf{e}_{1,k,\ell}^\top \mathbf{u}) \in B_2^{\text{out},n}(s_{k,\ell}), s_{k,\ell} \in (0, n], (k, \ell) \in \mathcal{J}\}.$$

Then $A^M \subseteq A \subseteq A^{M,\epsilon}$. Moreover, denote by $\mathcal{V}(A^M)$ the set consisting of M unit vectors that are outward normal to the facets of A^M . Then A^M can be alternatively characterized by

$$A^M = \cap_{\mathbf{v} \in \mathcal{V}(A^M)} \{\mathbf{w} \in \mathbb{R}^{2np} : \mathbf{w}^\top \mathbf{v} \leq S_A(\mathbf{v})\}, \quad S_A(\mathbf{v}) = \sup\{\mathbf{w}^\top \mathbf{v} : \mathbf{w} \in A^M\}.$$

Analogously, characterize $A^{M,\epsilon}$ by

$$A^{M,\epsilon} = \cap_{\mathbf{v} \in \mathcal{V}(A^M)} \{\mathbf{w} \in \mathbb{R}^{2np} : \mathbf{w}^\top \mathbf{v} \leq S_A(\mathbf{v}) + \epsilon_{\mathbf{v}}\}, \quad S_A(\mathbf{v}) = \sup\{\mathbf{w}^\top \mathbf{v} : \mathbf{w} \in A^M\},$$

where $\epsilon_{\mathbf{v}} \leq n^{-1}$ for large enough n . Thus our class $\mathcal{A}$ is a subclass of $\mathcal{A}^{\text{si}}(1, 3)$ (see [Chernozhukov et al., 2017, Section 3.1]). In the notation of Lemma SA-34, the ambient dimension is $m_0 = 2np$, the approximating polytopes have at most $M = n^2|\mathcal{J}| \leq (m_0 n)^3$ facets, $a = 1$, and $d = 3$. Since $m_0 \geq 3$ for all large n , the lemma's dimension condition is harmless. We check its conditions (M.1'), (M.2') and (E.1'). Let $\mathbf{v} \in \mathcal{V}(A^M)$. The definition of A^M implies $\mathbf{v} = v_{0,k,\ell} \mathbf{e}_{0,k,\ell} + v_{1,k,\ell} \mathbf{e}_{1,k,\ell}$ for some $(k, \ell) \in \mathcal{J}$, and $v_{0,k,\ell}^2 + v_{1,k,\ell}^2 = 1$. Let $\mathbf{v} \in \mathcal{V}(A^M)$.

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \mathbb{E}[|\mathbf{v}^\top \mathbf{C}_i|^2] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E} \left[\left(v_{0,k,\ell} \frac{n}{\sqrt{k(n-k)}} (\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n}) \sqrt{1-\xi} \frac{1-d_i}{1-\xi} \varepsilon_i(0) \right. \right. \\ & \quad \left. \left. + v_{1,k,\ell} \frac{n}{\sqrt{k(n-k)}} (\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n}) \sqrt{\xi} \frac{d_i}{\xi} \varepsilon_i(1) \right)^2 \right] \\ &= \frac{1}{n} \left(\frac{n}{\sqrt{k(n-k)}} \right)^2 \sum_{i=1}^n \left\{ v_{0,k,\ell}^2 \mathbb{E} \left[\left((\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n}) \sqrt{1-\xi} \frac{1-d_i}{1-\xi} \varepsilon_i(0) \right)^2 \right] \right. \\ & \quad \left. + v_{1,k,\ell}^2 \mathbb{E} \left[\left((\mathbf{1}(\#\pi^\ell(i) \leq k) - \frac{k}{n}) \sqrt{\xi} \frac{d_i}{\xi} \varepsilon_i(1) \right)^2 \right] \right\} \\ &\geq \min\{\mathbb{V}[\varepsilon_i(0)], \mathbb{V}[\varepsilon_i(1)]\}, \end{aligned}$$

which verifies (M.1'). The fact that only two entries of $\mathbf{v}$ are nonzero and $v_{0,k,\ell}^2 + v_{1,k,\ell}^2 = 1$ implies that

$$n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{v}^\top \mathbf{C}_i|^3] \leq 4n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{e}_{0,k,\ell}^\top \mathbf{C}_i|^3] + 4n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{e}_{1,k,\ell}^\top \mathbf{C}_i|^3] \lesssim \sqrt{n/r_n},$$

where the last inequality is from the calculation in Equation (SA-16), and this verifies (M.2') for the third moment. Moreover,

$$\begin{aligned} n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{v}^\top \mathbf{C}_i|^4] &\leq 8n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{e}_{0,k,\ell}^\top \mathbf{C}_i|^4] + 8n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{e}_{1,k,\ell}^\top \mathbf{C}_i|^4] \\ &\leq \sqrt{n/r_n} 8n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{e}_{0,k,\ell}^\top \mathbf{C}_i|^3] + \sqrt{n/r_n} 8n^{-1} \sum_{i=1}^n \mathbb{E}[|\mathbf{e}_{1,k,\ell}^\top \mathbf{C}_i|^3] \\ &\lesssim n/r_n. \end{aligned}$$

The same logic shows that $\mathbb{E}[\exp(|\mathbf{v}^\top \mathbf{C}_i|/(K\sqrt{n/r_n}))] \leq 2$, where K is an absolute constant. Putting these bounds together, we verify conditions (M.2') and (E.1') with $B_n = \sqrt{n/r_n}$. As in the proof of Theorem SA-1,

Lemma SA-34 is applied conditional on the covariate orderings with $X_i = \mathbf{C}_i$, with Gaussian analogues Y_i denoted by $\mathbf{D}_i$, and with the set class $\mathcal{A}$ above; the bound is then unconditioned. Hence there exist mean-zero random vectors $\mathbf{D}_i \sim N(\mathbf{0}, \mathbb{E}[\mathbf{C}_i \mathbf{C}_i^\top])$ such that

$$\sup_{A \in \mathcal{A}} |\mathbb{P}(n^{-1/2} \sum_{i=1}^n \mathbf{C}_i \in A) - \mathbb{P}(n^{-1/2} \sum_{i=1}^n \mathbf{D}_i \in A)| \lesssim \left(\frac{\log^7(n)}{r_n} \right)^{1/6}. \quad (\text{SA-38})$$

Step 3: Gaussian to Gaussian Approximation

For any $k_1, k_2 \in [n], \ell_1, \ell_2 \in [p]$, we have $\text{Cov}[\mathbf{e}_{0,k_1,\ell_1}^\top \mathbf{C}_i, \mathbf{e}_{1,k_2,\ell_2}^\top \mathbf{C}_i] = 0$. The same calculation as *Multivariate Case Step 2* for the proof of Theorem SA-1 allows us to replace $\mathbf{D}_i$ by another mean-zero Gaussian random vector $\mathbf{Z}_i$ such that

$$\text{Cov}[\mathbf{e}_{t_1,k_1,\ell_1}^\top \mathbf{Z}_i, \mathbf{e}_{t_2,k_2,\ell_2}^\top \mathbf{Z}_i] = \begin{cases} \text{Cov}[\mathbf{e}_{t_1,k_1,\ell_1}^\top \mathbf{D}_i, \mathbf{e}_{t_2,k_2,\ell_2}^\top \mathbf{D}_i], & \text{if } \ell_1 = \ell_2, \\ 0, & \text{otherwise.} \end{cases}$$

It remains to show $\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i$ is close to $\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{D}_i$, measured by probabilities of the sets in $\mathcal{A}$ defined at Equation (SA-37). Fix $A \in \mathcal{A}$ and use the polygonal approximation above. Since A^M is an intersection of M halfspaces, membership in A^M is equivalent to

$$\left(\mathbf{v}^\top n^{-1/2} \sum_{i=1}^n \mathbf{Z}_i \right)_{\mathbf{v} \in \mathcal{V}(A^M)} \leq \mathbf{t}_A$$

for a deterministic vector $\mathbf{t}_A \in \mathbb{R}^M$, and analogously for $\mathbf{D}_i$. Hence it is enough to show

$$\sup_{\mathbf{t} \in \mathbb{R}^M} \left| \mathbb{P}((\mathbf{v}^\top (\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i))_{\mathbf{v} \in \mathcal{V}(A^M)} \leq \mathbf{t}) - \mathbb{P}((\mathbf{v}^\top (\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{D}_i))_{\mathbf{v} \in \mathcal{V}(A^M)} \leq \mathbf{t}) \right| = o(1).$$

The definition of $\mathcal{A}$ in Equation (SA-37) implies that for any $A \in \mathcal{A}$ and $\mathbf{v} \in \mathcal{V}(A^M)$, there are $(k, \ell) \in \mathcal{J}$ and coefficients $v_{0,k,\ell}^2 + v_{1,k,\ell}^2 = 1$ such that $\mathbf{v} = v_{0,k,\ell} \mathbf{e}_{0,k,\ell} + v_{1,k,\ell} \mathbf{e}_{1,k,\ell}$. The treated and control coordinates are uncorrelated, so

$$\text{Cov} \left[\mathbf{e}_{0,k,\ell}^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i \right), \mathbf{e}_{1,k,\ell}^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i \right) \right] = \text{Cov} \left[\mathbf{e}_{0,k,\ell}^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{D}_i \right), \mathbf{e}_{1,k,\ell}^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{D}_i \right) \right] = 0,$$

and hence

$$\min_{\mathbf{v} \in \mathcal{V}(A^M)} \mathbb{V} \left[\mathbf{v}^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i \right) \right] \gtrsim 1.$$

Together with the same covariance comparison argument as Equation (SA-19), we know

$$\begin{aligned} & \max_{\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}(A^M)} \left| \text{Cov} \left[\mathbf{v}_1^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i \right), \mathbf{v}_2^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{Z}_i \right) \right] \right. \\ & \quad \left. - \text{Cov} \left[\mathbf{v}_1^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{D}_i \right), \mathbf{v}_2^\top \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{D}_i \right) \right] \right| = O \left(\sqrt{\frac{\log n}{r_n}} \right). \end{aligned}$$

This covariance bound holds on a permutation event with probability tending to one. On that event, Lemma SA-35 is applied to the projected Gaussian vectors

$$\mathbf{Z}_1 = \left(\mathbf{v}^\top n^{-1/2} \sum_{i=1}^n \mathbf{Z}_i \right)_{\mathbf{v} \in \mathcal{V}(A^M)}, \quad \mathbf{Z}_2 = \left(\mathbf{v}^\top n^{-1/2} \sum_{i=1}^n \mathbf{D}_i \right)_{\mathbf{v} \in \mathcal{V}(A^M)}$$

in dimension $M = n^2 |\mathcal{J}|$. For nonempty $\mathcal{J}$, $M \geq 3$ for all large n ; the empty case is trivial. The projected variances are bounded below by a positive constant, the maximal covariance discrepancy is $\Delta_n = O(\sqrt{\log n / r_n})$,

and the polygonal approximation changes each halfspace threshold by at most $\epsilon_{\mathbf{v}} \leq n^{-1}$. Applying the Gaussian to Gaussian comparison conditionally on the permutation event and then unconditioning, the complement contributes only $o(1)$. Applied to the M projected coordinates, Lemma SA-35 gives a bound of order $(\Delta_n \log^2 M)^{1/2} = O(\log(M)(\log n/r_n)^{1/4})$ for the difference over A^M . The same bound applies to $A^{M,\epsilon}$, and Nazarov's inequality controls the probability of the ϵ enlargement shell by a constant multiple of $\epsilon\sqrt{\log M}$ because of the variance lower bound. Since $M \lesssim n^3$ and $\epsilon \leq n^{-1}$, this shell term is $o(1)$. Therefore

$$\sup_{A \in \mathcal{A}} |\mathbb{P}(n^{-1/2} \sum_{i=1}^n \mathbf{Z}_i \in A) - \mathbb{P}(n^{-1/2} \sum_{i=1}^n \mathbf{D}_i \in A)| = O\left(\frac{\log^{5/4}(n)}{r_n^{1/4}}\right). \quad (\text{SA-39})$$

Step 4: Ornstein–Uhlenbeck Process Calculations

Returning to Equation (SA-35), consider

$$\mathcal{J}^{\text{Gauss}}(k, \ell) = (1 - \xi) \frac{k(n-k)}{n} (\tilde{\mu}_{L,0}(k, \ell) - \tilde{\mu}_{R,0}(k, \ell))^2 + \xi \frac{k(n-k)}{n} (\tilde{\mu}_{L,1}(k, \ell) - \tilde{\mu}_{R,1}(k, \ell))^2,$$

with

$$\begin{aligned} \tilde{\mu}_{L,0}(k, \ell) &= \frac{1}{k} \sum_{i \leq k} u_{\pi_\ell(i)}, & \tilde{\mu}_{L,1}(k, \ell) &= \frac{1}{k} \sum_{i \leq k} v_{\pi_\ell(i)}, \\ \tilde{\mu}_{R,0}(k, \ell) &= \frac{1}{n-k} \sum_{i > k} u_{\pi_\ell(i)}, & \tilde{\mu}_{R,1}(k, \ell) &= \frac{1}{n-k} \sum_{i > k} v_{\pi_\ell(i)}. \end{aligned}$$

Here (u_i) and (v_i) are independent Gaussian sequences with variances $\sigma_0^2/(1-\xi)$ and σ_1^2/ξ , respectively, where $\sigma_d^2 = \mathbb{V}[\varepsilon_i(d)]$. Equivalently, the scaled control and treated bridge coordinates appearing in $\mathcal{J}^{\text{Gauss}}$ have marginal variances σ_0^2 and σ_1^2 . The restriction of the threshold supremum to $z \in [-n, n]$ is without loss up to $o(1)$. The criteria are nonnegative, so thresholds $z < 0$ contribute nothing to the displayed lower bound. Moreover the Darling–Erdős normalization below implies that all relevant Gaussian maxima are $O_{\mathbb{P}}(\log \log n)$, and hence the probability that any displayed maximum exceeds n is $o(1)$. With the deterministic buffer convention just described, Equations (SA-38) and (SA-39) imply that

$$\begin{aligned} & \sup_{z \in [-n, n]} \mathbb{P} \left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{prox}}(k, j) < z - v_n, \max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{prox}}(k, 1) < z - v_n \right) \\ & - \mathbb{P} \left(\max_{\substack{j \neq 1 \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{prox}}(k, j) < z + v_n, \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{prox}}(k, 1) < z + v_n \right) \\ & = \sup_{z \in [-n, n]} \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) < z - v_n \right)^{p-1} \mathbb{P} \left(\max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{Gauss}}(k, 1) < z - v_n \right) \\ & - \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) < z + v_n \right)^{p-1} \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) < z + v_n \right) + o(1). \end{aligned}$$

The same argument as [Csörgö and Horváth, 1997, (A.4.25) to (A.4.37)] shows that there exist two independent standard Brownian bridges over $[0, 1]$, $B_{n,L}$ and $B_{n,R}$, for each n , such that

$$\begin{aligned} & \left| \max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) - \sup_{t \in [s_n/n, 1-s_n/n]} \left\{ \sigma_0^2 \frac{B_{n,L}^2}{t(1-t)} + \sigma_1^2 \frac{B_{n,R}^2}{t(1-t)} \right\} \right| = \epsilon_n, \\ & \left| \max_{k \in [s_n, n-s_n] \setminus [n^a, n^b]} \mathcal{J}^{\text{Gauss}}(k, 1) - \sup_{t \in [s_n/n, 1-s_n/n] \setminus [n^{a-1}, n^{b-1}]} \left\{ \sigma_0^2 \frac{B_{n,L}^2}{t(1-t)} + \sigma_1^2 \frac{B_{n,R}^2}{t(1-t)} \right\} \right| = \epsilon_n, \end{aligned}$$

with $\epsilon_n = o_{\mathbb{P}}(1)$. Let $\{U_L(t) : t \in \mathbb{R}\}$ and $\{U_R(t) : t \in \mathbb{R}\}$ be two independent O-U processes with $\mathbb{E}[U_j(t)] = 0$ and $\mathbb{E}[U_j(s)U_j(t)] = e^{-|s-t|/2}$, $j = L, R$. Then

$$\left\{ \left(\frac{B_{n,L}}{\sqrt{t(1-t)}}, \frac{B_{n,R}}{\sqrt{t(1-t)}} \right) : t \in [0, 1] \right\} \stackrel{d}{=} \{(U_L(\log(t/(1-t))), U_R(\log(t/(1-t)))) : t \in [0, 1]\}.$$

Take

$$Q_\Lambda(t) = \sigma_0^2 U_L(t)^2 + \sigma_1^2 U_R(t)^2, \quad \Lambda = (\sigma_0^2, \sigma_1^2).$$

Then a time change and stationarity of the O-U process implies

$$\begin{aligned} & \mathbb{P}\left(\sup_{t \in [1/n, 1-1/n] \setminus [n^{a-1}, n^{b-1}]} \left\{ \sigma_0^2 \frac{B_{n,L}^2}{t(1-t)} + \sigma_1^2 \frac{B_{n,R}^2}{t(1-t)} \right\} \leq y\right) \\ &= \mathbb{P}\left(\sup_{\substack{-\log(n-1) \leq s < \log \frac{n^{a-1}}{1-n^{a-1}} \\ \text{or } \log \frac{n^{b-1}}{1-n^{b-1}} < s \leq \log(n-1)}} Q_\Lambda(s) \leq y\right) \\ &= \mathbb{P}\left(\sup_{\substack{0 \leq s < \log \frac{n^{a-1}(n-1)}{1-n^{a-1}} \\ \text{or } \log \frac{n^{b-1}(n-1)}{1-n^{b-1}} < s \leq 2 \log(n-1)}} Q_\Lambda(s) \leq y\right), \end{aligned}$$

and

$$\mathbb{P}\left(\sup_{t \in [1/n, 1-1/n]} \left\{ \sigma_0^2 \frac{B_{n,L}^2}{t(1-t)} + \sigma_1^2 \frac{B_{n,R}^2}{t(1-t)} \right\} \leq y\right) = \mathbb{P}\left(\sup_{0 \leq s < 2 \log(n-1)} Q_\Lambda(s) \leq y\right).$$

Lemma SA-38 gives thresholds $w_n(u; \Lambda)$ and $\kappa_\Lambda > 0$ such that, for each fixed $c > 0$ and bounded u ,

$$\mathbb{P}\left(\sup_{0 \leq t < c \log(n)} Q_\Lambda(t) \leq w_n(u; \Lambda) + \epsilon_n\right) = \exp\{-c\kappa_\Lambda e^{-u}\} + o(1).$$

For finite grids, the events $\{\sup_{s \in I} Q_\Lambda(s) \leq y\}$ are symmetric convex events in the underlying Gaussian vector; the continuous-time statement follows by monotone approximation over dense grids. Therefore, Gaussian correlation inequality [Latała and Matlak, 2017, Remark 3 (i)] and stationarity of the O-U process imply

$$\begin{aligned} & \mathbb{P}\left(\sup_{\substack{0 \leq s < \log \frac{n^{a-1}(n-1)}{1-n^{a-1}} \\ \text{or } \log \frac{n^{b-1}(n-1)}{1-n^{b-1}} < s \leq 2 \log(n-1)}} Q_\Lambda(s) < w_n(u; \Lambda) + \epsilon_n\right) \\ & \geq \mathbb{P}\left(\sup_{0 \leq s < \log \frac{n^{a-1}(n-1)}{1-n^{a-1}}} Q_\Lambda(s) < w_n(u; \Lambda) + \epsilon_n\right) \\ & \quad \cdot \mathbb{P}\left(\sup_{0 < s \leq \log(n^{1-b}(n-1)(1-n^{b-1}))} Q_\Lambda(s) < w_n(u; \Lambda) + \epsilon_n\right) \\ & = \exp\{-(2 - (b-a))\kappa_\Lambda e^{-u}\} + o(1). \end{aligned}$$

Because $v_n = o(1)$, replacing $w_n(u; \Lambda)$ by $w_n(u; \Lambda) \pm v_n$ changes the u scale threshold by $o(1)$ uniformly over bounded u intervals, and the limiting distribution in Lemma SA-38 is continuous in u . Let $\alpha = (b-a)/2$. After the change of variable $s = 2\kappa_\Lambda e^{-u} \in (0, \infty)$, the limiting objective below is

$$e^{-(p-\alpha)s} - e^{-ps}.$$

It is maximized at $s^* = \alpha^{-1} \log\{p/(p-\alpha)\}$ and equals

$$\frac{\alpha}{p} \left(1 - \frac{\alpha}{p}\right)^{p/\alpha-1} = \frac{b-a}{2p} \left(1 - \frac{b-a}{2p}\right)^{\frac{2p}{b-a}-1}.$$

Choosing the corresponding u^* in the bounded threshold window, we obtain

$$\begin{aligned}
& \liminf_{n \rightarrow \infty} \sup_{z \in [-n, n]} \left\{ \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) < z - v_n \right)^{p-1} \mathbb{P} \left(\max_{\substack{k \notin [n^a, n^b] \\ k \in [s_n, n-s_n]}} \mathcal{J}^{\text{Gauss}}(k, 1) < z - v_n \right) \right. \\
& \quad \left. - \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) < z + v_n \right)^{p-1} \mathbb{P} \left(\max_{k \in [s_n, n-s_n]} \mathcal{J}^{\text{Gauss}}(k, 1) < z + v_n \right) \right\} \\
& \geq \sup_{u \in \mathbb{R}} \exp\{-2(p-1)\kappa_\Lambda e^{-u}\} \left(\exp\{-(2-(b-a))\kappa_\Lambda e^{-u}\} - \exp\{-2\kappa_\Lambda e^{-u}\} \right) \\
& = \frac{b-a}{2p} \left(1 - \frac{b-a}{2p} \right)^{\frac{2p}{b-a}-1} \\
& \geq \frac{b-a}{2pe}.
\end{aligned}$$

The same argument applied to the reflected ordering gives the right boundary counterpart. Thus, for any $0 < a < b < 1$ and $\ell \in [p]$, we have

$$\liminf_{n \rightarrow \infty} \mathbb{P}(n^a \leq \hat{i}_{\text{SSE}} \leq n^b, \hat{j}_{\text{SSE}} = \ell) \geq \frac{b-a}{2pe}, \quad \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i}_{\text{SSE}} \leq n - n^a, \hat{j}_{\text{SSE}} = \ell) \geq \frac{b-a}{2pe}.$$

SA-5.27 Proof of Theorem SA-26

Lemma SA-25, summed over the left and right boundary events and over coordinates, gives the uniform statement. The boundary statement follows from the same order statistic step as in the DIM proof, and the constants absorb the directional transfer probability, the split coordinate factor, and the treatment probability factors.

SA-5.28 Proof of Theorem SA-27

Set $a = b/2$. The only additional step relative to the DIM proof is to obtain the same type of root child event for the tree grown by the SSE rule. Lemma SA-25, summed over the two boundary sides and over coordinates, gives constants $c_{\text{root}}, q_{\text{root}} > 0$ and an event $\mathcal{E}_{\text{SSE}, n}$ with

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\mathcal{E}_{\text{SSE}, n}) \geq q_{\text{root}}(b-a),$$

such that, on $\mathcal{E}_{\text{SSE}, n}$, the root split selected by the SSE rule has a child $\mathbf{t}_n$ satisfying

$$M_n = n(\mathbf{t}_n) \in [n^a, n^b], \quad n_0(\mathbf{t}_n) \wedge n_1(\mathbf{t}_n) > 0, \quad M_n \Delta(\mathbf{t}_n)^2 \geq c_{\text{root}}^2 \log \log n.$$

The event records both the imbalanced size of the selected root child and a large CATE contrast on that child. The latter conclusion is the directional transfer from the bivariate SSE split criterion to the scalar contrast $\Delta(\mathbf{t}_n)$.

Let $\mathcal{P}(\mathbf{t}_n)$ be the terminal descendants of $\mathbf{t}_n$ in the final tree grown by the SSE rule. The descendants are again sample rectangles, they partition $\mathbf{t}_n$, and validity gives positive treated and control counts in every terminal node. Since the tree has depth at most K , $|\mathcal{P}(\mathbf{t}_n)| \leq 2^K$. The assumed depth condition implies the leaf count condition in Lemma SA-40 with $a = b/2$. Applying Corollary SA-41 with $c_0 = c_{\text{root}}$ gives

$$\mathbb{P} \left(\mathcal{E}_{\text{SSE}, n}, \max_{\mathbf{t}' \in \mathcal{P}(\mathbf{t}_n)} |\Delta(\mathbf{t}')| \geq \frac{c_{\text{root}}}{2} n^{-b/2} \sqrt{\log \log n} \right) \geq \mathbb{P}(\mathcal{E}_{\text{SSE}, n}) - o(1).$$

For every terminal descendant $\mathbf{t}'$ and every $\mathbf{x} \in \mathbf{t}'$,

$$\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau = \Delta(\mathbf{t}'),$$

because Definition SA-6 computes terminal treatment effects on a tree grown by the SSE rule using the DIM estimator. Taking the supremum over $\mathbf{x} \in \mathcal{X}$ and taking the liminf gives

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_{\text{SSE}}(\mathbf{x}) - \tau| \geq \frac{c_{\text{root}}}{2} n^{-b/2} \sqrt{\log \log n} \right) \geq q_{\text{root}}(b-a) = \frac{q_{\text{root}}}{2} b.$$

Thus the theorem holds with $c_{\text{SSE}} = c_{\text{root}}/2$ and $q_{\text{SSE}} = q_{\text{root}}/2$.

SA-5.29 Proof of Theorem SA-28

The empirical risk minimization property (SA-33) still holds for the fitted partition. Conditional on that partition, the terminal estimator is the same leafwise least squares treatment effect coefficient used in the DIM proof. The deterministic comparison with the causal prediction error and the entropy bound from Theorem SA-19 therefore yield the displayed $2^K \log^4(n) \log(np)/n$ rate.

SA-5.30 Proof of Theorem SA-29

The proof of Theorem SA-20 applies after replacing Theorem SA-16 by Theorem SA-24. The treatment-arm positivity factor is absorbed into the constant C_2 .

SA-5.31 Proof of Theorem SA-30

Conditional on the SSE construction fold, the honest estimation fold is independent of the selected partition and the terminal estimator is again the leafwise least squares treatment effect coefficient. Thus the conditional fixed partition argument from Theorem SA-21 applies without modification: the causal prediction class has two constants per leaf, the number of leaves is bounded by 2^K , and the truncation step is carried out on the independent estimation fold. This gives the same $2^K \log^5(n)/n$ expectation and probability bounds after unconditioning.

SA-5.32 Proof of Lemma SA-31

First, consider the honest estimator. This part uses only independence of the estimation fold from the constructed partition; the joint central symmetry assumption is not needed until the NSS argument below. Conditional on the tree partition and on the estimation fold covariates, the treatment assignments and potential outcome errors used for final estimation are independent of the tree construction. Let $\mathbf{t}_{\text{HON}}(\mathbf{x})$ be the terminal node containing $\mathbf{x}$. For IPW, the estimator is set to zero when $n(\mathbf{t}_{\text{HON}}(\mathbf{x})) = 0$, and otherwise has conditional expectation τ . Therefore

$$\mathbb{E}[\tilde{\tau}_{\text{IPW}}(\mathbf{x}; K)] = \tau \mathbb{P}(n(\mathbf{t}_{\text{HON}}(\mathbf{x})) > 0) = \tau - \tau \mathbb{P}(n(\mathbf{t}_{\text{HON}}(\mathbf{x})) = 0).$$

For DIM and SSE, the terminal node estimator is set to zero when either treatment arm is absent, and otherwise has conditional expectation τ . Thus

$$\mathbb{E}[\tilde{\tau}_l(\mathbf{x}; K)] = \tau \mathbb{P}(n_0(\mathbf{t}_{\text{HON}}(\mathbf{x})) > 0, n_1(\mathbf{t}_{\text{HON}}(\mathbf{x})) > 0), \quad l \in \{\text{DIM}, \text{SSE}\},$$

which gives the stated empty cell bias expression.

Finally, we consider NSS under Assumption SA-2. We first treat DIM and SSE under the joint central symmetry assumption on $(\varepsilon_i(0), \varepsilon_i(1))$. We will use an induction assumption.

Base case: $K = 1$. Because μ_0 and μ_1 are constant, the DIM and SSE splitting criteria from Definition SA-5 can be written as

$$\begin{aligned} \text{DIM : } \quad & \frac{n(\mathbf{t}_L)n(\mathbf{t}_R)}{n(\mathbf{t})} \left(\frac{1}{n_1(\mathbf{t}_L)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_L} d_i \varepsilon_i(1) - \frac{1}{n_0(\mathbf{t}_L)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_L} (1 - d_i) \varepsilon_i(0) \right. \\ & \quad \left. - \frac{1}{n_1(\mathbf{t}_R)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_R} d_i \varepsilon_i(1) + \frac{1}{n_0(\mathbf{t}_R)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_R} (1 - d_i) \varepsilon_i(0) \right)^2, \end{aligned} \quad (\text{SA-40})$$

and

$$\begin{aligned} \text{SSE : } \quad & \frac{n_1(\mathbf{t}_L)n_1(\mathbf{t}_R)}{n_1(\mathbf{t})} \left(\frac{1}{n_1(\mathbf{t}_L)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_L} d_i \varepsilon_i(1) - \frac{1}{n_1(\mathbf{t}_R)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_R} d_i \varepsilon_i(1) \right)^2 \\ & + \frac{n_0(\mathbf{t}_L)n_0(\mathbf{t}_R)}{n_0(\mathbf{t})} \left(\frac{1}{n_0(\mathbf{t}_L)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_L} (1 - d_i) \varepsilon_i(0) - \frac{1}{n_0(\mathbf{t}_R)} \sum_{i:\mathbf{x}_i \in \mathbf{t}_R} (1 - d_i) \varepsilon_i(0) \right)^2. \end{aligned} \quad (\text{SA-41})$$

Denote the vector $\boldsymbol{\varepsilon} = (\varepsilon_1(0), \varepsilon_1(1), \dots, \varepsilon_n(0), \varepsilon_n(1))$. Joint central symmetry and independence from covariates and treatment assignments imply $\boldsymbol{\varepsilon} \stackrel{d}{=} -\boldsymbol{\varepsilon}$ conditional on $(\mathbf{X}, \mathbf{d})$. For the DIM and SSE criteria, for

any fixed covariate array $\mathbf{X}$, treatment assignment $\mathbf{d} = (d_1, \dots, d_n)$, and valid candidate split $\mathbf{t}_L, \mathbf{t}_R$, $\boldsymbol{\varepsilon} = \mathbf{u}$ and $\boldsymbol{\varepsilon} = -\mathbf{u}$ give the same criterion value. The valid candidate set itself depends only on covariates and treatment counts, so it is unchanged by the sign flip. With deterministic tie breaking, the selected split is therefore invariant to the global sign flip $\boldsymbol{\varepsilon} \mapsto -\boldsymbol{\varepsilon}$. More formally, every event generated by the selected split is sign invariant conditional on $(\mathbf{X}, \mathbf{d})$; hence for every Borel set B and every such event A with positive probability,

$$\mathbb{P}(\boldsymbol{\varepsilon} \in B \mid A, \mathbf{X}, \mathbf{d}) = \mathbb{P}(-\boldsymbol{\varepsilon} \in B \mid A, \mathbf{X}, \mathbf{d}).$$

Thus, conditional on $(\mathbf{X}, \mathbf{d})$ and on the selected split, $\boldsymbol{\varepsilon}$ remains symmetrically distributed around zero. On the event that the terminal node denominator required by the DIM or SSE terminal estimator is positive, the centered terminal node estimator is odd in $\boldsymbol{\varepsilon}$ and has conditional expectation zero; on the complementary event the estimator is defined to be zero, which is also sign invariant. This gives the stated empty cell expression for DIM and SSE, and exact unbiasedness when the relevant denominators are positive almost surely.

Induction step: $K \geq 2$. Each leaf node $\mathbf{t}$ in layer $K - 1$ is further partitioned into $\mathbf{t}_L$ and $\mathbf{t}_R$ such that Equations (SA-40) and (SA-41) are maximized over valid candidates. The induction hypothesis is that, conditional on $\mathbf{X}, \mathbf{d}$, and all leaves in layer $K - 1$, $\boldsymbol{\varepsilon}$ is symmetrically distributed around zero. The same sign invariance and deterministic tie breaking argument applies within each leaf, so conditioning on the layer K partition preserves central symmetry of $\boldsymbol{\varepsilon}$. The centered terminal node estimator is odd on the event that its denominators are positive and is set to zero otherwise, which proves the same empty cell expression and the exact unbiasedness statement for DIM and SSE under positive terminal denominators.

For IPW, put $\tilde{\varepsilon}_i = y_i(d_i - \xi) / \{\xi(1 - \xi)\} - \tau$. The IPW split criterion is the CART variance maximization criterion applied to the pseudo outcomes $\tau + \tilde{\varepsilon}_i$, so the constant τ cancels from every candidate split. If $(\tilde{\varepsilon}_1, \dots, \tilde{\varepsilon}_n)$ is centrally symmetric conditional on the treatment assignments, then, because the transformed residuals are independent of the covariates conditional on the treatment assignments under the stated symmetry condition, the same deterministic tie breaking and sign invariance argument applies to the IPW partition, now with $\tilde{\varepsilon}$ in place of $\boldsymbol{\varepsilon}$. Conditional on the selected partition, the centered terminal IPW estimator is the average of $\tilde{\varepsilon}_i$ over the selected terminal node, and is set to zero if that node is empty. Hence the displayed IPW empty cell expression and the exact unbiasedness statement under positive terminal counts follow.

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